THE PERSONALIZED MATH SERIES

FOURTH EDITION / PRACTICE + EXAMS

Mathematics for Quantitative Political Science

From derivative rules to defensible empirical research

01 / Learn the logic 02 / Work the math 03 / Apply it

33 chapters + full problem sets + reviews + Z-table

Calculus / Probability / Inference / Econometrics / Game theory

October 2026 • Personalized research methods edition

Preface: why this book exists

This is the revised fourth edition of a *working textbook*, not a reading list. It pairs full derivations with quick checks, visual explanations, a printed normal-distribution table, an offline interactive learning lab, and occasional informal remarks—because mathematics is allowed to be interesting. It was designed around three diagnostic sessions: elementary statistical reasoning was substantially stronger than calculus and matrix derivations; introductory regression and event-study interpretation were familiar, but omitted-variable bias, endogeneity and difference-in-differences needed explicit teaching. Game theory was declared unfamiliar and is therefore taught from first principles in a separate supplement rather than used to assess your initial ability.

A fast trick in an orange box is useful for checking the result, but not a substitute for understanding the assumptions. Visuals are schematic and are drawn directly in LaTeX. The PDF has clickable contents, navigation, and links to the formula and Z-score references. A separately included HTML companion supplies live sliders and instant probability calculations; ordinary PDF readers do not provide those live controls.

The book follows one recurring research problem: *How are American and allied semiconductor export-control announcements associated with returns of Japanese and European firms, and under what assumptions could this association be interpreted causally?* Every example uses **illustrative or simulated numbers**, never real company observations. Results are exercises in methods, not empirical findings.

What you will be able to do: read an equation with its units and indices; carry out essential derivatives and matrix operations; derive OLS; state statistical assumptions precisely; estimate expected and abnormal returns; build a transparent event study; distinguish correlation from credible causal identification; use R to reproduce calculations; and explain limitations in graduate research writing. You will not become an expert in every method merely by finishing one book. Independent implementation and critical reading remain goals that require practice.

How to study, with only 2–3 hours a week: read for 30 minutes, reproduce a derivation for 45 minutes, answer multiple-choice checkpoints for 30 minutes, try the R example for 30 minutes, and spend 15 minutes recording mistakes. Checkpoints are short and have an answer key in the appendix. Cover the key until you have made your selections. For stronger diagnostic validity, use additional new questions in later review rather than remembering old answer letters. Skip quickly through secure material and spend the saved time on optimization, inverses, and OLS.

New in the fourth edition: practice with feedback

Every core chapter and every game-theory chapter now adds a separate printable *Practice Studio* with four further problems of increasing difficulty. Most chapters therefore contain six or seven multiple-choice questions including the original checkpoint. Mark your answer before

consulting the fully worked solution companion. Mathematical problems are not graded by guesswork: the new answer key shows intermediate calculations, unit checks, assumptions and interpretation, with longer reasoning for Challenge questions.

Six core review tests and a game-theory review assess several chapters together; a cumulative 24-question final examination covers 20 core mathematics/research-methods questions and four optional game-theory questions. Those four are a supplementary track, not a prerequisite for your econometric research. Use the review tests after the corresponding chapters. Do not attempt the final exam before taking the practice sections seriously.

Difficulty legend: Foundation tests a single idea; **Application** combines concepts or units; **Challenge** requires a derivation, careful interpretation, or identification reasoning. A correct answer is only the beginning: try explaining why the other options fail.

Three useful ways to read a mathematics chapter

First pass—the story. Read the intuition, look at the picture, and explain the idea in everyday language. For example, "a partial derivative changes one input while keeping the rest fixed."

Second pass—the calculation. Reproduce a worked example without looking at the intermediate steps. The orange *Fast lane* boxes provide shortcuts and sanity checks; check that the shortcut agrees with the full method.

Third pass—the researcher. Ask what each symbol means in a real data table. What is observed? What is modeled? Which assumption makes an interpretation legitimate? Try the small R examples and complete the multiple-choice checkpoints before opening the answer key.

When an equation seems like a secret spell, stop and trace it back to the preceding one. Your task is to understand its *journey*, not memorize its shape.

Notation and conventions

i indexes firms or observations; t indexes dates. Y is an outcome and X an explanatory variable, neither automatically causal. R_{it} denotes a stock return in *decimal units* unless a percentage sign is explicitly shown. Thus -0.04 means -4 percentage points in a return difference, not a 4% price difference. Hatted quantities are estimators or fitted values; bold lowercase vectors are columns and bold uppercase letters are matrices. $\mathbb{E}[\cdot]$ and $\mathrm{Var}(\cdot)$ denote expectation and variance. All prices, treatment assignments, policy examples and corporate observations in exercises are constructed for teaching.

Study plan: 48 core weeks, at approximately 2.5 hours/week

Weeks	Chapters	Focus
1–7	1–4	Algebra, derivatives, integration, partial derivatives and optimization
8–12	5–7	Matrices, rank, projections and quadratic forms
13–20	8–12	Probability refresh, distributions, sampling and inference

Weeks	Chapters	Focus
21–32	13–17	OLS derivations, omitted variables, exogeneity, uncertainty and diagnostics
33–44	18–23	Event studies, factor models, panels, difference-in-differences, time series
45–48	24–25	Reproducible replication and research-ready interpretation

The game theory supplement follows Chapters 5, 8 and 9 (or runs in parallel if your class requires it) and requires *additional* study hours. The weekly calendar is a planning tool, not a mastery guarantee.

START HERE / JUMP TO A REFERENCE

STANDARD NORMAL Z-TABLE | FORMULA ATLAS | HOW TO READ A Z-SCORE

Use the first two links like bookmarks while solving the chapter exercises. The printed tables use the *left-tail* convention $\Phi(z) = P(Z \leq z)$; other textbooks sometimes print area from zero to z instead.

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Part I

Mathematical Foundations

CHAPTER 1

Algebra, Functions, and Mathematical Notation

Intuition

An equation is a compressed argument. Before solving it, identify which symbols are variables, which are constants, and which units they represent. Algebra allows us to rewrite the argument without changing its meaning. Your diagnostic showed that recalling an algebraic rule and carrying it out reliably are not the same skill.

1.1 Equations and equivalent operations

Consider $\frac{2x-3}{4} + \frac{x+1}{2} = 5$. Multiply *every term* by four:

$$\begin{aligned} 2x - 3 + 2(x+1) &= 20 \\ 2x - 3 + 2x + 2 &= 20 \\ 4x - 1 &= 20 \\ \boxed{x = 21/4} & \end{aligned}$$

A common error is to multiply only one fraction. Verification by substitution is often faster than debugging a long solution: $(2(21/4) - 3)/4 + (21/4 + 1)/2 = 5$.

1.2 Powers, logarithms, and domain restrictions

The laws $a^m a^n = a^{m+n}$, $\log(ab) = \log a + \log b$ (for $a, b > 0$) and $\log(a^b) = b \log a$ are indispensable when estimating log-linear economic relationships. Solve

$$\ln x + \ln(x - 2) = \ln 8.$$

The domain is $x > 2$; combining logs gives $x(x-2) = 8$, so $(x-4)(x+2) = 0$. Only $\boxed{x = 4}$ is admissible; $x = -2$ violates the original logarithms.

Semiconductor research connection

If $\ln(\text{sales}_{it}) = a + bD_{it} + u_{it}$ and D is a zero-one indicator, the exact multiplicative comparison in modeled sales is e^b, and the exact percentage comparison is $100(e^b - 1)\%$. The approximation $100b\%$ is useful only when b is small. This is not automatically a causal effect.

1.3 Indices, summation, means, and units

For observations $x_1, \ldots, x_n$, $\sum_{i=1}^{n} x_i$ means add each x_i from the first through the nth. The sample mean is $\bar{x} = n^{-1} \sum_i x_i$. For $x = (2, 4, 6, 8)$, $\bar{x} = 5$, and

$$\sum_{i=1}^{4} (x_i - \bar{x})^2 = 9 + 1 + 1 + 9 = 20.$$

The sum measures total squared dispersion about the sample mean; dividing by $n-1$ produces the conventional unbiased sample-variance estimator under independent sampling.

1.4 Functions and slopes

A function $f : X \to Y$ maps each permitted input to exactly one output. The slope of $f(x) = a + bx$ is b; $f(x + h) - f(x) = bh$. For a curved function, slope can depend on the current input. An equation such as $R_{it} = \alpha + \beta D_{it} + \varepsilon_{it}$ is a model of conditional relations, not proof of a policy mechanism.

Worked example

Let $f(x) = 2 + 0.8x$. Changing x from 2 to 3 changes f from 3.6 to 4.4, an increase of 0.8 *units*, not necessarily 80%. Always check the unit of x, $f(x)$, and the coefficient.

```
x <- c(2,4,6,8)
mean(x)
sum((x - mean(x))^2)
log(1.1)    # natural logarithm
exp(log(1.1))
```

A faster route: simplify, solve, verify

Fast lane: a useful calculation trick

Before touching a complicated fraction, ask: *what is the smallest common denominator?* Multiplying every term by it turns a fraction equation into an ordinary equation. Keep parentheses until after multiplication. Finally, substitute the answer into the *original* equation; it catches lost signs in seconds.

Worked example

Suppose $\frac{x-1}{3} + \frac{x+2}{6} = 4$. Multiply by 6 to obtain $2(x-1) + (x+2) = 24$, so $3x = 24$ and $x = 8$. The fast check: $(8-1)/3 + (8+2)/6 = 7/3 + 5/3 = 4$. No twenty-line calculation required.

Fast lane: a useful calculation trick

Mental check for summations: $\sum_i (x_i - \bar{x}) = 0$ *exactly.* If your list of deviations does not sum to zero, something went wrong before you even squared a number. Also use $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$ as a second route to the same result.

REAL LIFE / The coffee receipt

You buy three coffees and a sandwich. The bill is 1,950 yen, and the sandwich costs 750 yen. The unknown coffee price is not mysterious: $3x+750 = 1950$. Solving gives $x = 400$ yen. A regression equation works with the same discipline: label every symbol and do operations to the whole equation, not just the convenient part.

> *"An equation is a receipt with some prices covered up; algebra uncovers them without changing the total.*

Multiple-choice checkpoint

1. **[Foundation]** Solve $3(x-2) = x+4$.
   A. $x = 3$ B. $x = 4$ C. $x = 5$ D. $x = 6$.
2. **[Application]** If $b = 0.10$ in a log-outcome model with a binary regressor, the exact model-implied percentage difference is closest to
   A. 10.00% B. 10.52% C. 0.10% D. 11.50%.
3. **[Challenge]** For $x = (1, 3, 5)$, what is $\sum_i (x_i - \bar{x})^2$?
   A. 4 B. 6 C. 8 D. 9.

Answers and explanations: Appendix A, Chapter 1.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Solve $5(x-1) = 2x + 10$ for x.

A. 3 **B.** 5
C. 7 **D.** 15

P2. Foundation. A commuter cycles 8 km in 32 minutes. What is average speed in km/h?

A. 4 **B.** 12
C. 15 **D.** 20

P3. Application. Solve $\ln x + \ln(x-1) = \ln 6$ over the real numbers.

A. $x = 2$ **B.** $x = 3$
C. $x = -2$ **D.** $x = 6$

P4. Challenge. A model predicts $\widehat{Y} = 5 + 0.3X$, where Y is measured in thousands of yen and X in hours. Increasing X by 4 hours changes the prediction by how much?

A. 0.3 yen **B.** 1.2 thousand yen
C. 5.3 thousand yen **D.** 20 thousand yen

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 2

Differentiation: Rules You Can Reconstruct

Intuition

A derivative measures how fast an output changes when an input moves by a small amount. You remembered polynomial differentiation, but your handwritten work showed uncertainty with product and chain rules. We will derive those rules before applying them.

2.1 Derivatives from first principles

The derivative, when it exists, is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

For $f(x) = x^2$, expand $(x+h)^2 = x^2 + 2xh + h^2$. The numerator is $2xh + h^2$, and division by h gives $2x + h$. Taking the limit yields $f'(x) = 2x$. This is the logic behind differentiation, rather than merely a rule to memorize.

2.2 Rules and what each one means

The power rule is $\frac{d}{dx}x^p = px^{p-1}$ for real powers where defined. Linearity gives $(af + bg)' = af' + bg'$. The product and chain rules are

$$\begin{aligned} (u(x)v(x))' &= u'(x)v(x) + u(x)v'(x), \\ [g(h(x))]' &= g'(h(x))h'(x), \\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2} \quad (v \neq 0). \end{aligned}$$

Why two terms in the product rule? When both factors change, $\Delta(uv) = u\Delta v + v\Delta u + \Delta u \Delta v$. After dividing by Δx, the final cross-product vanishes in the limit for differentiable factors.

Worked example

For $f(x) = x^2 e^x$, both x^2 and e^x change:

$$f'(x) = 2xe^x + x^2 e^x = e^x(2x + x^2).$$

For $g(x) = (3x^2 + 1)^4$, identify outer z^4 and inner $z = 3x^2 + 1$:

$$g'(x) = 4(3x^2 + 1)^3 \times 6x = 24x(3x^2 + 1)^3.$$

2.3 Exponential and logarithmic derivatives

Since $(e^x)' = e^x$, we have $(e^{h(x)})' = e^{h(x)}h'(x)$. Since $(\ln x)' = 1/x$ for $x > 0$, $(\ln h(x))' = h'(x)/h(x)$ where $h(x) > 0$. A common example is $f(x) = \ln(1 + x^2)$, giving $2x/(1 + x^2)$.

2.4 Marginal interpretation and elasticities

If revenue is $V(q)$, $V'(q)$ is marginal revenue. If $C(q)$ is cost, $C'(q)$ is marginal cost. The elasticity of positive $y = f(x)$ with respect to positive x is

$$\frac{d\ln y}{d\ln x} = \frac{x}{y}\frac{dy}{dx}.$$

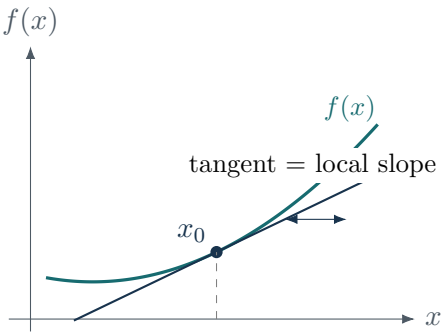
An elasticity is unitless, unlike a derivative. Confusing percentage and level changes can cause large interpretation errors in applied papers.

```
f <- function(x) x^2 * exp(x)
h <- 1e-5
(f(2 + h) - f(2 - h)) / (2 * h)
exp(2) * (2*2 + 2^2)  # analytic answer
```

Research caution

A numerical derivative checks your algebra but cannot replace understanding. Very large steps cause approximation error; extremely small steps can introduce floating-point cancellation.

A picture of a derivative: the slope you see when you zoom in



This tangent is not the curve itself. It is the *best local linear approximation*. If a derivative is 0.48 at x_0, then a small increase Δx predicts about $0.48\Delta x$ additional units of f.

Fast lane: a useful calculation trick

Spot the outermost operation first. For $(3x^2 + 1)^4$, the final operation is "raise something to the fourth power"; use the chain rule. For x^2e^x, the final operation is multiplication; use the product rule. Label the pieces u and v before writing any derivatives.

Worked example

Try two routes to $\frac{d}{dx}(x^2e^x)$: by product rule, $2xe^x + x^2e^x$; by factoring afterward, $xe^x(x + 2)$. The second form is faster for finding stationary points, while the first is faster for checking differentiation.

REAL LIFE / The speedometer

A bicycle speedometer reports your speed at a moment, not your average speed over an entire commute. A derivative does the same for a changing quantity: it is the instantaneous slope. If distance is $d(t) = t^2$ meters, then speed is $d\prime(t) = 2t$ meters per second. In finance, a derivative can describe how modeled expected return changes when an exposure measure moves slightly, holding the model fixed.

> *"A derivative is the mathematical version of asking: how fast is this changing* right now*?*

Multiple-choice checkpoint

1. **[Foundation]** Differentiate x^3e^x.
   A. $3x^2e^x$ B. x^3e^x C. $e^x(3x^2 + x^3)$ D. $3x^2 + x^3$.

2. **[Application]** Differentiate $\ln(1 + 2x^2)$.
   A. $1/(1 + 2x^2)$ B. $4x/(1 + 2x^2)$ C. $2x/(1 + 2x^2)$ D. $4x$.

3. **[Challenge]** At $q = 4$, marginal cost for $C(q) = 5 + 3q + q^2$ is
   A. 7 B. 8 C. 11 D. 16.

Answers and explanations: Appendix A, Chapter 2.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Differentiate $f(x) = 7x^3 - 2x + 4$.

A. $21x^2 - 2$ **B.** $7x^2 - 2$
C. $21x^3 - 2$ **D.** $21x^2 + 4$

P2. Foundation. A ride costs $C(q) = 100 + 15q + q^2$ yen. What is marginal cost at $q = 5$?

A. 15 yen/unit **B.** 20 yen/unit
C. 25 yen/unit **D.** 125 yen/unit

P3. Application. Differentiate $g(x) = (2x + 1)^5$.

A. $5(2x + 1)^4$ **B.** $10(2x + 1)^4$
C. $(2x + 1)^6$ **D.** $10(2x + 1)^5$

P4. Challenge. Differentiate $h(x) = x^2 e^{3x}$; which expression is correct?

A. $2xe^{3x}$ **B.** $3x^2 e^{3x}$
C. $e^{3x}(2x + 3x^2)$ **D.** $e^{3x}(2x + x^2)$

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 3

Integration: Accumulation, Area, and Expectations

Intuition

Differentiation answers "how fast?" Integration answers "how much accumulated?" Your diagnostic exposed missing fluency here, so this chapter is intentionally slower.

3.1 Antiderivatives and the constant of integration

An antiderivative F of f satisfies $F' = f$. The indefinite integral denotes a family:

$$\int f(x)\,dx = F(x) + C.$$

For powers with $p \neq -1$, $\int x^p\,dx = x^{p+1}/(p+1) + C$. In particular,

$$\int (6x^2 - 4x + 3)\,dx = 2x^3 - 2x^2 + 3x + C.$$

The $+C$ is required because constants disappear under differentiation. For $x > 0$, $\int x^{-1}\,dx = \ln x + C$; more generally $\ln|x| + C$ for $x \neq 0$.

3.2 The fundamental theorem of calculus

If $F' = f$ and f is continuous on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

This definite integral is a signed accumulated quantity, not always a positive geometric area. For example,

$$\begin{aligned}\int_0^2 (3x^2 + 1)\,dx &= [x^3 + x]_0^2 \\ &= (8 + 2) - (0 + 0) = \boxed{10}.\end{aligned}$$

3.3 Substitution as the reverse chain rule

Given $\int 2xe^{x^2}\,dx$, set $u = x^2$, so $du = 2x\,dx$. Therefore

$$\int 2xe^{x^2}\,dx = \int e^u\,du = e^{x^2} + C.$$

The role of substitution is not to swap letters arbitrarily; it recognizes an inner function whose derivative appears nearby.

3.4 Probability density as an integral

For a continuous random variable with density f_X, probability is accumulated density:

$$\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\,dx.$$

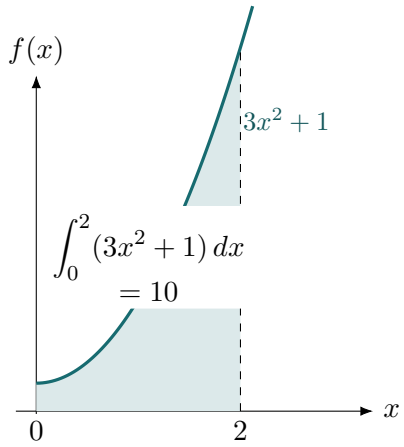
For X uniformly distributed on $[0, 2]$, $f_X(x) = 1/2$ inside the interval. Thus $\mathbb{P}(0.5 < X < 1.5) = \int_{.5}^{1.5} \frac{1}{2} dx = 0.5$. Likewise $\mathbb{E}[X] = \int x f_X(x)\,dx$ whenever the integral exists. We will return to this in Chapter 9.

Semiconductor research connection

Cumulative abnormal return is an *integer-time sum*, $CAR = \sum_t AR_t$, not a continuous integral. The conceptual connection is accumulation across time, but the correct mathematical operator depends on whether time is discrete or continuous.

```
f <- function(x) 3*x^2 + 1
integrate(f, lower = 0, upper = 2)$value
F <- function(x) x^3 + x
F(2) - F(0)
```

Area made visible



Fast lane: a useful calculation trick

Integration sanity check: If the integrand is positive throughout $[a, b]$, the definite integral must be positive. If it rises from 1 to 13 on an interval of width 2, an answer like -4 or 0.03 cannot be right. The approximate rectangle "width $\times$ typical height" is a useful order-of-magnitude test.

Worked example

A quick reverse-derivative check: $\int 8x^3\,dx = 2x^4 + C$ because differentiating $2x^4$ returns $8x^3$. For $\int_0^2 (3x^2+1)\,dx = 10$, this graphical area is positive and reasonably between the rough rectangle bounds 2 and 26.

REAL LIFE / The bike ride odometer

Your bicycle reports speed every moment, but the odometer cares about total distance. The integral adds up all the tiny pieces of motion. At a constant 4 meters per second for ten seconds, distance is $\int_0^{10} 4\,dt = 40$ meters; with changing speed, the same accumulation logic works under the curve. Later, a probability density uses area to represent probability.

"A derivative watches the speedometer; an integral reads the odometer.

Multiple-choice checkpoint

1. **[Foundation]** The derivative of $x^4/4 + 7$ is
   A. $4x^3$ B. x^3 C. x^4 D. $x^3 + 7$.
2. **[Application]** What is $\int_0^1 2x\,dx$?
   A. 0 B. 1/2 C. 1 D. 2.
3. **[Challenge]** Evaluate $\int 3e^{3x}\,dx$.
   A. $e^{3x} + C$ B. $3e^{3x} + C$ C. $e^x + C$ D. $9e^{3x} + C$.

Answers and explanations: Appendix A, Chapter 3.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Find an antiderivative of $6x^2 - 4x$.

A. $2x^3 - 2x^2 + C$ **B.** $12x - 4 + C$
C. $2x^3 - 4x^2 + C$ **D.** $6x^3 - 2x^2 + C$

P2. Foundation. Evaluate $\int_0^2 2x\,dx$.

A. 2 **B.** 4
C. 6 **D.** 8

P3. Application. If $f(x) = e^{2x}$, what is $\int_0^1 f(x)\,dx$?

A. $e^2 - 1$ **B.** $(e^2 - 1)/2$
C. $2(e^2 - 1)$ **D.** $e - 1$

P4. Challenge. A bicycle's speed is $v(t) = 3 + 2t$ meters per second for $0 \le t \le 4$ seconds. How far does it travel?

A. 16 m **B.** 20 m
C. 28 m **D.** 44 m

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 4

Partial Derivatives and Optimization

Intuition

Statistical estimation often means choosing parameters to make an objective as small as possible. Partial derivatives are the tools for changing one parameter while holding others fixed. You identified their meaning correctly in the first diagnostic but struggled to compute them in handwriting.

4.1 One-variable maxima and minima

For differentiable interior optima, a necessary condition is $f'(x^*) = 0$. If $f''(x^*) < 0$ it is a strict local maximum; if $f''(x^*) > 0$ a strict local minimum. A zero second derivative does not settle the question.

Worked example

Profit $\pi(q) = 100q - 2q^2 - 200$, with $q \geq 0$, has derivative $100 - 4q$. Setting it to zero gives $q^* = 25$. The second derivative is $-4 < 0$, and the maximum value is $\pi(25) = 2500 - 1250 - 200 = 1050$. Check the boundary $q = 0$, where profit is -200.

4.2 Holding other variables fixed

For $f(x, y) = 3x^2 + 2xy + 4y^2$, treat y as a constant when differentiating with respect to x:

$$f_x = 6x + 2y, \qquad f_y = 2x + 8y.$$

At $(x, y) = (1, 2)$ these are $f_x = 10$ and $f_y = 18$. In a regression coefficient vector, this is the basic meaning of changing one coefficient at a time in an objective function.

4.3 Gradients and Hessians

The gradient collects first derivatives, $\nabla f = (f_x, f_y)^{\mathsf{T}}$. The Hessian collects second derivatives:

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

When H_f is positive definite everywhere, f is strictly convex and any stationary point is the unique global minimizer. Negative definite gives strict concavity and a unique maximizer, under the corresponding domain conditions.

4.4 Two-parameter least squares, preview

For observations (x_i, y_i), consider $L(a, b) = \sum_i (y_i - a - bx_i)^2$. Differentiate with respect to both coefficients:

$$\frac{\partial L}{\partial a} = -2 \sum_i (y_i - a - bx_i),$$
$$\frac{\partial L}{\partial b} = -2 \sum_i x_i (y_i - a - bx_i).$$

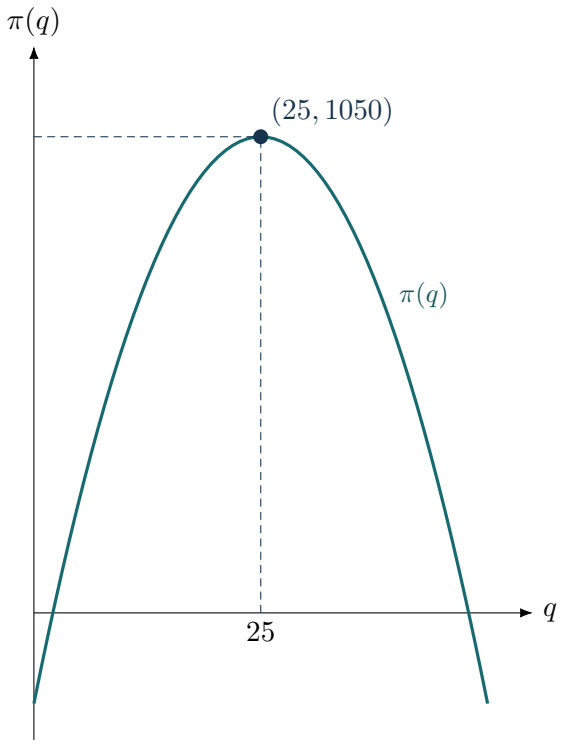
Set both equal to zero; solving yields $\hat{b} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / \sum_i (x_i - \bar{x})^2$ (provided the denominator is positive) and $\hat{a} = \bar{y} - \hat{b}\bar{x}$. Chapter 13 derives this carefully.

Research caution

An optimization condition identifies the best fit *within the specified objective*. It says nothing, by itself, about random sampling, omitted variables, or causal identification. This distinction directly corrects your Session 3 omitted-variable-bias mistake.

```
profit <- function(q) 100*q - 2*q^2 - 200
optimize(profit, interval=c(0, 100), maximum=TRUE)
# Compare the analytic solution: q=25; profit=1050
```

Optimization is a hill-climbing problem



Fast lane: a useful calculation trick

For $aq^2 + bq + c$ with $a < 0$, the maximum occurs at $q^* = -b/(2a)$—the vertex formula. Why does it work? Because $\pi'(q) = 2aq + b = 0$ at that point. Use the vertex formula as a *speed check*, not a substitute for learning derivatives and boundary checks.

Worked example

A partial derivative shortcut: in $3x^2 + 2xy + 4y^2$, when finding $\partial f/\partial x$, imagine y written in ink and x in pencil. Only the pencil changes: $6x + 2y$. For $\partial f/\partial y$, freeze x, giving $2x + 8y$.

REAL LIFE / The cafe that sells too much

Imagine a cafe with profit $\pi(q) = 20q - q^2$ in simplified units. Selling one more cup initially helps, but eventually added costs and diminishing gains take over. Setting $\pi\prime(q) = 0$ locates a candidate best quantity, here $q = 10$; checking curvature confirms a maximum. Regression will use a related idea in reverse: choose coefficients that make total prediction errors as small as possible.

> *"An optimum is not the point where effort disappears; it is where the next tiny change stops paying.*

Multiple-choice checkpoint

1. **[Foundation]** For $f(x, y) = x^2 + 3xy$, $\partial f/\partial x$ is
   A. $2x + 3y$ B. $2x + 3x$ C. $x + 3y$ D. $2x + 3$.
2. **[Application]** Which q maximizes $8q - q^2$ for $q \geq 0$?
   A. 2 B. 4 C. 8 D. 16.
3. **[Challenge]** A twice-differentiable one-variable objective satisfies $f'(2) = 0$ and $f''(2) > 0$. What can be concluded?
   A. Strict local maximum B. Inflection necessarily C. Strict local minimum D. Global maximum necessarily.

Answers and explanations: Appendix A, Chapter 4.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. For $f(x, y) = 2x^2 + 5xy + y^2$, calculate $\partial f/\partial x$.

A. $4x + 5y$ **B.** $4x + 5x$
C. $2x + 5y$ **D.** $4x + 2y$

P2. Foundation. Which value of $q \geq 0$ maximizes $\pi(q) = 12q - q^2$?

A. 3 **B.** 6
C. 12 **D.** 24

P3. Application. Minimize $L(b) = (4 - 2b)^2 + (1 - b)^2$ over b.

A. $b = 1$ **B.** $b = 9/5$
C. $b = 2$ **D.** $b = 3$

P4. Challenge. A researcher minimizes $L(a, b) = \sum_i (y_i - a - bx_i)^2$. Which pair of equalities must hold at an interior optimum?

A. $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$ **B.** $\sum_i e_i = 1$ and $\sum_i x_i = 0$
C. $e_i = 0$ for every i **D.** $a = 0$ and $b = 0$

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review A: Calculus and Algebra Checkpoint

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. Solve $4x + 9 = 2x + 17$.

A. 2 **B.** 4
C. 8 **D.** 13

R2. Foundation. Differentiate $5x^4 - 3x^2$.

A. $20x^3 - 6x$ **B.** $20x^4 - 6x$
C. $5x^3 - 3x$ **D.** $10x^3 - 6$

R3. Application. Evaluate $\int_0^3 2x\,dx$.

A. 3 **B.** 6
C. 9 **D.** 18

R4. Application. Find $\partial(x^2y + 3y^2)/\partial y$.

A. $2xy + 6y$ **B.** $x^2 + 6y$
C. $x^2 + 3y$ **D.** $2xy + 3$

R5. Challenge. For $f(q) = 18q - 3q^2$, find the maximizing q over $q \geq 0$.

A. 1 **B.** 2
C. 3 **D.** 6

R6. Challenge. Differentiate xe^{2x} at $x = 0$.

A. 0 **B.** 1
C. 2 **D.** 3

Answer key: *see worked solutions for Review A*.

CHAPTER 5

Vectors, Matrices, and Regression Data

Intuition

Matrices are convenient bookkeeping for many observations and many regressors. Most difficulty disappears when you check dimensions before computing. A 3×2 matrix times a 2×1 vector results in a 3×1 vector; reversing this order may not even be defined.

5.1 Dimensions, transpose, and dot products

A vector $\boldsymbol{x} = (x_1, \ldots, x_n)^\mathsf{T}$ is an $n \times 1$ column. Its transpose $\boldsymbol{x}^\mathsf{T}$ is $1 \times n$. The dot product is

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \sum_{i=1}^{n} x_i y_i.$$

For $\boldsymbol{x} = (1, 2, 3)^\mathsf{T}$ and $\boldsymbol{y} = (2, 4, 5)^\mathsf{T}$, the product is $2{+}8{+}15 = 25$, whereas $\boldsymbol{x}^\mathsf{T}\boldsymbol{x} = 1{+}4{+}9 = 14$. A vector norm is $\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}^\mathsf{T}\boldsymbol{x}}$.

5.2 Matrix multiplication by rows and columns

If A is $m \times k$ and B is $k \times n$, then AB is $m \times n$ with $(AB)_{ij} = \sum_{\ell=1}^{k} a_{i\ell} b_{\ell j}$.

Worked example

For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}\right)$,

$$AB = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}, \qquad BA = \begin{pmatrix} 2 & 4 \\ 10 & 14 \end{pmatrix}.$$

Matrix multiplication is not commutative in general: $AB \neq BA$. Compute each entry as one row–column dot product.

5.3 The regression design matrix

Suppose n firms each have one exposure x_i, and we fit an intercept and one slope. The design matrix is

$$X = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}_{n\times 2}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

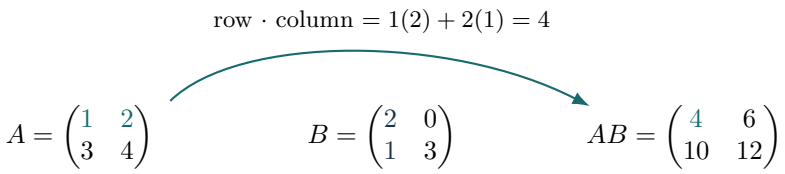
The equation $X\boldsymbol{\beta}$ creates n fitted responses. The column of ones creates the intercept; omitting it forces the fitted line through the origin.

5.4 Transpose identities worth knowing

When dimensions conform, $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$; $(A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}$; and $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$ is a scalar, hence equal to its own transpose $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{x}$. These identities make least-squares differentiation possible.

```
A <- matrix(c(1,3,2,4), nrow=2) # columns filled first
B <- matrix(c(2,1,0,3), nrow=2)
A %*% B
B %*% A
t(A)
x <- c(1,2,3); y <- c(2,4,5)
sum(x*y); drop(t(x) %*% y)
```

Matrix multiplication without the headache



Fast lane: a useful calculation trick

Shape before arithmetic: $(m \times k)(k \times n)$ produces $m \times n$. The two *inner* dimensions must match; the two *outer* dimensions tell you the answer's shape. If the dimensions disagree, stop immediately rather than multiplying the wrong numbers.

Worked example

For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}\right)$, write one output cell at a time: $AB = \left(\begin{smallmatrix} 1(2)+2(1) & 1(0)+2(3) \\ 3(2)+4(1) & 3(0)+4(3) \end{smallmatrix}\right) = \left(\begin{smallmatrix} 4 & 6 \\ 10 & 12 \end{smallmatrix}\right)$. Matrix multiplication is generally not commutative: $BA \neq AB$.

REAL LIFE / The shopping basket

A basket contains two apples, one sandwich and three bottles of water. Put quantities in $q = (2, 1, 3)$ and prices in $p = (180, 450, 120)$ yen. The total bill is the dot product $q \cdot p = 2(180) + 1(450) + 3(120) = 1170$ yen. Matrix notation is this same bookkeeping scaled up to thousands of firms and dates.

> *"A matrix is not a new species of mathematics; it is organized arithmetic with excellent filing habits.*

Multiple-choice checkpoint

1. [**Foundation**] A 4×3 matrix multiplied by a 3×2 matrix yields
A. 3×3 B. 4×2 C. 2×4 D. undefined.

2. [**Application**] With $x = (2, 3)^\top$ and $y = (1, 4)^\top$, $x^\top y$ equals
A. 10 B. 11 C. 14 D. 16.

3. [**Challenge**] A regression including an intercept and three regressors for 100 observations has a design matrix with dimensions
A. 100×3 B. 3×100 C. 100×4 D. 4×4.

Answers and explanations: Appendix A, Chapter 5.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. What is $(2, -1, 3) \cdot (4, 2, 1)$?

A. 5 **B.** 9
C. 11 **D.** 13

P2. Foundation. If A is 3×2 and B is 2×5, what are the dimensions of AB?

A. 2×3 **B.** 3×5
C. 2×5 **D.** 3×2

P3. Application. With $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, find Av.

A. $(4, 3)^\top$ **B.** $(2, 1)^\top$
C. $(3, 3)^\top$ **D.** $(4, 1)^\top$

P4. Challenge. In regression $y = X\beta + u$, X has $n = 100$ observations and $k = 4$ columns. What dimensions must β and $X^\top y$ have?

A. 4×1 and 4×1 **B.** 100×1 and 4×1
C. 4×4 and 100×1 **D.** 1×4 and 1×100

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 6

Systems, Determinants, Inverses, and Rank

Intuition

An inverse answers the question: can you undo a linear transformation? An inverse does not exist if different inputs produce the same output. In regression this happens when a predictor is an exact linear combination of other predictors.

6.1 Linear systems as matrices

The system $2x + y = 7$ and $x + 3y = 11$ is

$$\underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{z} = \underbrace{\begin{pmatrix} 7 \\ 11 \end{pmatrix}}_{b}.$$

Multiply the second equation by two and subtract the first: $5y = 15$, so $y = 3$, then $x = 2$. This is exactly what Gaussian elimination accomplishes in larger problems.

6.2 The 2×2 inverse rule and its meaning

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, define $\det A = ad - bc$. When $\det A \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Worked example

Take $A = \left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$. Since $\det A = 6 - 5 = 1$,

$$A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

Verify instead of relying on memory: $AA^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. In your handwritten diagnostic, recalling the inversion pattern without checking the product contributed to an error.

6.3 Rank, collinearity, and uniqueness

Rank is the number of linearly independent columns (equivalently rows) of a matrix. If X has k columns and $\text{rank}(X) = k$, then $X^\top X$ is invertible. If the second explanatory column is twice the first, rank is deficient. No amount of extra arithmetic can uniquely distinguish the two associated slopes from the sample.

Research caution

Even when a matrix is invertible, a very small determinant or a large condition number can indicate numerical instability. Practical R work should use `solve(A,b)` or QR-based regression methods rather than calculating A^{-1} and multiplying by b explicitly.

6.4 Eigenvalues, only what you need now

An eigenvector $\boldsymbol{v} \neq 0$ satisfies $A\boldsymbol{v} = \lambda\boldsymbol{v}$. A symmetric A is positive definite if all eigenvalues are positive, or equivalently $\boldsymbol{z}^\top A\boldsymbol{z} > 0$ for every nonzero $\boldsymbol{z}$. This links rank and curvature when minimizing quadratic regression objectives.

```
A <- matrix(c(2,5,1,3), 2, 2)
solve(A)
A %*% solve(A)
solve(A, c(7, 19))  # a separate illustrative system
qr(A)$rank
kappa(A)               # numerical conditioning
```

The two-by-two inverse: swap, flip, divide

Fast lane: a useful calculation trick

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, calculate $ad - bc$. If it is nonzero, **swap** the diagonal entries a, d, **flip** signs of the off-diagonal entries b, c, then **divide** by $ad - bc$:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Say *swap–flip–divide* out loud. It is silly enough to remember and useful enough to keep.

Worked example

Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$. The determinant is $2(3) - 1(5) = 1$. So $A^{-1} = \left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$. Check the top-left entry of AA^{-1}: $2(3) + 1(-5) = 1$; the top-right entry is $2(-1) + 1(2) = 0$. Repeat for the bottom row to verify the identity matrix.

Research caution

The mnemonic is for 2×2 matrices only. It does not generalize by swapping two entries in a larger matrix. For larger systems, use elimination or a suitable numerical factorization; in R, prefer `solve(A, b)` rather than explicitly computing `solve(A) %*% b`.

REAL LIFE / Undoing a recipe

You know the total cost of two mixed lunch sets but not each ingredient price. Two independent linear equations may let you recover each price: inverting a nonsingular coefficient matrix reverses the mixing operation. If both lunch sets contain ingredients in exactly the same proportions, the two equations repeat the same information and the determinant is zero. No clever calculator can recover unique prices from redundant evidence.

> *"An inverse can undo a transformation, but it cannot invent information that was never there.*

Multiple-choice checkpoint

1. **[Foundation]** What is the determinant of $\left(\begin{smallmatrix}1 & 2\\ 2 & 4\end{smallmatrix}\right)$?
A. 0 B. 2 C. 4 D. 8.

2. **[Application]** If X is 30×3 but its columns are linearly dependent, then
A. $X^{\mathsf{T}}X$ must be invertible B. $X^{\mathsf{T}}X$ is singular C. OLS errors are normal D. n becomes 3.

3. **[Challenge]** Solve $2x + y = 7$, $x + 3y = 11$.
A. $(1, 5)$ B. $(3, 1)$ C. $(2, 3)$ D. $(4, -1)$.

Answers and explanations: Appendix A, Chapter 6.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. What is the determinant of $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$?

A. 2 **B.** 4
C. 6 **D.** 8

P2. Foundation. Which matrix is the inverse of $\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$?

A. $\begin{pmatrix} 1 & 0 \\ 0 & 1/5 \end{pmatrix}$ **B.** $\begin{pmatrix} 1 & 0 \\ 0 & -5 \end{pmatrix}$
C. $\begin{pmatrix} 0 & 1 \\ 5 & 0 \end{pmatrix}$ **D.** $\begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$

P3. Application. Solve $2x + y = 8$ and $x - y = 1$.

A. $(x, y) = (2, 4)$ **B.** $(3, 2)$
C. $(4, 0)$ **D.** $(1, 6)$

P4. Challenge. A design matrix has two identical columns. Which consequence follows for ordinary $(X^\top X)^{-1}$?

A. It is guaranteed to exist **B.** It is singular, so the ordinary inverse does not exist
C. Its determinant must equal one **D.** The outcome vector becomes zero

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 7

Projection Geometry and Quadratic Forms

Intuition

Regression is easiest to understand geometrically: predictions are the closest point to the outcome vector within the space spanned by the regressors. The leftover residual is perpendicular to that space.

7.1 Projection onto a single line

Let $\boldsymbol{x}$ be a nonzero explanatory vector and $\boldsymbol{y}$ the outcomes. The fitted vector through the origin is $\hat{\boldsymbol{y}} = \hat{\beta}\boldsymbol{x}$. To choose the closest fit, minimize its squared Euclidean distance:

$$L(\beta) = \|\boldsymbol{y} - \beta\boldsymbol{x}\|^2 = (\boldsymbol{y} - \beta\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \beta\boldsymbol{x}).$$

Expanding the quadratic gives

$$L = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\beta\boldsymbol{x}^\mathsf{T}\boldsymbol{y} + \beta^2\boldsymbol{x}^\mathsf{T}\boldsymbol{x}.$$

The derivative is $-2\boldsymbol{x}^\mathsf{T}\boldsymbol{y} + 2\beta\boldsymbol{x}^\mathsf{T}\boldsymbol{x} = 0$. Thus

$$\hat{\beta} = \frac{\boldsymbol{x}^\mathsf{T}\boldsymbol{y}}{\boldsymbol{x}^\mathsf{T}\boldsymbol{x}}, \quad \boldsymbol{x}^\mathsf{T}(\boldsymbol{y} - \hat{\beta}\boldsymbol{x}) = 0.$$

The final equality says the residual is orthogonal to the fitted line. It does *not* say residuals are independent of exposure in the population.

7.2 Projections in multiple dimensions

For full-column-rank X, the matrix

$$P_X = X(X^\mathsf{T}X)^{-1}X^\mathsf{T}$$

projects onto the column space of X; fitted values are $\hat{\boldsymbol{y}} = P_X\boldsymbol{y}$. The residual maker $M_X = I - P_X$ produces $\hat{\boldsymbol{e}} = M_X\boldsymbol{y}$. Both P_X and M_X are symmetric and idempotent ($P_X^2 = P_X$). Orthogonality is $X^\mathsf{T}\hat{\boldsymbol{e}} = 0$.

7.3 Why squared errors?

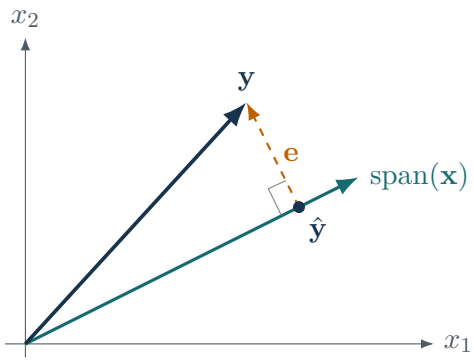
Squaring gives a differentiable, nonnegative objective and penalizes larger residuals more strongly. With a column of ones in X, residuals sum to zero in-sample: $\sum_i \hat{e}_i = 0$. This is an algebraic property of the estimator, not proof that the true disturbances have mean zero conditional on regressors.

Worked example

For $\boldsymbol{x} = (1, 2, 3)^\mathsf{T}$ and $\boldsymbol{y} = (2, 4, 5)^\mathsf{T}$, the scalar projection coefficient is $25/14 \approx 1.786$. Predicted values are approximately $(1.786, 3.571, 5.357)$; residuals are approximately $(0.214, 0.429, -0.357)$. Their dot product with $\boldsymbol{x}$ is zero up to rounding.

```
x <- c(1,2,3); y <- c(2,4,5)
b <- sum(x*y)/sum(x*x)
e <- y - b*x
sum(x*e)              # approximately zero
P <- outer(x,x)/sum(x^2)
stopifnot(max(abs(P %*% P - P)) < 1e-10)
```

See OLS as dropping a perpendicular



OLS chooses the fitted vector $\hat{\mathbf{y}}$ lying in the space generated by regressors, minimizing the length of residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$. At an interior optimum $X^\mathsf{T}\mathbf{e} = 0$: the residual is orthogonal to every regressor column. The drawing is a two-dimensional analogy, not a depiction of a particular dataset.

Fast lane: a useful calculation trick

To check a hand-calculated regression with an intercept, residuals should sum to zero (within rounding): $\sum_i \hat{e}_i = 0$. To check any standard full-rank OLS fit, compute $X^\mathsf{T}\hat{\mathbf{e}}$—its entries should be numerically close to zero.

REAL LIFE / A shadow on the wall

Shine a light so that a three-dimensional object casts a shadow on a flat wall. The shadow retains the part that lies in the wall plane and discards the perpendicular part. OLS does something similar: it projects the observed outcome vector onto the space spanned by the model regressors. The fitted values are the shadow; residuals are what the model

cannot express in that space.

"Prediction is a shadow of the data; the residual is the part that refuses to lie flat.

Multiple-choice checkpoint

1. **[Foundation]** If e is the OLS residual from a full-rank regression on X, which is always true in the fitted sample?
A. $X^\top e = 0$ B. $\mathbb{E}[e \mid X] = 0$ in the population C. $e = 0$ D. e is independent of X.

2. **[Challenge]** What does $M_X = I - P_X$ do to y?
A. Scales every outcome by n B. Produces OLS residuals C. Produces coefficient standard errors D. Finds eigenvalues.

Answers and explanations: Appendix A, Chapter 7.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. If $v = (3, 4)^\top$, what is its Euclidean norm?

A. 5 **B.** 7
C. 12 **D.** 25

P2. Foundation. If fitted values are $\hat{y} = (2, 3)^\top$ and observed values are $y = (4, 4)^\top$, what is the residual vector?

A. $(6, 7)^\top$ **B.** $(2, 1)^\top$
C. $(-2, -1)^\top$ **D.** $(2, 3)^\top$

P3. Application. For OLS with full-column-rank X, what is the geometric relationship between residuals e and every column of X?

A. They are parallel **B.** They are orthogonal
C. They have identical lengths **D.** They are always zero

P4. Challenge. A vector $y = (3, 4)^\top$ is projected onto the horizontal axis. Which projected vector and residual are correct?

A. $\hat{y} = (3, 0)^\top$, $e = (0, 4)^\top$ **B.** $\hat{y} = (0, 4)^\top$, $e = (3, 0)^\top$
C. $\hat{y} = (3, 4)^\top$, $e = (0, 0)^\top$ **D.** $\hat{y} = (0, 0)^\top$, $e = (3, 4)^\top$

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review B: Matrix Fluency Checkpoint

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. Find $(1, 2, 1) \cdot (3, 0, 4)$.

A. 3 **B.** 4
C. 7 **D.** 11

R2. Foundation. If X is 80×5, what is the shape of $X^\top y$ when y is 80×1?

A. 80×1 **B.** 5×1
C. 5×5 **D.** 80×5

R3. Application. Compute determinant of $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

A. 8 **B.** 10
C. 12 **D.** 14

R4. Application. Solve $x + y = 7$ and $x - y = 1$.

A. $(3, 4)$ **B.** $(4, 3)$
C. $(5, 2)$ **D.** $(6, 1)$

R5. Challenge. A least-squares residual is orthogonal to the regressor $x = (1, 1)^\top$. Which residual could satisfy this?

A. $(2, 2)^\top$ **B.** $(1, -1)^\top$
C. $(0, 3)^\top$ **D.** $(4, 1)^\top$

R6. Challenge. Suppose $X^\top X = \mathrm{diag}(1, 9)$ and $X^\top y = (3, 18)^\top$. Find OLS $\hat{\beta}$.

A. $(3, 2)^\top$ **B.** $(1/3, 1/2)^\top$
C. $(3, 18)^\top$ **D.** $(9, 162)^\top$

Answer key: *see worked solutions for Review B.*

Part II

Probability and Mathematical Statistics

CHAPTER 8

Probability Rules, Conditioning, and Bayes

Intuition

You scored strongly on introductory probability, so this is a compressed review. The aim is to make probability calculations fluent enough that later regression uncertainty and Bayesian models do not feel mysterious.

8.1 Events, complements and intersections

For events A and B, union means A *or* B (including both), and intersection means A *and* B. The inclusion–exclusion identity is

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The intersection is subtracted because simple addition double-counts it. Complements satisfy $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

8.2 Conditioning changes the denominator

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The denominator is the size of the conditioning event, not the original population. For 60 students, 24 studying Python and 12 studying both Python and R, $\mathbb{P}(R \mid P) = 12/24 = 1/2$. Independent events satisfy $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, equivalently $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ when $\mathbb{P}(B) > 0$. *Mutually exclusive* means $\mathbb{P}(A \cap B) = 0$; two non-null mutually exclusive events cannot be independent.

8.3 Total probability and Bayes' theorem

If D and D^c partition the sample space, then

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c).$$

Bayes' theorem rearranges conditional probability:

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)}.$$

Worked example

With prevalence $\mathbb{P}(D) = .10$, sensitivity $\mathbb{P}(+ \mid D) = .80$, and false-positive probability $\mathbb{P}(+ \mid D^c) = .20$,

$$\mathbb{P}(D \mid +) = \frac{.8(.1)}{.8(.1) + .2(.9)} = \frac{.08}{.26} \approx .3077.$$

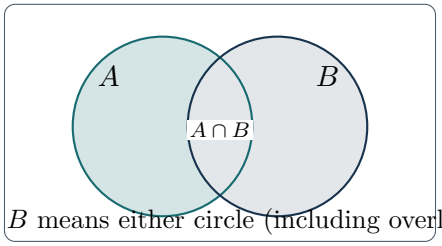
A positive test does not mean 80% posterior probability. This was one of your correct Session 2 responses.

8.4 A political-science application without misleading causality

Let A be a policy announcement and R a negative stock return. $\mathbb{P}(R \mid A)$ is an association conditional on announcements, not necessarily $\mathbb{P}(R \mid do(A))$, which refers to a causal intervention. Selection, expectations and simultaneous news can separate the two quantities.

```
pD <- .10; sensitivity <- .80; false_positive <- .20
p_positive <- sensitivity*pD + false_positive*(1-pD)
pD_given_positive <- sensitivity*pD/p_positive
pD_given_positive
```

One diagram to stop mixing up $A \cap B$ and $A \cup B$



$A \cup B$ means either circle (including overlap)

Fast lane: a useful calculation trick

The overlap is counted twice when you add $P(A) + P(B)$, so subtract $P(A \cap B)$ once. For conditional probability, redraw the sample space as "only the cases where B happened"; the denominator is then $P(B)$, not $P(A) + P(B)$.

Worked example

If $P(A) = 0.45$, $P(B) = 0.35$, and $P(A \cap B) = 0.15$, then $P(A \cup B) = 0.45 + 0.35 - 0.15 = 0.65$. Quick reasonableness check: a union must be at least as large as either event and at most 1.

REAL LIFE / Spam filters and rare events

Imagine a news classifier that correctly flags almost every rare policy announcement. A flagged document is not automatically a genuine policy event because the database may contain vastly more ordinary news. Bayes' theorem combines the prior rarity with the detector's sensitivity and false-positive rate. The denominator matters: conditioning changes which pile of documents you are looking at.

"A 90% accurate signal can still fool you if the thing you seek is rare.

Multiple-choice checkpoint

1. **[Foundation]** If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, then $P(A \cup B) =$
A. 0.3 B. 0.7 C. 0.9 D. 1.1.

2. **[Challenge]** If $P(A \mid B) = P(A)$ and $P(B) > 0$, the events are
A. independent B. mutually exclusive C. identical D. complementary.

Answers and explanations: Appendix A, Chapter 8.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. In a survey, $P(A) = 0.4$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.

A. 0.2 **B.** 0.6

C. 0.8 **D.** 1.0

P2. Foundation. If 30 of 50 students use Python and 12 of those 30 also use R, what is $P(R \mid \text{Python})$?

A. 0.24 **B.** 0.30

C. 0.40 **D.** 0.60

P3. Application. Two independent events have probabilities 0.2 and 0.3. Find the probability that both occur.

A. 0.06 **B.** 0.10

C. 0.50 **D.** 0.60

P4. Challenge. A screening system flags 90% of truly relevant articles and 10% of irrelevant articles; only 5% of all articles are relevant. What proportion of flagged articles is relevant (approximately)?

A. 9% **B.** 32%

C. 50% **D.** 90%

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 9

Random Variables, Expectations, and Distributions

Intuition

A probability attaches weights to possible outcomes. Expected value computes a weighted average; variance quantifies uncertainty around it. You did well on these calculations, so the emphasis here is on properties needed for econometric proofs.

9.1 Discrete distributions

For a discrete random variable, $p(x) = \mathbb{P}(X = x)$ is nonnegative and sums to one. If moments exist,

$$\mathbb{E}[X] = \sum_x x p(x), \quad \operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For $X \in \{1, 3, 5\}$ with equal probabilities, $\mathbb{E}[X] = 3$ and $\operatorname{Var}(X) = [4{+}0{+}4]/3 = 8/3$. Standard deviation is $\sqrt{8/3}$, expressed in the original units rather than squared units.

9.2 Linearity versus variance

For constants a, b, $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$, but $\operatorname{Var}(a + bX) = b^2 \operatorname{Var}(X)$. For two variables,

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y),$$

where $\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$. If X and Y are independent, covariance is zero, but zero covariance does not imply independence in general.

Worked example

Let $\operatorname{Var}(X) = 4$, $\operatorname{Var}(Y) = 9$ and $\operatorname{Cov}(X, Y) = 2$. Then $\operatorname{Var}(X + Y) = 4 + 9 + 4 = 17$. Ignoring covariance would underestimate uncertainty. This reappears when abnormal returns are correlated across firms or days.

9.3 Bernoulli, binomial, normal

A Bernoulli variable with success probability p has mean p and variance $p(1-p)$. For n independent Bernoulli trials, $X \sim \text{Binomial}(n, p)$,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mathbb{E}X = np, \quad \text{Var}\, X = np(1-p).$$

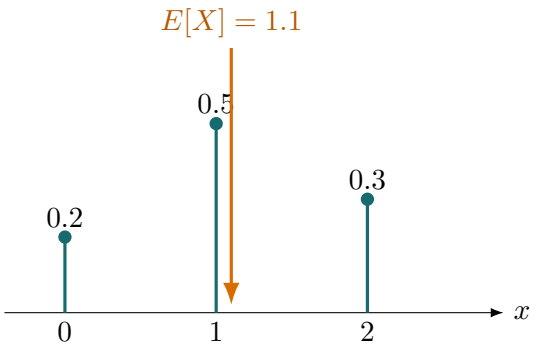
For a normal variable $X \sim N(\mu, \sigma^2)$, standardization $Z = (X - \mu)/\sigma$ has distribution $N(0, 1)$. You should recognize the density shape and learn to read probabilities, not memorize normal-table decimals.

9.4 Continuous expectations

For density f_X integrating to one, $\mathbb{E}[g(X)] = \int g(x) f_X(x)\, dx$ when defined. This connects the integral chapter to the random-variable chapter. For $X \sim \text{Uniform}(0, 2)$, $\mathbb{E}X = \int_0^2 x(1/2)dx = 1$.

```
x <- 0:2; p <- c(.2,.5,.3)
mu <- sum(x*p)
variance <- sum((x-mu)^2*p)
c(mu=mu, variance=variance)
dbinom(3, size=4, prob=.5)
pnorm(1.96) - pnorm(-1.96)
```

Expected value as a balance point



The mean is a weighted average, not necessarily one of the outcomes. Here $E[X] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$.

Fast lane: a useful calculation trick

For a short discrete distribution, make *three rows*: x, $p(x)$, and $x\,p(x)$. Add the last row for $E[X]$. Then use $\text{Var}(X) = E[X^2] - E[X]^2$ as the fast route to variance, and verify that the answer is nonnegative. For binary $X \sim \text{Bernoulli}(p)$: memorize the useful pair $E[X] = p$, $\text{Var}(X) = p(1-p)$.

REAL LIFE / An uncertain dinner budget

Your dinner costs 1,000 yen with probability 0.7 and 3,000 yen with probability 0.3. The expected cost is $0.7(1000) + 0.3(3000) = 1600$ yen. That does not mean any meal costs exactly 1,600 yen; expectation is a long-run weighted average. Variance, meanwhile, describes how much actual costs scatter around that average.

> *"The expectation tells you the average bill; the variance tells you how nervous your wallet should be.*

Multiple-choice checkpoint

1. **[Foundation]** If $\mathbb{E}[X] = 2$ and $\mathrm{Var}(X) = 3$, then $\mathrm{Var}(4 + 2X) =$
A. 6 B. 12 C. 16 D. 20.

2. **[Application]** If $X \sim \mathrm{Binomial}(3, 1/2)$, $P(X = 2) =$
A. 1/8 B. 1/4 C. 3/8 D. 1/2.

3. **[Challenge]** If $\mathrm{Cov}(X, Y) < 0$, what does the sign describe?
A. Certain causality B. Negative linear association about means C. Identical variance D. Both variables must be negative.

Answers and explanations: Appendix A, Chapter 9.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A variable equals 0 with probability 0.25 and 4 with probability 0.75. What is its expectation?

A. 1 **B.** 2
C. 3 **D.** 4

P2. Foundation. If $\mathrm{Var}(X) = 9$, calculate $\mathrm{Var}(2X + 5)$.

A. 14 **B.** 18
C. 36 **D.** 81

P3. Application. A fair coin is flipped 3 times. What is the probability of exactly 2 heads?

A. $1/8$ **B.** $1/4$
C. $3/8$ **D.** $1/2$

P4. Challenge. Suppose $E[X] = 2$, $E[Y] = 5$, and $\mathrm{Cov}(X, Y) = -1$. Find $E[3X - 2Y + 7]$.

A. -1 **B.** 3
C. 7 **D.** 13

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 10

Sampling Distributions, Standard Errors, and the CLT

Intuition

A sample mean is a random number before the data arrive. If you sampled different groups of firms, you would obtain different means. A standard error describes that sampling variation of an estimator, not the spread of individual observations.

10.1 Why the sample mean is unbiased under random sampling

For independent, identically distributed $X_1, \dots, X_n$ with $\mathbb{E}[X_i] = \mu$, the sample mean is $\bar{X} = n^{-1} \sum X_i$. By linearity,

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum \mathbb{E}[X_i] = \mu.$$

If each X_i has finite variance σ^2 and observations are independent,

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2} \sum \mathrm{Var}(X_i) = \frac{\sigma^2}{n}, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Worked example

When $\sigma = 12$ and $n = 36$, the standard error is $12/6 = 2$. Increasing sample size fourfold halves this standard error. This result requires the assumed independence; repeated returns of the same firm often violate it.

10.2 Why $n - 1$ occurs in sample variance

Define $s^2 = (n-1)^{-1} \sum (X_i - \bar{X})^2$. Under iid sampling with finite variance, $\mathbb{E}[s^2] = \sigma^2$. The denominator $n-1$ compensates for estimating the population mean from the same sample: deviations around $\bar{X}$ must sum to zero, leaving $n-1$ independent deviations in a linear-algebra sense.

10.3 Central limit theorem and standardization

Under iid observations with finite, positive variance,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

For sufficiently large samples, the distribution of the standardized *mean* is approximately normal, even when individual observations are not normal. The approximation can be poor for some heavy-tailed or highly skewed data at modest sample sizes. Dependence requires alternative conditions and variance formulas.

10.4 Law of large numbers versus CLT

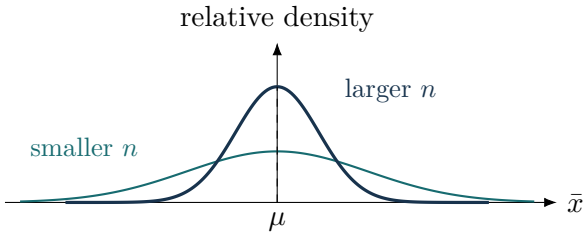
The law of large numbers concerns concentration or convergence of $\bar{X}$ toward μ. The CLT characterizes the scaled fluctuations around μ. They answer different questions: consistency versus approximate inference.

```
set.seed(2601)
B <- 4000
means <- replicate(B, mean(rexp(36, rate=1)))
mean(means)           # near population mean 1
sd(means)             # near population SE 1/sqrt(36)
quantile(means, c(.025,.975))
```

Research caution

You cannot mechanically replace n by the number of firm-days and claim independence. Firms share market shocks; observations within a firm can be serially dependent. Later chapters discuss clustering and time-series corrections.

Why more observations reduce uncertainty—but not proportionally



Fast lane: a useful calculation trick

Because $SE(\bar{X}) = \sigma/\sqrt{n}$ under iid sampling, **quadruple** n to halve the standard error. Doubling n reduces it by only $1/\sqrt{2} \approx 0.707$, not by half. This is the *square-root law*: sampling precision has diminishing returns.

Research caution

The curves are illustrative. Additional observations do not fix nonrandom sampling, confounding, or strong dependence; $\sigma/\sqrt{n}$ is not a universal standard-error formula for panel or time-series data.

REAL LIFE / Tasting a giant pot of soup

If you want to know whether a pot of soup is salty, sampling one spoonful is noisy. Taking more independent, representative spoonfuls improves precision, but each additional

spoon adds less than the previous one. Standard error decreases with the square root of sample size, not linearly. Sampling many near-identical spoonfuls from the same corner, however, may provide less independent information than the raw count suggests.

"*More data help; more* independent information *helps more.*

Multiple-choice checkpoint

1. **[Foundation]** If population SD is 10 and iid sample size is 100, $SE(\bar{X}) =$
   A. 0.1 B. 1 C. 10 D. 100.
2. **[Challenge]** If independent sample size increases from 25 to 100, the standard error of the mean with fixed σ is multiplied by
   A. 1/4 B. 1/2 C. 2 D. 4.

Answers and explanations: Appendix A, Chapter 10.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A population standard deviation is 15 and a simple random sample contains 25 independent observations. What is $SE(\bar{X})$?

A. 0.6 **B.** 3

C. 5 **D.** 15

P2. Foundation. If sample size quadruples under independent identical sampling, what happens to $SE(\bar{X})$?

A. It quadruples **B.** It doubles

C. It halves **D.** It stays unchanged

P3. Application. In 100 independent Bernoulli trials with success probability 0.4, what is $\mathrm{Var}(\bar{X})$?

A. 0.0024 **B.** 0.024

C. 0.24 **D.** 2.4

P4. Challenge. Two firms' daily returns are strongly correlated because they react to the same news. A researcher treats 200 firm-day observations as independent. What is the major danger?

A. The point estimate must equal zero **B.** Conventional standard errors may be too small

C. Correlation guarantees a positive causal effect **D.** Sample size automatically doubles

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 11

Confidence Intervals and Hypothesis Tests

Intuition

Your statistical diagnostics were strong, but you missed one critical threshold comparison. Statistical significance is a statement about a testing procedure under assumptions; it is neither the chance that the null is true nor evidence of causal identification.

11.1 The role of an estimated standard error

If $\hat{\theta}$ estimates θ, many large-sample procedures use

$$Z = \frac{\hat{\theta} - \theta_0}{\widehat{SE}(\hat{\theta})} \overset{\text{approximately}}{\sim} N(0,1) \quad \text{under } H_0 : \theta = \theta_0.$$

A two-sided large-sample 95% confidence interval is $\hat{\theta} \pm 1.96\widehat{SE}(\hat{\theta})$. For $\hat{\theta} = 4$, estimated SE 1.5, the interval is $[1.06, 6.94]$. The procedure, used repeatedly under its assumptions, covers the fixed true parameter about 95% of the time. It does not assign a 95% probability to a fixed parameter after observing this one interval.

11.2 Reading the printed Z table

For any $Z \sim N(0,1)$, the number in our **Z-score table** is the *area to the left* of z: $\Phi(z) = P(Z \le z)$. When you want the right tail, use $1 - \Phi(z)$. For a negative score, symmetry gives $\Phi(-z) = 1 - \Phi(z)$. The standard normal is continuous, so $P(Z = z) = 0$ and $P(Z < z) = P(Z \le z)$.

ONE NUMBER, THREE QUESTIONS

Suppose the table reports $\Phi(1.23) = 0.8907$.

$P(Z \le 1.23) = 0.8907$	left of 1.23,
$P(Z > 1.23) = 1 - 0.8907 = 0.1093$	right of 1.23,
$P(\lvert Z \rvert \ge 1.23) = 2(0.1093) = 0.2186$	both outer tails.

Fast lane: the table is a cumulative probability, not a density height. If your area increases as you move right, you are looking at the correct convention. **Open the full table and illustrated guide.**

11.3 Computing a two-sided p-value

For a standard normal reference,

$$p = 2\{1 - \Phi(|z|)\}.$$

Worked example

For $\hat{\beta} = 3.0$ and $SE = 1.2$, the test of $H_0 : \beta = 0$ has $z = 2.5$ and $p \approx .0124$. Reject at 5% but not at 1%, because $2.5 > 1.96$ and $2.5 < 2.576$. Your earlier incorrect answer rejected at both thresholds; the distinction matters when reading published regression tables.

11.4 What a t test changes

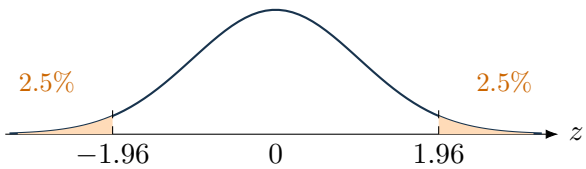
If a normal model has unknown variance and a suitable variance estimator with ν degrees of freedom, a t_ν reference accounts for estimating uncertainty. For small samples, t tails are wider than standard-normal tails. A confidence interval then uses $t_{\nu,1-\alpha/2}$ instead of 1.96. In complex regressions, approximate reference distributions and degrees-of-freedom choices depend on the estimator and dependence structure.

11.5 Type I errors, power, and multiple testing

A Type I error rejects a true null; its nominal rate is α under a valid test. Power is the probability of rejecting under a specified alternative. Testing 100 firms or event windows increases the chance of at least one false positive. For independent tests each at 5%, the probability of at least one rejection when all nulls hold is $1 - (.95)^{100}$, approximately 99.4%. Independence is a simplifying assumption, not a description of correlated securities.

```
beta <- 3; se <- 1.2
z <- beta/se
p <- 2*pnorm(-abs(z))
c(z=z, p=p, lower=beta-1.96*se, upper=beta+1.96*se)
```

A confidence interval is a method with repeated-sampling coverage



Fast lane: a useful calculation trick

For a two-sided normal test remember three landmarks: $|z| \approx 1.645$ (10%), 1.960 (5%), 2.576 (1%). A z score of 2.50 crosses the 5% threshold *but not* the 1% threshold—exactly the distinction that tripped you up in Session 2. These are *normal* critical values; small-sample t critical values differ.

Worked example

If $\hat{\beta} = 3$ and $\widehat{SE} = 1.2$, the two-sided z statistic for $H_0 : \beta = 0$ is 2.5, and $p \approx 0.0124$. Reject at 5%, not at 1%. The approximate 95% CI is $3 \pm 1.96(1.2) = [0.648, 5.352]$ and excludes zero.

REAL LIFE / The airport metal detector

An airport detector can make two mistakes: flag a harmless passenger or miss a prohibited item. Statistical testing also needs a stated decision rule and recognizes false positives and false negatives. A small p-value means the observed statistic is unusual under a specified null model; it does not tell you the probability that your preferred research story is true.

"A p-value can question a null hypothesis; it cannot certify your whole explanation.

Multiple-choice checkpoint

1. **[Foundation]** If $z = 2.3$ in a two-sided normal test, then
   A. reject at 1% and 5% B. reject at 5%, not 1% C. reject at neither D. p-value is 23%.
2. **[Application]** Under the usual repeated-sampling interpretation, a 95% confidence interval means
   A. 95% probability the fixed parameter lies in this observed interval B. 95% of future data points lie inside C. the method covers the fixed true parameter in about 95% of repeated samples D. the null is true with 5% probability.
3. **[Challenge]** If $\hat{\beta} = 2$, $SE = .5$, an approximate normal 95% interval is
   A. $[1.02, 2.98]$ B. $[1.5, 2.5]$ C. $[0, 4]$ D. $[.02, 3.98]$.

Answers and explanations: Appendix A, Chapter 11.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A normal z statistic equals 2.1 for a two-sided test. At the 5% level, what decision uses the standard normal reference?

A. Reject, since $2.1 > 1.96$ **B.** Reject, since $2.1 > 2.576$

C. Do not reject, since $2.1 < 2.576$ **D.** Do not reject because z is positive

P2. Foundation. For an estimate 5 with standard error 1, what is the approximate 95% normal confidence interval?

A. $[4, 6]$ **B.** $[3.04, 6.96]$

C. $[1.96, 5]$ **D.** $[5, 6.96]$

P3. Application. A two-sided normal-test p-value is 0.03. Which statement is correct?

A. Significant at 1% only **B.** Significant at 5% but not 1%

C. 3% probability the null is true **D.** 97% chance of a causal effect

P4. Challenge. A researcher tries 20 event windows and reports only the smallest p-value, without disclosing the search. What is the primary concern?

A. The estimator must be unbiased **B.** Multiplicity can inflate the chance of a false discovery

C. All p-values become exactly zero **D.** Event windows become independent by definition

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 12

Conditional Expectation and Linear Prediction

Intuition

Regression tries to explain how outcomes vary with information. Conditional expectation is the average outcome among observations with a specified value of a predictor; a regression line approximates that relationship in a restricted mathematical form.

12.1 Conditional expectation and the law of iterated expectations

$\mathbb{E}[Y \mid X = x]$ is the mean of Y in the subpopulation where $X = x$, when this conditional expectation is well defined. A crucial identity is

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y].$$

For discrete X, it follows by weighting each conditional mean by the probability of its X group. Also,

$$\mathbb{E}[Y - \mathbb{E}(Y \mid X) \mid X] = 0.$$

This is a population orthogonality fact about the true conditional expectation, not automatically an empirical regression condition.

12.2 Best linear prediction

Suppose we choose a, b to minimize $\mathbb{E}[(Y - a - bX)^2]$, assuming finite second moments and $\mathrm{Var}(X) > 0$. Differentiation gives two population normal equations:

$$\begin{aligned} \mathbb{E}[Y - a - bX] &= 0, \\ \mathbb{E}[X(Y - a - bX)] &= 0. \end{aligned}$$

Solve the first for $a = \mathbb{E}Y - b\mathbb{E}X$; substitute into the second to obtain

$$\boxed{b = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad a = \mathbb{E}Y - b\mathbb{E}X.}$$

This is the *best linear predictor* under squared error. It is a linear approximation even if the conditional expectation is curved.

12.3 The difference between an association and a structural effect

Suppose X denotes policy exposure and Y a return. The slope of the best linear predictor summarizes a covariance ratio. It is causal only if further assumptions justify treating variation in X as exogenous with respect to counterfactual outcomes. A statistically strong b can result from firm size, expected demand, or policy selection.

Worked example

If $\mathrm{Var}(X) = 4$ and $\mathrm{Cov}(X, Y) = 1.2$, the population best-linear-prediction slope is 0.3. If $\mathbb{E}X = 2$ and $\mathbb{E}Y = 5$, the intercept is $5 - 0.3(2) = 4.4$. The fitted conditional *linear prediction* is $4.4 + 0.3X$.

```
set.seed(14)
x <- rnorm(1000)
y <- 1 + 0.5*x + x^2 + rnorm(1000)
coef(lm(y ~ x))
# A line approximates a curved conditional mean.
```

The important trick: condition first, then average

Fast lane: a useful calculation trick

Remember the law of iterated expectations as **"average the averages"**: $E[Y] = E[E[Y \mid X]]$. In a two-group example, calculate each group's mean and weight it by the group's probability. This is especially handy when you know group means but not every individual observation.

Worked example

Suppose 40% of illustrative firms belong to group A with expected return 2%, and 60% are in group B with expected return -1%. The pooled expectation is $0.4(2\%) + 0.6(-1\%) = 0.2\%$. It is not the unweighted average $(2\% - 1\%)/2 = 0.5\%$ unless the groups are equally likely.

REAL LIFE / The crowded train

The average commute time on rainy days may be longer than on dry days, but the weather is mixed with other factors such as traffic, holidays and train delays. A conditional expectation asks for the average commute *within a given condition*. A regression is a systematic approximation to such conditional patterns. Causal statements need an additional argument about why the compared groups are otherwise comparable.

> *"Conditioning tells us whom we compared; identification tells us why that comparison means anything.*

Multiple-choice checkpoint

1. **[Foundation]** When $\mathrm{Var}(X) = 2$ and $\mathrm{Cov}(X, Y) = -1$, the best-linear-prediction slope equals
   A. -2 B. $-.5$ C. .5 D. 2.

2. **[Challenge]** If a regression coefficient is statistically significant, it is therefore causal.
A. Always B. Only if $n > 100$ C. Not without additional identification assumptions D. Only when $R^2 > 0.5$.

Answers and explanations: Appendix A, Chapter 12.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Suppose $E[Y \mid X = x] = 2 + 0.5x$. What is the conditional mean when $x = 6$?

A. 3 **B.** 5

C. 6 **D.** 8

P2. Foundation. If $\mathrm{Cov}(X, Y) = 6$ and $\mathrm{Var}(X) = 3$, what is the slope of the best linear predictor of Y from X?

A. 0.5 **B.** 2

C. 3 **D.** 18

P3. Application. Which condition is stronger than $\mathrm{Cov}(X, u) = 0$ and commonly used for the exogeneity discussion?

A. $E[u \mid X] = 0$ **B.** $\mathrm{Var}(X) = 0$

C. $E[X] = 0$ **D.** $P(X = 0) = 1$

P4. Challenge. A regression predicts higher returns for high-exposure firms, but exposure is chosen based on expected future demand. Which interpretation is justified by the regression alone?

A. Exposure necessarily causes high returns with exposure **B.** Predicted returns differ conditionally

C. Demand cannot confound the estimate **D.** The difference is experimentally identified

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review C: Probability and Statistical Inference Review

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. Suppose $P(A) = 0.3$, $P(B) = 0.4$, $P(A \cap B) = 0.1$. What is $P(A \cup B)$?

A. 0.1 **B.** 0.6

C. 0.7 **D.** 0.8

R2. Foundation. A random variable takes values 1 and 5 with equal probability. What is $E[X]$?

A. 2 **B.** 3

C. 4 **D.** 5

R3. Application. With standard deviation $\sigma = 20$ and independent sample size $n = 100$, what is $SE(\bar{X})$?

A. 0.2 **B.** 2

C. 10 **D.** 20

R4. Application. An estimate is 10 with normal standard error 2. Which is the approximate 95% confidence interval?

A. $[8, 12]$ **B.** $[6.08, 13.92]$

C. $[5, 15]$ **D.** $[9.5, 10.5]$

R5. Challenge. An event has prevalence 20%, a detector is positive for 90% of events and 10% of nonevents. Find $P(\text{event} \mid +)$.

A. 0.18 **B.** 0.50

C. 0.6923 **D.** 0.90

R6. Challenge. An estimate is 2.4 with SE 1.0. Under a two-sided standard normal test of zero, which decision is appropriate?

A. Reject at both 5% and 1% **B.** Reject at 5% but not 1%

C. Reject at neither 5% nor 1% **D.** The null is true with 2.4% probability

Answer key: see worked solutions for Review C.

Part III

Regression and Econometrics

CHAPTER 13

Simple Regression: Deriving Ordinary Least Squares

Intuition

Your earlier diagnostic asked you to minimize $\sum_i (y_i - \beta x_i)^2$ and you left the derivation blank. This chapter closes that specific gap. Understanding how OLS *optimizes* is separate from knowing whether OLS is unbiased or whether a coefficient is causal.

13.1 Through-the-origin model, every step

Begin with the fitted rule $\hat{y}_i = \beta x_i$ and errors $e_i(\beta) = y_i - \beta x_i$. Define

$$L(\beta) = \sum_{i=1}^{n} (y_i - \beta x_i)^2.$$

Expand each square, then separate sums:

$$\begin{aligned} L(\beta) &= \sum_i (y_i^2 - 2\beta x_i y_i + \beta^2 x_i^2) \\ &= \sum_i y_i^2 - 2\beta \sum_i x_i y_i + \beta^2 \sum_i x_i^2. \end{aligned} \tag{13.1}$$

Differentiate with respect to the *parameter* β, treating the observed x_i, y_i as constants:

$$\frac{dL}{d\beta} = -2 \sum_i x_i y_i + 2\beta \sum_i x_i^2.$$

Set this to zero and solve:

$$\boxed{\hat{\beta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}}.$$

The second derivative is $2 \sum_i x_i^2 > 0$ provided at least one $x_i \neq 0$. Therefore the minimizer is unique.

Worked example

For $x = (1, 2, 3)$ and $y = (2, 4, 5)$,

$$\sum x_i y_i = 2 + 8 + 15 = 25, \quad \sum x_i^2 = 1 + 4 + 9 = 14, \quad \boxed{\hat{\beta} = 25/14 \approx 1.786}.$$

Do not mistakenly use $\sum x_i$ in the denominator. The residuals $y_i - \hat{\beta} x_i$ are $(3/14, 6/14, -5/14)$; the *weighted* sum $\sum_i x_i e_i = 3/14 + 12/14 - 15/14 = 0$.

13.2 Adding an intercept

For a more realistic line, estimate $\hat{y}_i = a + b x_i$ by minimizing

$$L(a, b) = \sum_i (y_i - a - b x_i)^2.$$

The two first-order conditions are

$$\sum_i (y_i - a - b x_i) = 0,$$
$$\sum_i x_i (y_i - a - b x_i) = 0.$$

From the first, $a = \bar{y} - b\bar{x}$. Substitute into the second:

$$\sum_i x_i \left[(y_i - \bar{y}) - b(x_i - \bar{x}) \right] = 0.$$

Since $\sum_i (x_i - \bar{x}) = 0$, centering the first factor is valid. We obtain

$$\boxed{\hat{b} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{a} = \bar{y} - \hat{b}\bar{x}.}$$

The denominator must be positive. The fitted line goes through $(\bar{x}, \bar{y})$. With an intercept, the residuals sum to zero; without an intercept, that need not hold.

13.3 Unbiasedness is a different theorem

Suppose the *true* data-generating process is $y_i = \beta x_i + \varepsilon_i$. For the through-origin estimator,

$$\hat{\beta} = \beta + \frac{\sum_i x_i \varepsilon_i}{\sum_i x_i^2}.$$

Conditional on the entire observed regressor vector $\boldsymbol{x}$, if $\mathbb{E}[\varepsilon_i \mid \boldsymbol{x}] = 0$ for all i, then $\mathbb{E}[\hat{\beta} \mid \boldsymbol{x}] = \beta$. If this conditional-mean restriction fails, minimizing squared errors still works algebraically but need not recover the true parameter on average.

Semiconductor research connection

If exposure is measured only by whether a company appears on a policy list, that assignment may reflect firm characteristics. An OLS coefficient on that indicator is not automatically a policy effect. We must examine assignment mechanisms and simultaneous news.

```
x <- c(1,2,3); y <- c(2,4,5)
b0 <- sum(x*y)/sum(x^2)
coef(lm(y ~ 0 + x)) # exactly through the origin
coef(lm(y ~ x))     # includes an intercept
all.equal(unname(coef(lm(y ~ 0 + x))[1]), b0)
```

The best algebraic shortcut for simple OLS

Fast lane: a useful calculation trick

Once you know *why* OLS minimizes squared errors, the practical formula for a regression with intercept is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

A hand-computation shortcut uses $S_{xy} = \sum xy - n\bar{x}\bar{y}$ and $S_{xx} = \sum x^2 - n\bar{x}^2$. The two versions are identical; one is conceptually clearer, the other sometimes faster on a calculator.

Worked example

For $x = (1, 2, 3)$ and $y = (2, 4, 5)$, $\bar{x} = 2$, $\bar{y} = 11/3$, $S_{xy} = 3$, $S_{xx} = 2$. Hence $\hat{\beta}_1 = 3/2$ and $\hat{\beta}_0 = 11/3 - 3 = 2/3$. Predicted y values are $13/6$, $11/3$, and $31/6$. The residuals sum to zero—an immediate check on the intercept fit.

Semiconductor research connection

Before interpreting a dummy-variable coefficient measured in *decimal returns*, convert it to percentage *points*: -0.04 corresponds to a predicted -4-percentage-point return difference. It does not imply a 4% reduction in the underlying *share price*. Units are part of the answer.

REAL LIFE / Fitting a ruler to scattered points

Suppose you plot the time you spent studying and your quiz score across ten occasions. The points do not sit on one straight line, so you choose a line that stays close to all of them. OLS measures vertical gaps, squares them, and chooses the line minimizing their sum. Squaring keeps positive and negative misses from cancelling, while strongly penalizing very large misses.

> *"OLS is a precise answer to the modest question: which line misses these points the least, by squared vertical distance?*

Multiple-choice checkpoint

1. **[Foundation]** For through-origin OLS, $\sum_i x_i y_i = 12$ and $\sum_i x_i^2 = 8$ imply $\hat{\beta} =$
   A. 0.67 B. 1.5 C. 4 D. 20.

2. **[Application]** In OLS with an intercept, which sample relationship always holds?
   A. $\sum_i \hat{e}_i = 0$ B. $\hat{\beta}$ is causal C. $\text{Var}(\hat{e}) = 0$ D. errors are independent.

3. **[Challenge]** Which additional assumption supports conditional unbiasedness of OLS in a correctly specified linear model?
   A. Large fitted values B. $R^2 > 0.7$ C. Zero conditional mean of errors D. Normally distributed X.

Answers and explanations: Appendix A, Chapter 13.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Observed $y = (2, 4)$ and fitted values $\hat{y} = (1, 5)$. What is the residual sum of squares?

A. 0 **B.** 1

C. 2 **D.** 4

P2. Foundation. In the through-origin model $y_i = \beta x_i + u_i$, suppose $\sum x_i y_i = 18$ and $\sum x_i^2 = 9$. Find $\hat{\beta}$.

A. $1/2$ **B.** 2

C. 9 **D.** 162

P3. Application. In an OLS regression with an intercept, $\bar{x} = 3$, $\bar{y} = 10$, and $\hat{\beta}_1 = 2$. Find the intercept.

A. 4 **B.** 6

C. 8 **D.** 16

P4. Challenge. With data $(x, y) = (1, 1), (2, 3), (3, 5)$, what is the simple OLS slope with an intercept?

A. 1 **B.** $3/2$

C. 2 **D.** 3

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 14

Multiple Regression and the Matrix OLS Derivation

Intuition

Scalar OLS becomes matrix OLS when we estimate several coefficients simultaneously. The derivation is the same minimization with vectors replacing individual observations. This chapter is the bridge from undergraduate mathematics to graduate econometrics.

14.1 Data shapes and model

Let $\boldsymbol{y} \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times k}$ (including an intercept column if desired), $\boldsymbol{\beta} \in \mathbb{R}^k$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$. Then

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \hat{\boldsymbol{y}} = X\hat{\boldsymbol{\beta}}.$$

k counts parameters, not observations. $X^\mathsf{T}X$ is $k \times k$; $X^\mathsf{T}\boldsymbol{y}$ is $k \times 1$.

14.2 Step-by-step derivation

Minimize the squared Euclidean residual length:

$$L(\boldsymbol{\beta}) = (\boldsymbol{y} - X\boldsymbol{\beta})^\mathsf{T}(\boldsymbol{y} - X\boldsymbol{\beta}).$$

Expand, applying $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$:

$$\begin{aligned} L(\boldsymbol{\beta}) &= \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - \boldsymbol{y}^\mathsf{T}X\boldsymbol{\beta} - \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\boldsymbol{y} + \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}X\boldsymbol{\beta} \\ &= \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\boldsymbol{y} + \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}X\boldsymbol{\beta}. \end{aligned}$$

The middle two expressions are equal because each is a scalar and they are transposes of each other. For symmetric A, $\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\mathsf{T}A\boldsymbol{\beta}) = 2A\boldsymbol{\beta}$, so

$$\nabla L = -2X^\mathsf{T}\boldsymbol{y} + 2X^\mathsf{T}X\boldsymbol{\beta}.$$

Setting the gradient to zero gives the *normal equations*:

$$X^\mathsf{T}X\hat{\boldsymbol{\beta}} = X^\mathsf{T}\boldsymbol{y}.$$

If X has full column rank, $X^\mathsf{T}X$ is invertible and

$$\boxed{\hat{\boldsymbol{\beta}} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}\boldsymbol{y}.}$$

The Hessian is $2X^\mathsf{T}X$, which is positive definite under full column rank. Thus the solution is the unique global minimum.

14.3 A concrete two-regressor interpretation

Suppose $R_i = \beta_0 + \beta_1 D_i + \beta_2 S_i + \varepsilon_i$, where D_i is restriction exposure and S_i is firm size. $\hat{\beta}_1$ is the fitted difference in R for D changing by one unit *holding S fixed*. This "partial" interpretation uses variation in D not linearly predicted by the other columns. It does not ensure true comparability between exposed and unexposed firms.

14.4 Frisch–Waugh–Lovell intuition

Partition $X = [X_1 \ \boldsymbol{x}_2]$, where X_1 includes all controls and possibly an intercept. Let $M_1 = I - X_1(X_1^\mathsf{T} X_1)^{-1} X_1^\mathsf{T}$. The coefficient on $\boldsymbol{x}_2$ is

$$\hat{\beta}_2 = \frac{(M_1 \boldsymbol{x}_2)^\mathsf{T} (M_1 \boldsymbol{y})}{(M_1 \boldsymbol{x}_2)^\mathsf{T} (M_1 \boldsymbol{x}_2)}.$$

In words: remove the linear component explained by controls from both outcome and treatment regressor; regress what remains. This is the partial-regression theorem, given the required rank conditions.

Research caution

Writing $(X^\mathsf{T} X)^{-1}$ is mathematically illuminating; explicitly forming it is not the preferred numerical algorithm. Regression software typically uses stable decompositions such as QR. A highly collinear design can cause large uncertainty even when inversion is formally possible.

```
set.seed(29)
n <- 80
D <- rbinom(n, 1, .5)
S <- rnorm(n)
Y <- .02 - .03*D + .01*S + rnorm(n, sd=.02)
X <- model.matrix(~ D + S)
manual <- solve(crossprod(X), crossprod(X,Y))
automatic <- coef(lm(Y ~ D + S))
stopifnot(isTRUE(all.equal(drop(manual), unname(automatic))))
```

Matrix OLS: the derivation in four memorable moves

Fast lane: a useful calculation trick

1. Residual: $e = y - X\beta$. **2. Square:** $L = e^\mathsf{T} e$.
3. Differentiate: $\nabla_\beta L = -2X^\mathsf{T}(y - X\beta)$.
4. Solve: $X^\mathsf{T} X\hat{\beta} = X^\mathsf{T} y$. If X has full column rank, $\hat{\beta} = (X^\mathsf{T} X)^{-1} X^\mathsf{T} y$. This is *one* reusable derivation rather than an isolated formula to memorize.

Worked example

Check the dimensions before every equality. If X is $n \times k$ and y is $n \times 1$, then $X^\mathsf{T} X$ is $k \times k$, $(X^\mathsf{T} X)^{-1} X^\mathsf{T} y$ is $k \times 1$, and $X\hat{\beta}$ is $n \times 1$. Dimensions are surprisingly effective at catching algebra errors.

Research caution

Computationally, never implement OLS by hand-inverting $X^\top X$ unless the goal is purely pedagogical: explicit inversion can be unstable. In R use `lm()`, `qr.solve(X, y)`, or another numerically appropriate solver.

REAL LIFE / The research spreadsheet gets crowded

With one predictor you can sketch a line. With company size, industry exposure, currency movement, market return and year indicators, the same calculation becomes a many-column problem. Matrix OLS is the compact bookkeeping that handles all coefficients simultaneously. The notation looks intimidating only until you check the dimensions of every multiplication.

> *"When regression grows more columns, the arithmetic does not change species; it changes stationery.*

Multiple-choice checkpoint

1. [**Foundation**] If X is 100×4, then $X^\top X$ is
   A. 100×100 B. 4×4 C. 100×4 D. 4×100.
2. [**Application**] The OLS normal equations are
   A. $X^\top(y - X\hat{\beta}) = 0$ B. $X(y - X\hat{\beta}) = 0$ C. $X\hat{\beta} = y$ always D. $X^\top X = I$ always.
3. [**Challenge**] Full column rank is needed for the conventional matrix inverse formula because
   A. it makes data normal B. it eliminates errors C. it guarantees $X^\top X$ is invertible D. it guarantees exogeneity.

Answers and explanations: Appendix A, Chapter 14.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. If X has n rows and $k = 3$ columns, what are the dimensions of $X^\top X$?

A. $n \times n$ **B.** 3×3
C. $n \times 3$ **D.** $3 \times n$

P2. Foundation. The first-order condition for minimizing $(y - Xb)^\top (y - Xb)$ is which equation?

A. $X^\top (y - X\hat{b}) = 0$ **B.** $y^\top X = 0$
C. $X\hat{b} = 0$ **D.** $\hat{b}^\top \hat{b} = 1$

P3. Application. If $X^\top X = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ and $X^\top y = (6, 8)^\top$, find $\hat{\beta}$.

A. $(3, 2)^\top$ **B.** $(12, 32)^\top$
C. $(1/3, 1/2)^\top$ **D.** $(6, 8)^\top$

P4. Challenge. When can the usual OLS formula $(X^\top X)^{-1} X^\top y$ be used with a unique coefficient solution?

A. Whenever $n \geq 1$ **B.** When X has full column rank
C. Only when all residuals are zero **D.** Only when y is normally distributed

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 15

Endogeneity and Omitted-Variable Bias

Intuition

Your only wrong answer in Session 3 was the idea that minimizing squared errors makes OLS unbiased. This is the most important conceptual correction in the book. Best fit, unbiasedness and causal identification are three different claims.

15.1 From the estimator to its error

Assume $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u}$ with full-column-rank X. Substitute this into the OLS formula:

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}(X\boldsymbol{\beta} + \boldsymbol{u}) \\ &= \boldsymbol{\beta} + (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}\boldsymbol{u}.\end{aligned}$$

Thus conditional unbiasedness is ensured when $\mathbb{E}[\boldsymbol{u} \mid X] = 0$ (under suitable existence conditions), since

$$\mathbb{E}[\hat{\boldsymbol{\beta}} \mid X] = \boldsymbol{\beta} + (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}\mathbb{E}[\boldsymbol{u} \mid X] = \boldsymbol{\beta}.$$

A common weaker moment target is $\mathbb{E}[X_i u_i] = 0$ in appropriate settings; this supports consistency with regularity conditions. Nonzero correlation between X and the disturbance is a familiar sign of endogeneity. Importantly, even an unbiased coefficient need not equal a *causal* estimand unless the model's interpretation and assignment assumptions are justified.

15.2 Deriving omitted-variable bias

Suppose the data-generating relationship is

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + u, \quad \operatorname{Cov}(X, u) = 0, \quad \operatorname{Var}(X) > 0.$$

We wrongly regress Y only on an intercept and X. The population linear-projection slope is

$$\begin{aligned}\tilde{\beta}_1 &= \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)} \\ &= \frac{\beta_1 \operatorname{Var}(X) + \beta_2 \operatorname{Cov}(X, Z) + \operatorname{Cov}(X, u)}{\operatorname{Var}(X)} \\ &= \beta_1 + \beta_2 \frac{\operatorname{Cov}(X, Z)}{\operatorname{Var}(X)}.\end{aligned}$$

So the omitted-variable distortion is

$$\boxed{\tilde{\beta}_1 - \beta_1 = \beta_2 \frac{\mathrm{Cov}(X, Z)}{\mathrm{Var}(X)}.}$$

Two ingredients are required: the omitted Z affects Y ($\beta_2 \neq 0$) and is related to X ($\mathrm{Cov}(X, Z) \neq 0$). When both are positive, the distortion is upward.

Worked example

If $\beta_1 = -0.02$, $\beta_2 = 0.04$, $\mathrm{Cov}(X, Z) = 0.3$, and $\mathrm{Var}(X) = 0.6$, then the omitted-variable term is $0.04(0.3/0.6) = 0.02$, and the population reduced-regression slope is zero. A real negative conditional effect can be masked by a positive confounder in this constructed example.

15.3 Five sources of endogeneity or identification difficulty

Confounding: a common cause affects exposure and returns. **Reverse causation:** outcomes affect exposure decisions. **Measurement error:** recorded exposure differs from the target construct, with bias depending on the error process. **Selection:** included firms or periods are not comparable to excluded ones. **Simultaneous shocks:** earnings or market information coincides with policy news. These mechanisms are not automatically fixed by adding more observations, control variables, or factor regressions.

15.4 Potential outcomes and an explicit causal target

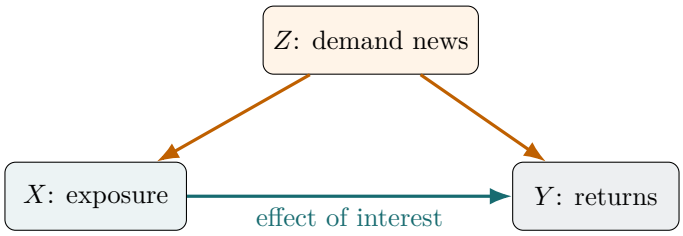
Write $Y_i(1)$ as the return under an announcement exposure and $Y_i(0)$ as the return under no announcement exposure, with the treatment and timing clearly defined. The average treatment effect is $ATE = \mathbb{E}[Y(1) - Y(0)]$. We cannot observe both outcomes for the same unit under the same circumstances. Causal identification provides assumptions under which observed comparisons can recover an estimand involving these missing counterfactuals. In public-policy event studies, defining the treatment at the level of the *information shock* is itself nontrivial.

Semiconductor research connection

For export-control news, distinguish announcement date, implementation date, and dates when investors may already have anticipated the policy. Exogeneity of a narrow announcement-day innovation is a separate question from the exogeneity of firms' long-run exposure.

```r
set.seed(9)
n <- 5000
Z <- rnorm(n)
X <- 0.7*Z + rnorm(n)
Y <- -0.02*X + 0.05*Z + rnorm(n, sd=.2)
coef(lm(Y ~ X))["X"]        # omits a common driver
coef(lm(Y ~ X + Z))["X"]    # includes that driver
```

A causal diagram worth drawing before any regression



If a pre-existing factor Z affects both the explanatory variable and the outcome, the observed X–Y association can combine the effect of X with a back-door path $X \leftarrow Z \rightarrow Y$. The diagram illustrates a hypothetical confounder, not a claim about a specific real-world announcement.

Fast lane: a useful calculation trick

Bias sign in one line: in simple OVB, *sign of bias = sign of effect of Z on Y × sign of correlation between X and Z*. Positive times positive means upward bias; positive times negative means downward bias. If either relevant factor is zero, this simplified bias is zero.

Worked example

Suppose $\beta_Z = 2$, $\mathrm{Cov}(X, Z) = 3$, and $\mathrm{Var}(X) = 6$. The omitted-variable bias is $2(3/6) = 1$: the population short-regression slope is one unit above the structural β_X under the stated model. OLS's algebra can be perfect while the identification argument is wrong.

REAL LIFE / Ice cream and sunburn

Ice-cream sales and sunburn cases may rise together. It would be absurd to conclude that buying ice cream causes sunburn merely from this correlation: hot weather influences both. Omitted-variable bias is the formal version of that story. In political economy, expected industry demand could similarly influence both exposure to trade restrictions and subsequent firm performance.

> *"The largest coefficient in the room may be standing on an invisible variable's shoulders.*

Multiple-choice checkpoint

1. **[Foundation]** Z has a positive effect on Y and positive correlation with X. Omitting Z tends to bias the slope of X
   A. downward B. upward C. to exactly zero D. not at all because OLS minimizes SSR.
2. **[Application]** Which condition represents a standard conditional-mean exogeneity assumption?
   A. $\mathbb{E}[u \mid X] = 0$ B. $\mathbb{E}[X] = 0$ C. $R^2 = 1$ D. $X^\mathsf{T}X = 0$.
3. **[Challenge]** A 95% significant association is necessarily a causal effect.
   A. Yes B. Yes if sample is large C. No, identification requires additional assumptions D. Yes with robust standard errors.

Answers and explanations: Appendix A, Chapter 15.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. True coefficient on X is 2, but omitting a positively related confounder creates bias $+0.5$. What is the resulting population regression slope?

A. 1.5 **B.** 2.0

C. 2.5 **D.** 4.0

P2. Foundation. Which description matches endogeneity in a simple regression?

A. A regressor correlates with the error term **B.** The outcome has positive variance

C. The sample mean is nonzero **D.** The model includes an intercept

P3. Application. In the omitted-variable expression, $\beta_Z = -3$ and $\mathrm{Cov}(X, Z)/\mathrm{Var}(X) = 2$. What is the bias from omitting Z?

A. -6 **B.** -1.5

C. $+1.5$ **D.** $+6$

P4. Challenge. A firm announces earnings unexpectedly on the same day a government announces export controls. Why may a negative event-study return not isolate the policy effect?

A. OLS forces a negative coefficient **B.** Simultaneous earnings news can also affect returns

C. Negative returns are mathematically impossible **D.** Changing standard errors automatically solves confounding

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 16

Regression Uncertainty, Heteroskedasticity, and Robust Standard Errors

Intuition

A coefficient is a point estimate. To decide how much uncertainty surrounds it, we need the distribution of estimation error. Robust standard errors change estimates of uncertainty; they do *not* repair omitted-variable bias.

16.1 Classical conditional variance of OLS

Under $\mathbb{E}[\boldsymbol{u} \mid X] = 0$ and $\mathrm{Var}(\boldsymbol{u} \mid X) = \sigma^2 I_n$, the sampling variance of OLS conditional on X is

$$\begin{aligned}\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) &= \mathrm{Var}\left((X^\mathsf{T}X)^{-1}X^\mathsf{T}\boldsymbol{u} \mid X\right) \\ &= (X^\mathsf{T}X)^{-1}X^\mathsf{T}(\sigma^2 I)X(X^\mathsf{T}X)^{-1} \\ &= \boxed{\sigma^2(X^\mathsf{T}X)^{-1}}.\end{aligned}$$

If X has k linearly independent columns, an estimator of σ^2 under standard assumptions is $\hat{\sigma}^2 = \sum_i \hat{u}_i^2/(n-k)$. A conventional coefficient standard error is the square root of the relevant diagonal entry of $\hat{\sigma}^2(X^\mathsf{T}X)^{-1}$.

16.2 Heteroskedasticity and the sandwich

If $\mathrm{Var}(\boldsymbol{u} \mid X) = \Omega$ is not a scalar multiple of the identity, the correct variance expression becomes

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\mathsf{T}X)^{-1}X^\mathsf{T}\Omega X(X^\mathsf{T}X)^{-1}.$$

Under independent observations with potentially different error variances, the HC0 sample estimate is

$$\widehat{V}_{HC0} = (X^\mathsf{T}X)^{-1}\left(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i^\mathsf{T}\hat{u}_i^2\right)(X^\mathsf{T}X)^{-1}.$$

The left and right "bread" matrices surround the central "meat" of estimated error variability. HC1, HC2 and HC3 apply various finite-sample adjustments; no one method guarantees perfect inference in all samples.

16.3 When residuals correlate within a firm

In panel data, u_{it} can be serially dependent within firm i. A firm-clustered variance estimator aggregates score contributions by firm, not by observation:

$$\widehat{V}_{cl} = (X^\top X)^{-1} \left[\sum_{g=1}^{G} (X_g^\top \hat{\boldsymbol{u}}_g)(X_g^\top \hat{\boldsymbol{u}}_g)^\top \right] (X^\top X)^{-1},$$

possibly with finite-cluster corrections. Clustering at the firm level does not automatically handle common shocks across firms on the same day; two-way clustering or another design-specific approach may be appropriate. A small number of clusters creates additional problems.

16.4 What robust standard errors do and do not change

Changing covariance estimators leaves the original OLS $\hat{\beta}$ unchanged. It can change standard errors, t statistics, confidence intervals, and p-values. It does not solve omitted variables, reverse causality, or measurement error. Similarly, heteroskedasticity may invalidate conventional OLS standard errors without invalidating coefficient consistency under suitable exogeneity and moment assumptions.

```r
# Requires one optional CRAN package for robust covariance.
# install.packages("sandwich")
set.seed(5)
n <- 400; x <- rnorm(n)
y <- 1 + 2*x + rnorm(n, sd=0.5+abs(x))
fit <- lm(y ~ x)
summary(fit)$coef
if (requireNamespace("sandwich", quietly=TRUE)) {
  sqrt(diag(sandwich::vcovHC(fit, type="HC3")))
}
```

Robust does not mean magically unbiased

Fast lane: a useful calculation trick

When you change from conventional to heteroskedasticity-robust standard errors in a fixed OLS fit, remember this split: **same point estimates, different uncertainty estimate**. This works because the *fitted coefficients* solve the same least-squares problem; only the estimator of their covariance matrix changes.

Worked example

Suppose $\hat{\beta} = 0.04$. Conventional $SE = 0.015$ gives $t \approx 2.67$; robust $SE = 0.025$ gives $t = 1.60$. The magnitude 0.04 has not changed at all, yet the evidence against $\beta = 0$ is much weaker under the robust uncertainty estimate. Robust SE do not cure endogeneity or measurement error.

REAL LIFE / A noisy measuring tape

Two measuring tapes can report the same length but have very different accuracy. Two regression procedures can likewise produce exactly the same slope yet attach different uncertainty to it. Robust standard errors change how we estimate the noise around an estimator; they do not change the slope calculated by ordinary least squares from the same data.

> *"A standard error is the warning label on an estimate, not a repair kit for a bad research design.*

Multiple-choice checkpoint

1. [**Foundation**] Replacing conventional OLS standard errors with HC3 without refitting the model changes
A. only the sample size B. coefficient values necessarily C. uncertainty estimates, potentially tests D. the causal validity automatically.

2. [**Challenge**] Under homoskedastic independent disturbances conditional on X, $\mathrm{Var}(\hat{\beta} \mid X)$ is
A. $\sigma^2(X^\mathsf{T}X)^{-1}$ B. $\sigma^2 X^\mathsf{T}X$ C. $\sigma^2 I_n$ D. $X^\mathsf{T}y$.

Answers and explanations: Appendix A, Chapter 16.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A coefficient estimate is 0.6 with standard error 0.2. What is its t statistic for testing zero?

A. 0.12 **B.** 0.33

C. 3 **D.** 6

P2. Foundation. Replacing conventional OLS standard errors with heteroskedasticity-robust ones, while keeping the same sample and OLS estimator, changes which element?

A. Estimated slopes automatically **B.** The outcome observations

C. The reported uncertainty and test statistics **D.** Number of firms

P3. Application. Under independent homoskedastic errors with variance σ^2, the conditional variance of OLS is which matrix?

A. $\sigma^2(X^\top X)^{-1}$ **B.** $\sigma^2 X^\top X$

C. $(X^\top y)^{-1}$ **D.** $\sigma^2 I_n$

P4. Challenge. A policy event affects many firms on one date, causing correlated residuals across firms. Which concern is most directly raised?

A. More observations guarantee independent information **B.** Naive firm-by-firm standard errors may be overstated or understated because cross-firm covariance matters

C. Dependence proves there is no mean effect **D.** Every coefficient necessarily becomes biased

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 17

Model Diagnostics and Honest Interpretation

Intuition

A fitted model can be numerically correct yet substantively misleading. Diagnostics ask whether our specification, uncertainty estimates, and causal story are plausible enough for the intended conclusion.

17.1 Residual structure, leverage, and influential observations

For regression matrix X of full column rank, the hat matrix $H = X(X^\mathsf{T}X)^{-1}X^\mathsf{T}$ maps y to fitted values. The diagonal h_{ii} measures leverage, arising from unusual regressor combinations. A high-leverage point is not necessarily influential unless its outcome also pulls the fit. Residual-versus-fitted plots can reveal nonlinearity, changing spread, and outliers; they do not establish the absence of confounding.

17.2 Fit is not identification

A model's coefficient of determination for an intercept-containing OLS regression is

$$R^2 = 1 - \frac{\sum_i \hat{e}_i^2}{\sum_i (y_i - \bar{y})^2}.$$

It measures reduction of in-sample squared error relative to predicting the sample mean. High R^2 may reflect strong shared trends or fixed effects; low R^2 does not automatically invalidate a credible causal design. Additional regressors cannot increase training SSR under OLS but may overfit prediction and obscure identification.

17.3 Multicollinearity and uncertainty

When two regressors move together, it becomes harder to estimate their separate partial associations. Perfect collinearity prevents unique OLS coefficients; near collinearity can inflate variances. For a two-regressor illustration, the variance of the slope on x_1 under homoskedasticity is related to $1/[\sum(x_{1i} - \bar{x}_1)^2(1 - R^2_{1|2})]$, where $R^2_{1|2}$ measures the fit of x_1 on the other regressors. As that fit approaches one, uncertainty grows.

17.4 A structured specification audit

Before reporting coefficients, record: outcome and unit; treatment/exposure construction; timing; sample selection; missingness; intended estimand; covariates chosen *before* outcomes are inspected; uncertainty estimator; alternative comparison windows; confounding events; sensitivity analyses; and limitations. Changes after seeing results must be labeled exploratory.

Semiconductor research connection

For an export-control event, checking an earnings-calendar overlap may be more important to interpretation than optimizing R^2. A factor model accounts for systematic co-movement; it cannot magically remove unexpected firm-specific earnings news on the same day.

```
set.seed(11)
x <- rnorm(100); z <- 0.95*x + rnorm(100, sd=.1)
y <- 1 + 2*x + rnorm(100)
fit <- lm(y ~ x + z)
summary(fit)
plot(fitted(fit), resid(fit), xlab="Fitted", ylab="Residual")
abline(h=0, lty=2)
hatvalues(fit)[1:5]
```

A five-second diagnostic for any fitted regression

Fast lane: a useful calculation trick

When an output table looks impressive, ask in this order: **units, sample, model, assumptions, uncertainty**. A small p-value says something conditional on the model and its inference assumptions; it does not answer whether treatment assignment was as-good-as-random. This short checklist is more valuable than memorizing a long list of significance stars.

Semiconductor research connection

For daily financial returns, ask whether residuals cluster by firm, by announcement date, or both; an apparent sample of hundreds of firm-days may contain only a handful of independent news shocks. Standard errors must respect the dependence structure supported by the research design.

REAL LIFE / A perfect practice test

Imagine memorizing 100 practice questions and scoring 100%, then failing a new exam. Excellent fit to what you have already seen says little about new situations. In-sample R^2 is useful for describing explained variation, but it does not prove out-of-sample validity or causation. Check residual patterns, multicollinearity, changing samples and robustness.

"A model that memorizes yesterday may be a poor guide to tomorrow.

Multiple-choice checkpoint

1. [**Foundation**] Does very high R^2 ensure that the coefficient on export-control exposure is causal?
A. Yes B. Only with 100 observations C. No D. Only for daily returns.

2. [**Challenge**] Near-perfect correlation between two predictors often leads to
A. more precise separate slopes B. greater coefficient uncertainty C. automatic randomization D. zero residuals.

Answers and explanations: Appendix A, Chapter 17.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A model achieves $R^2 = 0.95$. What is justified?

A. Its causal identification is proven **B.** It explains 95% of the sample outcome variation in the usual intercept-based definition

C. All residuals are zero **D.** Every slope is statistically significant

P2. Foundation. If observed $y = (2, 6)$ and predictions are $\hat{y} = (3, 5)$, compute the sum of squared residuals.

A. 0 **B.** 1

C. 2 **D.** 4

P3. Application. Two regressors are nearly linearly dependent. What typically happens to estimates of their separate slopes?

A. Their standard errors may become large **B.** Their coefficients must sum to one

C. OLS is automatically unbiased **D.** Their standard errors always go to zero

P4. Challenge. A model fits the estimation sample exceptionally well but predicts poorly in later periods. What is a defensible interpretation?

A. High in-sample fit does not guarantee out-of-sample generalization **B.** The algebra of OLS is wrong

C. Future data must be corrupted **D.** Goodness of fit establishes policy causation

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review D: Regression and Identification Review

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. In a through-origin regression, $\sum x_i y_i = 30$ and $\sum x_i^2 = 15$. What is OLS slope?

A. $1/2$ **B.** 2

C. 15 **D.** 45

R2. Foundation. A dummy regression coefficient is -0.03 for outcome returns in decimals. What is its conditional predicted difference?

A. -0.03 percentage points **B.** -3 percentage points

C. -30 percentage points **D.** -3 percent of the price level

R3. Application. If the omitted-variable coefficient is $+2$ and its association ratio is -0.4, what is the bias?

A. $+0.8$ **B.** -0.8

C. -2.4 **D.** $+2.4$

R4. Application. A coefficient is -0.8 with standard error 0.4. What is its two-sided normal z statistic testing zero?

A. -2 **B.** -0.5

C. 0.5 **D.** 2

R5. Challenge. Why doesn't a low residual sum of squares prove exogeneity?

A. Fitting minimizes observed errors but does not restrict correlation of regressors with unobserved causes **B.** Low fit implies no causal effect

C. Sum of squared residuals cannot be negative **D.** The number of regressors must equal sample size

R6. Challenge. A regressor is exactly a sum of two other design-matrix columns. What is the algebraic problem?

A. Exact collinearity makes $X^\top X$ singular **B.** OLS must achieve $R^2 = 1$

C. Every coefficient is significant **D.** Robust standard errors become zero

Answer key: see worked solutions for Review D.

Part IV

Applications to Political Science Research

CHAPTER 18

Event-Study Design and Return Measurement

Intuition

An event study compares actual security returns with a benchmark for what returns might have been absent unusual information. The hardest part is not subtraction. It is specifying the event, its timing, the expected-return benchmark, and the assumptions needed for interpretation.

18.1 The central estimand and event time

Index firms by i and exchange-specific trading dates by t. For an event e define event time $\tau = 0$ as the first relevant trading session when information could be reflected in that market. An event window $[\tau_1, \tau_2]$ might be $[-1, +1]$ or $[0, +1]$; different windows answer different timing questions. Define simple return

$$R_{it} = \frac{P_{it}^{adj}}{P_{i,t-1}^{adj}} - 1,$$

where P^{adj} accounts appropriately for splits and distributions. Log return is $r_{it} = \ln(P_{it}^{adj}/P_{i,t-1}^{adj})$. Simple returns accumulate multiplicatively over time; log returns add over time. The choice must stay explicit throughout the paper.

18.2 Estimation window versus event window

An *estimation window* uses pre-event trading days to fit a benchmark such as $R_{it} = \alpha_i + \beta_i R_{mt} + u_{it}$. An *event window* uses these pre-estimated coefficients to predict expected returns surrounding the policy news. Leaving a separation or buffer between estimation and event windows can reduce information leakage. For example, estimation $[-140, -21]$ and event $[-1, +1]$ are *illustrative choices*, not universal standards; the appropriate choice depends on trading liquidity, sample availability, earlier announcements and robustness.

18.3 Constructing the event calendar

A transparent machine-readable event dataset should record an event identifier, announcement timestamp with timezone, issuing jurisdiction, policy instrument, affected products, whether the action is a proposal or final rule, public-information source, implementation date and

anticipated implementation, exchange-specific first-response date, overlapping corporate announcements, and coding uncertainty. Distinguish US announcements, ally announcements and coordinated events to avoid treating intertwined policy actions as independent shocks.

Worked example

Suppose an announcement is released after a Japanese exchange closes, while European shares are still trading. The Japanese first-response trading date may be the next session; a European first-response date may be the same calendar day. Both can have event time $\tau = 0$ despite different civil dates. Define times in local zones and convert to a common instant for matching.

18.4 Sampling and anticipation

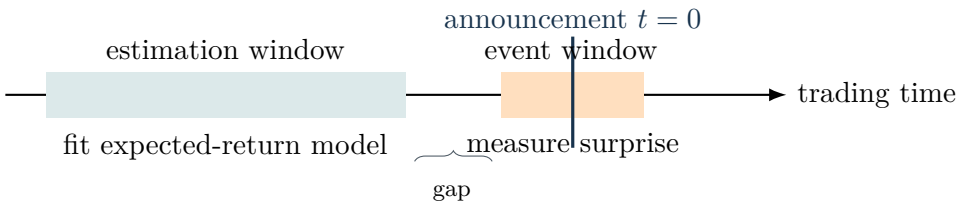
Choose firms using pre-specified industry role and listing criteria rather than observed returns. Firms may be exposed directly via controlled technology, indirectly via customer demand, or through supply-chain substitution. These are different mechanisms requiring different exposure definitions. If a restriction was widely anticipated, announcement-day abnormal returns can be close to zero even when the policy has important long-run economic effects.

Research caution

Abnormal return is *model-relative*. A nonzero value is not automatically a causal export-control effect. Earnings announcements, guidance, macroeconomic news, AI-market developments, and leaks can coincide. For cross-border firms, use suitable local market returns, currencies and trading calendars.

```
prices <- c(100, 102, 99, 100)
ret_simple <- prices[-1] / prices[-length(prices)] - 1
ret_log <- diff(log(prices))
stopifnot(isTRUE(all.equal(prod(1 + ret_simple) - 1,
                           tail(prices, 1)/prices[1] - 1)))
```

See the trading-day windows before writing an equation



Fast lane: a useful calculation trick

Build the event calendar *before* calculating returns. For each announcement store both its timestamp and the local market close. News after close may first affect the *next* trading session. Never equate the calendar date with market event day without checking the exchange calendar and time zone.

Semiconductor research connection

The diagram is schematic; estimation windows, gaps, and event windows must be chosen and justified for the actual event study. Do not infer event timing from this cartoon.

REAL LIFE / The announcement arrived at 5 p.m.

A Japanese trading session ends before an American policy announcement published late at night in Tokyo. A calendar date is not automatically a trading date: the first observed market reaction may appear in the next session. Event-study design starts with a timestamp, exchange calendar, information-release mechanism and well-justified event window. Getting the clock wrong can make good mathematics estimate the wrong event.

"In an event study, the clock is part of the model.

Multiple-choice checkpoint

1. **[Foundation]** A Japanese company receives material US policy news after Tokyo's exchange closes. Which event-day rule is most defensible?
   A. Always use announcement calendar day regardless of exchange B. Use first plausible Japanese trading session that can incorporate the information C. Always use implementation day D. Use whichever date produces largest CAR.
2. **[Challenge]** The estimation window is primarily for
   A. fitting the expected-return benchmark B. choosing significant firms C. computing policy implementation lag only D. proving causal exogeneity.

Answers and explanations: Appendix A, Chapter 18.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Which interval normally estimates a firm's expected-return model before an event?

A. Event window **B.** Estimation window

C. Post-policy sales period only **D.** Policy eligibility window

P2. Foundation. An announcement happens after Tokyo Stock Exchange trading closes on Monday. Which is the earliest plausible regular trading session for the market reaction?

A. Monday's already completed session **B.** Tuesday's session

C. Previous Friday **D.** No trading session could respond

P3. Application. A market model predicts $+1\%$ and a firm's realized return is -2%. What is abnormal return?

A. -3 percentage points **B.** -1 percentage point

C. $+1$ percentage point **D.** $+3$ percentage points

P4. Challenge. Why might an anticipated policy announcement generate little abnormal return on its official date?

A. Prices may already have incorporated the expected information **B.** Every policy has exactly zero economic effect

C. Markets cannot anticipate policy **D.** An event study measures annual sales instead of returns

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 19

Expected Returns: Market Models and Risk Factors

Intuition

A stock falling 3% on an announcement day may not be unusual if the entire market fell 3%. Expected-return models attempt to subtract the predictable component, but every model leaves estimation and specification uncertainty.

19.1 Mean-adjusted and market-adjusted benchmarks

A simple mean-adjusted model predicts the stock's own estimation-window average: $\hat{R}_{it} = \bar{R}_{i,est}$. A market-adjusted model uses $\hat{R}_{it} = R_{mt}$, implicitly imposing zero intercept and unit market beta. A fitted market model is more flexible:

$$R_{it} = \alpha_i + \beta_i R_{mt} + u_{it} \quad \text{(estimation window)},$$
$$\widehat{R}_{it} = \hat{\alpha}_i + \hat{\beta}_i R_{mt} \quad \text{(event window)}.$$

These are statistical benchmarks, not guaranteed counterfactual outcomes.

19.2 CAPM-style excess returns

Subtract a risk-free rate measured at the matching frequency and (where appropriate) currency:

$$R_{it} - R_{ft} = \alpha_i + \beta_i (R_{mt} - R_{ft}) + u_{it}.$$

The intercept in a time-series asset-pricing regression is often called alpha, but its financial interpretation depends on model assumptions. Be alert to simple versus log returns and whether data providers use adjusted closing prices or total-return indices.

19.3 Fama–French three-factor regression

A commonly used factor regression is

$$R_{it} - R_{ft} = \alpha_i + b_i (R_{mt} - R_{ft}) + s_i SMB_t + h_i HML_t + u_{it}.$$

SMB is a size factor, and HML a value-related factor. They are *return series*, not firm-level size or book-to-market columns. Factor and risk-free data must use consistent geography, sampling frequency, units and timestamps. A daily factor of 0.1 may mean *0.1 percentage points* in one data file but 0.1 in decimal returns elsewhere, a 100-fold difference.

19.4 Factor estimation and uncertainty

Estimate coefficients on pre-event data, then predict conditional benchmark returns in the event window. This generates abnormal returns

$$AR_{it} = (R_{it} - R_{ft}) - \left[\hat{\alpha}_i + \hat{b}_i(R_{mt} - R_{ft}) + \hat{s}_i SMB_t + \hat{h}_i HML_t\right].$$

When subtracting matched R_{ft} from actual and benchmark simple returns, the expression agrees with an appropriately defined total-return comparison. Estimation uncertainty in factor betas affects event-study inference; treating the predicted benchmark as known exactly can understate variance.

Worked example

Let a fitted model have $\hat{\alpha} = 0$, $\hat{b} = 1.2$, $\hat{s} = .4$, $\hat{h} = -.2$. If market excess return is -1%, $SMB = 0.5\%$, and $HML = -0.5\%$, fitted excess return is $1.2(-1) + .4(.5) - .2(-.5) = -0.9\%$. If actual excess return is -3.0%, abnormal excess return is -2.1 percentage points. Units must be consistent.

Semiconductor research connection

A model fitted to US traded factors may not automatically be appropriate for a Japanese stock. Using a market model or suitable regional factor series is a model-selection decision to defend empirically, not a universal hierarchy in which more factors always solve the problem.

```
set.seed(30)
n <- 120
market_excess <- rnorm(n, sd=.01)
SMB <- rnorm(n, sd=.005)
HML <- rnorm(n, sd=.005)
firm_excess <- 1.2*market_excess + .4*SMB - .2*HML +
  rnorm(n, sd=.006)
fit <- lm(firm_excess ~ market_excess + SMB + HML)
coef(fit)
```

How to read a factor model in one breath

Fast lane: a useful calculation trick

Read the excess-return regression left to right: **what the firm earned above cash = constant component + market exposure + size exposure + value exposure + unexplained shock**. A coefficient describes sensitivity to a factor, not necessarily a causal mechanism. Multiplying each factor realization by its estimated loading gives a quick contribution calculation.

Worked example

In a purely illustrative three-factor model, suppose $R_f = 0.01\%$, $\alpha = 0$, $\beta_M = 1.2$, $\beta_S = 0.4$, $\beta_H = -0.2$, with market excess return -1%, $SMB = 0.5\%$, and $HML = -0.5\%$. Predicted excess return is $1.2(-1) + 0.4(0.5) - 0.2(-0.5) = -0.9\%$. Add R_f to get predicted raw return -0.89%. Notice all terms use compatible *percentage-point*

units.

REAL LIFE / Cycling in a strong wind

If you cycle slowly during a strong headwind, the ride may be disappointing without any problem with your bicycle. Similarly, a firm's return may be poor simply because the entire market is falling. A factor model estimates the return attributable to broad systematic exposures before looking for an unusual residual. It does not automatically filter out company-specific headlines.

"Before blaming the bicycle, subtract the headwind.

Multiple-choice checkpoint

1. [**Foundation**] If actual return is -4% and model-predicted return is -1%, abnormal return is
A. -5 pp B. -3 pp C. $+3$ pp D. -4 pp.

2. [**Challenge**] SMB_t in the Fama–French three-factor regression is
A. a firm's number of employees B. a time-varying size-factor return C. an intercept dummy D. the stock price in yen.

Answers and explanations: Appendix A, Chapter 19.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. The risk-free daily return is 0.01% and firm return is 0.31%. What is the firm's excess return?

A. 0.30% **B.** 0.32%

C. 0.31% **D.** 3.0%

P2. Foundation. In a three-factor model, what does SMB typically represent?

A. The return difference between small- and large-cap portfolios **B.** The consumer price index

C. A government subsidy dummy **D.** The firm's sales growth

P3. Application. Predicted excess return equals $\alpha + \beta_m MKT + s\,SMB + h\,HML$. Let $\alpha = 0.1\%$, $\beta_m = 1$, $MKT = -1\%$, $s = 0$, $h = 0$. What is predicted excess return?

A. $+1.1\%$ **B.** -0.9%

C. -1.0% **D.** $+0.1\%$

P4. Challenge. A factor model fits past Japanese equity returns well. Why might its abnormal-return estimate still be distorted on an export-control announcement date?

A. A benchmark model cannot fail after estimation **B.** A coincident firm-specific news shock is not automatically removed by market, size and value factors

C. Fitted returns always equal realized returns **D.** The model proves an export-control effect

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 20

Abnormal Returns, CAR, and Event-Study Inference

Intuition

Subtracting the benchmark is the beginning of analysis, not the end. When multiple firms and event dates overlap, residual correlation matters. A tiny conventional standard error can be misleading if a large number of nominal observations share one policy shock.

20.1 From an event-day surprise to a cumulative measure

With model-predicted return $\widehat{R}_{it}$, define

$$AR_{it} = R_{it} - \widehat{R}_{it}, \qquad CAR_i(a,b) = \sum_{\tau=a}^{b} AR_{i\tau}.$$

For daily AR values $+2\%, -1\%, +0.5\%$, the sum is $\boxed{+1.5\%}$. A CAR is a *sum of daily abnormal simple returns*; it is not exactly the compounded abnormal wealth gain. Define separately a buy-and-hold abnormal return if that is your estimand.

20.2 Across-firm aggregation

For N_e firms associated with an event e, average abnormal return is $AAR_{e\tau} = N_e^{-1} \sum_i AR_{ie\tau}$ and cumulative average abnormal return is $CAAR_e(a,b) = \sum_{\tau=a}^{b} AAR_{e\tau}$. Averaging firms does not create N_e independent experiments when all share a single event announcement.

20.3 Variance with and without dependence

For two daily abnormal returns A_1, A_2,

$$\mathrm{Var}(A_1 + A_2) = \mathrm{Var}(A_1) + \mathrm{Var}(A_2) + 2\,\mathrm{Cov}(A_1, A_2).$$

A naive test sets covariance to zero. If consecutive days share a market microstructure effect or firms move together due to industry news, this can be false. A factor model also introduces parameter-estimation uncertainty, and event-time overlap across events requires careful handling. Inference methods may include well-justified event-level aggregation, cluster-robust approaches, block/bootstrap methods, or randomization/permutation tests when their assumptions hold. None is universally correct.

Worked example

Suppose two ARs each have SD 1% and correlation 0.5. Their CAR variance in *squared percentage-point units* is $1^2+1^2+2(.5)(1)(1)=3$, so SD is $\sqrt{3}\approx 1.732$ percentage points. Treating days as independent would give $\sqrt{2}\approx 1.414$, understating uncertainty.

20.4 Windows, multiplicity, and sensitivity

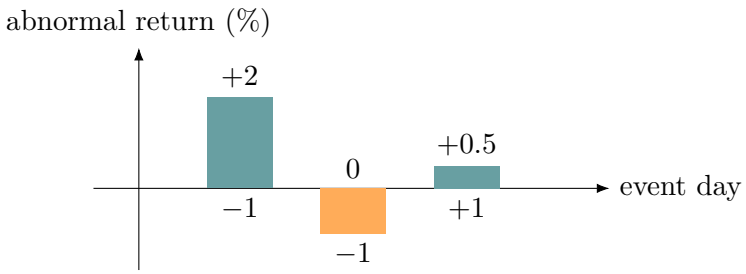
Pre-specify primary windows and estimates where possible. Test $[-1,+1]$, $[0,+1]$ and related windows transparently as robustness analysis, not a hunt for stars. Record missing holidays, nontrading securities, and stale-price issues. Show the distribution of firm outcomes as well as a pooled coefficient; a large reaction from one firm can dominate the mean.

Semiconductor research connection

If export controls are announced jointly by several countries, counting the same economic shock as three statistically independent country events exaggerates effective sample size. Consider grouping linked announcements, and report how inference changes under alternative definitions.

```
ar <- c(.02, -.01, .005)
CAR <- sum(ar)
compound_actual_minus_benchmark <- prod(1 + ar) - 1
c(CAR=CAR, compounded_AR_product=compound_actual_minus_benchmark)
# This product of (1+AR) is NOT generally a correctly
# defined buy-and-hold abnormal return: calculate actual
# and benchmark compounded returns separately instead.
```

CAR: adding surprises, and adding their uncertainty



Fast lane: a useful calculation trick

For a *CAR* defined as the sum of daily abnormal simple returns, add the abnormal-return percentages directly: $2\% - 1\% + 0.5\% = 1.5\%$. But **do not add standard errors**. If day-level abnormal returns are independent, variances add; with dependence, covariances must be included. CAR is an arithmetic event-study statistic, not the same as the compounded buy-and-hold abnormal return.

Worked example

If three daily abnormal returns have equal standard deviation 1% and are independent, $SE(CAR) = \sqrt{3}(1\%) \approx 1.73\%$. If positively correlated, the SE is larger; three days do not magically give three independent pieces of news.

REAL LIFE / Three days, one story

A firm may fall sharply on announcement day and partly recover the next day. An event window sums abnormal returns to capture a short sequence rather than treating every day in isolation. But correlated daily shocks also affect uncertainty: adding numbers is easy; adding their variances correctly may require covariance terms.

> *"CAR is the running score, but statistical inference asks how noisy the scoreboard is.*

Multiple-choice checkpoint

1. **[Foundation]** AR values $-1\%, -2\%, +0.5\%$ have CAR equal to
   A. -3.5% B. -2.5% C. -1.5% D. $+2.5\%$.
2. **[Challenge]** Positive correlation across abnormal returns, ignored in the standard-error calculation, can
   A. understate variance of their average or sum B. guarantee unbiased causality C. change return units automatically D. remove confounding announcements.

Answers and explanations: Appendix A, Chapter 20.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Abnormal returns over three days are $1\%, -2\%, 0.5\%$. Find cumulative abnormal return.

A. -0.5% **B.** $+0.5\%$
C. -1.5% **D.** $+3.5\%$

P2. Foundation. Which operation defines an average cumulative abnormal return over N firms?

A. $\sum_i CAR_i$ **B.** $N \sum_i CAR_i$
C. $N^{-1} \sum_i CAR_i$ **D.** $\prod_i CAR_i$

P3. Application. Two event days have AR standard deviations 1 and 2 percentage points, with zero covariance. What is the standard deviation of their CAR?

A. 1 **B.** $\sqrt{5}$
C. 3 **D.** 5

P4. Challenge. Two abnormal returns have covariance $+0.6$ in squared percentage-point units. How does the covariance enter variance of their two-day CAR?

A. Subtract 1.2 **B.** Add 0.6
C. Add 1.2 **D.** Covariance never matters

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 21

Panel Data, Fixed Effects, and Clustered Inference

Intuition

A panel observes the same firms repeatedly. This allows us to control for stable differences between firms but introduces dependence across time and makes the source of identifying variation especially important.

21.1 Fixed-effects model

Let i index firms, t trading days or years, and D_{it} an exposure variable. A two-way fixed-effects regression is

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}.$$

α_i absorbs any *time-invariant* firm characteristic, observed or not: some persistent business model differences, fixed industry classifications, and stable governance features. λ_t absorbs shocks shared by all included firms on the same date. It does not remove time-varying unobserved shocks that affect firms differently.

21.2 The within transformation

For one-way firm FE and a balanced panel, take firm averages:

$$\bar{Y}_i = \alpha_i + \beta \bar{D}_i + \bar{u}_i.$$

Subtract to remove α_i:

$$Y_{it} - \bar{Y}_i = \beta(D_{it} - \bar{D}_i) + (u_{it} - \bar{u}_i).$$

The slope is estimated from *within-firm* variation. If $D_{it} = D_i$ never varies over time, then $D_{it} - \bar{D}_i = 0$ and its direct level effect is unidentified in a firm-FE model. With two-way fixed effects, date dummies remove additional common time variation.

21.3 What variation remains for your research?

Suppose all firms are coded $D_{it} = 1$ on every export-control announcement date and 0 otherwise. Date fixed effects perfectly absorb this common indicator, so a separate effect cannot be identified. To estimate heterogeneous exposure, one might instead use $D_{it} = \text{Exposure}_i \times \text{Announcement}_t$. The firm and date effects leave variation in *differential* responses

by exposure. Such an interaction identifies a causal difference only under additional assumptions.

21.4 Why cluster standard errors

Serial correlation in outcomes for the same firm means treating daily firm observations as independently sampled is risky. Cluster by firm for within-firm dependence; if large common daily shocks persist, consider appropriate multiway clustering or other methods. Effective cluster counts, unbalanced panels, heterogeneous policy shocks and sparse treatment are all relevant. More sophisticated staggered-adoption designs deserve separate treatment from a simple TWFE coefficient.

Worked example

Firm A has outcome $(2, 3)$ and Firm B $(10, 11)$ across two years. Their level difference is eight in each year, but both rise by one. Firm FE remove stable level differences; time FE remove the common one-unit increase. No independent differential treatment effect can be identified from these four outcome cells alone.

```
set.seed(41)
firms <- rep(1:20, each=10)
day <- rep(1:10, times=20)
exposure <- rep(runif(20), each=10)
shock <- as.integer(day>=6)
y <- rep(rnorm(20),each=10) + rep(rnorm(10),20) +
  -0.03*exposure*shock + rnorm(200,sd=.05)
fit <- lm(y ~ I(exposure*shock) + factor(firms) + factor(day))
coef(fit)["I(exposure * shock)"]
# Here shock alone is absorbed by date fixed effects.
```

Fixed effects: an intuitive subtraction trick

Fast lane: a useful calculation trick

The within transformation **subtracts each firm's average from each of its observations**. In $Y_{it} = \alpha_i + \beta X_{it} + u_{it}$, demeaning removes α_i because a fixed firm's constant intercept minus itself is zero. If X_{it} does not change within a firm, its demeaned value is also zero, so its effect cannot be separately estimated using firm FE alone.

Worked example

A firm has outcomes $Y = (8, 12)$ and exposures $X = (1, 3)$ in two periods. Demeaning gives $\tilde{Y} = (-2, 2)$ and $\tilde{X} = (-1, 1)$. The within slope is $\sum \tilde{X}\tilde{Y} / \sum \tilde{X}^2 = (2+2)/(1+1) = 2$. We learned the association using within-firm changes; whether this is causal still depends on the error assumptions.

REAL LIFE / A student's repeated quiz results

If one student always scores higher because of stable prior preparation, comparing raw levels across students confuses stable differences with changes over time. Firm fixed

effects apply the same idea: compare each firm to its own long-run average. This removes stable characteristics, but does not magically remove time-varying surprises.

"Fixed effects remove some old differences; they do not remove every new surprise.

Multiple-choice checkpoint

1. [**Foundation**] A predictor that is constant within every firm over time, in a model with firm fixed effects, is
A. identified by within-firm changes B. absorbed by firm effects C. equivalent to residual variance D. necessarily causal.

2. [**Application**] Date fixed effects absorb
A. all time-varying firm-specific shocks B. all omitted-variable bias C. shocks common to all firms on a date D. all firm-specific stable traits.

3. [**Challenge**] Firm-level clustering primarily allows for
A. arbitrary dependence within a firm cluster under its assumptions B. unlimited causality C. data to become iid D. increasing number of firms.

Answers and explanations: Appendix A, Chapter 21.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. A firm has outcomes $(4, 6)$ in two years. What are its within-firm demeaned outcomes?

A. $(4, 6)$ **B.** $(-1, 1)$

C. $(1, -1)$ **D.** $(0, 0)$

P2. Foundation. A firm-fixed-effect regression includes firm dummy variables. Which variation directly identifies a time-varying slope?

A. Changes within the same firm over time **B.** Only differences between firms that never change

C. Only the year labels **D.** Only constants

P3. Application. A year fixed effect for 2025 absorbs which shock most directly?

A. A shock common to all sampled firms in 2025 **B.** Only a single firm's 2025 hiring decision

C. All firm-specific time-varying shocks **D.** The entire past history of every company

P4. Challenge. A country's export-control exposure indicator is constant for each firm throughout the panel. What happens when you include firm fixed effects?

A. Exposure receives a larger coefficient **B.** The indicator is collinear with firm effects, so its separate constant slope cannot be identified

C. Fixed effects guarantee a causal interpretation **D.** Every firm receives identical treatment

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 22

Difference-in-Differences: Parallel Trends

Intuition

You explicitly marked difference-in-differences and parallel trends as new material. This chapter begins from a four-cell table and moves to potential outcomes. Unlike simple before–after comparisons, DiD subtracts changes experienced by a comparison group.

22.1 The four-cell calculation

For groups T (treated) and C (comparison) and periods 0 and 1,

$$\widehat{DiD} = (\bar{Y}_{T1} - \bar{Y}_{T0}) - (\bar{Y}_{C1} - \bar{Y}_{C0}).$$

Worked example

Treated firms fall from 10 to 7, a change of -3. Comparison firms fall from 8 to 7, a change of -1. Thus

$$\widehat{DiD} = (-3) - (-1) = \boxed{-2}.$$

The baseline outcome levels differ (10 versus 8), but this fact alone does not prevent DiD. The identifying assumption concerns untreated *changes*.

22.2 A potential-outcomes derivation

Let $Y_{gt}(0)$ denote the untreated potential outcome for group g in period t. In the absence of anticipation, assume treatment affects only group T in post-period 1. Write observed $Y_{T1} = Y_{T1}(1)$ and $Y_{T0} = Y_{T0}(0)$, while comparison outcomes are untreated. Define the ATT in period 1:

$$ATT = \mathbb{E}[Y_{T1}(1) - Y_{T1}(0) \mid T].$$

The *parallel trends* assumption is

$$\mathbb{E}[Y_{T1}(0) - Y_{T0}(0) \mid T] = \mathbb{E}[Y_{C1}(0) - Y_{C0}(0) \mid C].$$

Add and subtract the unobserved treated counterfactual $Y_{T1}(0)$ in the observed treated change. Under parallel trends, no anticipation and suitable stable group composition, the difference of observed changes equals the ATT. This is the key identification logic.

22.3 Regression equivalent with two periods and two groups

Define $T_i = 1$ for treated group and $Post_t = 1$ in period 1. The saturated regression

$$Y_{it} = \alpha + \gamma T_i + \lambda Post_t + \delta(T_i Post_t) + u_{it}$$

produces $\hat{\delta} = \widehat{DiD}$ in the simple two-group/two-period cell-mean setup, given the relevant group-period observations and weighting. With more periods, continuous exposure and repeated policy adoption, design and interpretation become more complex.

22.4 What can go wrong?

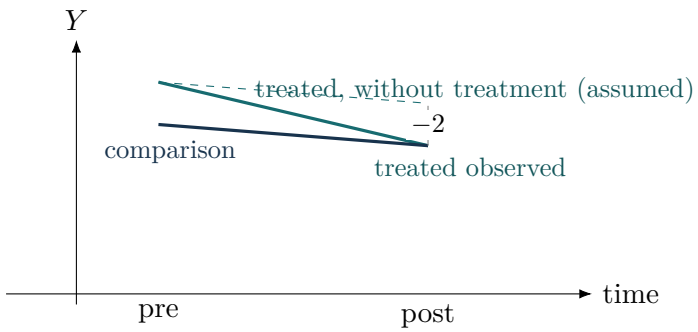
Differential pre-existing trends, policy anticipation, group-specific concurrent shocks, changing composition, spillovers, and treatment heterogeneity undermine naive causal interpretation. Plot pre-treatment trends to probe plausibility, but passing a pre-trend test *does not prove* parallel trends for the unobserved post-treatment counterfactual. Recent staggered-treatment settings may require estimators beyond simple TWFE due to heterogeneous treatment effects and comparisons involving previously treated units.

Semiconductor research connection

A DiD comparing Japanese and European firms before and after one rule is credible only if the comparison group offers a defensible counterfactual. Different currency regimes, exposure profiles and simultaneous national industrial policy may violate parallel trends. A carefully designed event study may be more feasible than DiD when treatment assignment is inseparable from firms' business models.

```
dat <- data.frame(group=rep(c("T","C"),each=2),
                  post=rep(c(0,1),2),
                  y=c(10,7,8,7))
coef(lm(y ~ I(group=="T")*post, data=dat))
# The interaction coefficient is -2 in this four-cell example.
```

The difference-in-differences picture



Fast lane: a useful calculation trick

Two subtractions: (treated after − treated before) − (control after − control before). In your diagnostic, treated 10 → 7 means −3 and controls 8 → 7 means −1, giving

$-3-(-1)=-2$. Write the two group changes separately before combining them; it avoids most sign mistakes.

Research caution

The dashed counterfactual line is *not observed*. It is justified by assumptions such as parallel untreated trends, not drawn from treated-group post-treatment data. Modern staggered-adoption designs require additional care beyond the basic two-by-two formula.

REAL LIFE / Two shops on different streets

A road project closes traffic near one cafe but not another. Both cafes face a citywide slowdown. Difference-in-differences compares how much each cafe changed, then subtracts the comparison cafe's change. Its crucial claim is about how the treated cafe would have moved without the project—a counterfactual assumption, not a directly visible fact.

> *"DiD does not demand equal starting points; it demands a defensible story about roads not taken.*

Multiple-choice checkpoint

1. **[Foundation]** Treated rises from 4 to 7; control rises from 2 to 3. DiD estimate is
   A. 1 B. 2 C. 3 D. 4.
2. **[Application]** Parallel trends requires that without treatment, the groups would have had
   A. the same outcome level B. the same number of firms C. the same expected change in outcomes D. identical observed characteristics.
3. **[Challenge]** Do non-significant pre-trends conclusively prove the post-treatment parallel-trends assumption?
   A. Yes B. No C. Only if each group has 10 firms D. Yes if R^2 is high.

Answers and explanations: Appendix A, Chapter 22.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Treated outcomes move from 8 to 13; comparison outcomes move from 4 to 7. Find two-period DiD.

A. 2 **B.** 3

C. 5 **D.** 8

P2. Foundation. Does parallel trends require equal pre-treatment outcome levels?

A. Yes, always **B.** No, it concerns counterfactual outcome changes without treatment

C. Only for randomized trials **D.** Only with three treated groups

P3. Application. In the model $Y = \alpha + \beta T + \gamma Post + \delta(T \times Post) + u$, which coefficient gives the simple two-group, two-period DiD contrast?

A. α **B.** β

C. γ **D.** δ

P4. Challenge. Treatment is announced months before implementation and firms adjust in advance. What threatens a conventional DiD interpretation using implementation as the treatment date?

A. Anticipation violates the no-early-treatment assumption **B.** The outcome must become normally distributed

C. Parallel trends become unnecessary **D.** Fixed effects cannot be included

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 23

Time-Series Dependence and Event-Time Problems

Intuition

Time makes observations dependent. Market returns on adjacent days, volatility clustering, predictable calendar effects and repeated policy announcements create structures that cross-sectional assumptions may not handle.

23.1 Autocovariance and stationarity

A weakly stationary process has a constant mean and covariance depending on lag rather than calendar time. Let $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h})$. Its lag-$h$ autocorrelation is $\rho(h) = \gamma(h)/\gamma(0)$ when variance is positive. Price levels and returns can behave very differently: equity prices are often persistent and nonstationary-like; returns are usually more appropriate objects for short-window event studies, but their volatility can vary over time.

23.2 AR(1) and dependence

Consider $X_t = \phi X_{t-1} + \varepsilon_t$, with iid mean-zero innovations of variance σ_ε^2. When $|\phi| < 1$, the stationary solution has

$$\mathrm{Var}(X_t) = \frac{\sigma_\varepsilon^2}{1-\phi^2}, \qquad \rho(h) = \phi^{|h|}.$$

If $\phi = 0.5$, the lag-two autocorrelation is 0.25. A naive iid standard error for averages disregards such persistence.

23.3 Variance of a sample mean with dependence

For a stationary sequence,

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\left(n\gamma(0) + 2\sum_{h=1}^{n-1}(n-h)\gamma(h)\right).$$

If all nonzero autocovariances vanish, this reduces to $\gamma(0)/n$. Positive serial dependence typically increases the variance of the mean. Heteroskedasticity and autocorrelation consistent (HAC) methods estimate a long-run variance under appropriate conditions and tuning choices.

23.4 Event calendar and time aggregation

An announcement at 18:00 Washington time can occur on the next civil date in Tokyo; market holiday schedules differ. A single daily data row must not be matched just by country-local date without checking when the information was available. Likewise a Japanese stock price series in yen and an index in US dollars may embed currency movements. Define returns, comparison indices, market calendars, and factor files before joining datasets.

```
set.seed(21)
n <- 250
x <- numeric(n); e <- rnorm(n)
for (t in 2:n) x[t] <- 0.5*x[t-1] + e[t]
acf(x, lag.max=12)
cor(x[-1], x[-n])
```

Dates, lags, and the off-by-one trap

Fast lane: a useful calculation trick

R_{t-1} means *one trading observation before* t in an equally spaced trading-day return series; it need not be one calendar day earlier. In R, sort explicitly by firm and trading date before constructing lags. A row shifted in a spreadsheet is not automatically an economically meaningful lag.

Semiconductor research connection

When policy announcements occur on weekends, holidays, or after an exchange closes, use the correct *first tradable session*. Document the rule in the data dictionary. A correct regression on misaligned dates can still answer the wrong question.

REAL LIFE / Your sleep schedule remembers yesterday

If you went to bed very late yesterday, tonight's sleep timing may be related to it. Many economic time series have similar persistence. Treating each day as a completely new independent draw can misrepresent uncertainty. Autoregressive models and dependence-aware standard errors try to respect the fact that time has memory.

> *"Yesterday is not always finished with today.*

Multiple-choice checkpoint

1. **[Foundation]** For stationary AR(1) with $\phi = 0.6$, lag-two autocorrelation is
   A. 0.12 B. 0.36 C. 0.60 D. 1.20.
2. **[Challenge]** Positive serial covariance usually makes the iid variance formula for a time-series sample mean
   A. too large B. unchanged by definition C. potentially too small D. exact only if firm returns are positive.

Answers and explanations: Appendix A, Chapter 23.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. In $Y_t = 0.7Y_{t-1} + u_t$, what is the autoregressive coefficient?

A. 0.3 **B.** 0.7

C. 1.0 **D.** 1.7

P2. Foundation. For a stationary AR(1) with $\phi = 0.5$, what is the theoretical lag-two autocorrelation?

A. 0.25 **B.** 0.50

C. 0.75 **D.** 1.00

P3. Application. For daily observations, why might serially correlated regression errors matter for inference?

A. Their covariance changes uncertainty calculations **B.** They automatically cause a zero slope

C. They guarantee policy causation **D.** They change the number of calendar days

P4. Challenge. A price level trends over time, but daily returns fluctuate around a stable mean. Why might returns be preferable to raw prices for a simple short-window event study?

A. Returns often remove much of the level scale and trend, aligning with expected-return benchmarks **B.** Returns eliminate every confounder

C. Price levels are always stationary **D.** Returns are identical across companies

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review E: Event Studies, Panels and Time Series

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. Realized return is +2% and benchmark predicts −1%. What is AR?

A. −3 pp **B.** −1 pp

C. +1 pp **D.** +3 pp

R2. Foundation. Three-day ARs are $-0.5\%, -1\%, +0.25\%$. What is CAR?

A. −1.25% **B.** −0.75%

C. +0.75% **D.** +1.25%

R3. Application. Treated firms change from 5 to 9; controls change from 6 to 8. What is two-period DiD?

A. 0 **B.** 2

C. 4 **D.** 6

R4. Application. A regressor stays unchanged within each firm for all years. Which statement is correct in a firm fixed-effects model?

A. Within transformation removes that regressor's direct variation **B.** It gets a unique coefficient regardless

C. It becomes the year dummy **D.** It necessarily predicts CAR perfectly

R5. Challenge. Returns of 10 firms react together to a single policy news release. Which uncertainty issue deserves attention?

A. Cross-firm covariance at the event date **B.** Only number of decimal places

C. Only the sign of coefficients **D.** Independence is guaranteed across ticker symbols

R6. Challenge. Why should a researcher not use an implementation date automatically as event day when policy was announced earlier?

A. Price reactions may occur at the earlier information arrival **B.** Implementation is never important

C. Returns can only change on weekdays **D.** All announcements are surprises

Answer key: see worked solutions for Review E.

Part V

Independent Research Practice

CHAPTER 24

Guided Replication: An Event Study in R

Intuition

The purpose of a replication is to make every transformation auditable. Rather than pressing a button to produce stars, trace the chain from prices to returns, from pre-event benchmarks to abnormal returns, and from estimates to qualified conclusions. This chapter creates *synthetic data* so results are reproducible without proprietary market databases.

24.1 Define the question and unit

Illustrative question: "How did returns of four simulated equipment firms depart from market-model predictions around one artificial export-control announcement?" Unit of data: firm–trading day. Estimation window: days −100 through −11. Event window: days −1 through +1. This is an instructional demonstration, *not* a credible causal estimate from observed policy data.

24.2 Generate and analyze simulated observations

The following code is intentionally complete and uses only base R. It can be copied into `synthetic_event_study.R` and run from a clean R session.

```
set.seed(20261015)
firms <- c("JP_A", "JP_B", "EU_A", "EU_B")
days <- -100:5
N <- length(days)
market <- rnorm(N, mean=.0003, sd=.012)
results <- data.frame()
for (i in seq_along(firms)) {
  alpha <- 0.0001
  beta <- c(.8, 1.1, 1.2, .95)[i]
  noise <- rnorm(N, sd=.009)
  return <- alpha + beta*market + noise
  return[days == 0] <- return[days == 0] +
    c(-.025, -.012, -.018, -.006)[i]
  firm_data <- data.frame(firm=firms[i], day=days,
                          market=market, ret=return)
  estimation <- subset(firm_data, day>=-100 & day<=-11)
  fit <- lm(ret ~ market, data=estimation)
  firm_data$expected <- predict(fit, newdata=firm_data)
```

```
  firm_data$AR <- firm_data$ret - firm_data$expected
  results <- rbind(results, firm_data)
}
event <- subset(results, day>=-1 & day<=1)
CAR <- aggregate(AR ~ firm, event, sum)
names(CAR)[2] <- "CAR_3day"
print(CAR)
print(aggregate(AR ~ day, event, mean))
# Expect negative CARs on average because we built in
# a negative event-day shock. This is simulation, not evidence.
```

24.3 Interpret, rather than merely calculate

If the average simulated CAR is negative, is it because the research design uncovered a policy effect? *No.* It is because we explicitly inserted negative day-zero shocks into synthetic returns. In real data we do not know the unobserved counterfactual. A credible methods section must explain why a fitted pre-event benchmark is informative, how event windows and exposure were selected, and what concurrent news could contaminate the event.

24.4 Diagnostics and extensions

Check the pre-event residual distribution, market-model fit, trading gaps, thinly traded observations and changes in coefficients across estimation windows. Compare $[-1, +1]$ with $[0, +1]$ and one-day windows, presenting the entire set instead of only significant results. For inference, remember that a single event with four firms is not four independent policy experiments. A cross-firm average CAR and its conventional standard error may be invalid if residuals move together.

24.5 A replication checklist

1. **Input audit**: source, timestamp, price adjustment, currency, firm identifiers.
2. **Event audit**: official announcement time, first tradable response time, anticipation, overlaps.
3. **Preprocessing**: sorted firm dates, return construction, join keys, missing-data flags.
4. **Estimation**: windows, regression specification, sufficient observations, fit diagnostics.
5. **Outputs**: expected returns, AR, CAR, aggregation, uncertainty, sensitivity.
6. **Reproducibility**: code version, software versions, data dictionary, no silent manual changes.

REAL LIFE / A recipe another person can cook

A recipe that says "cook until perfect cannot be reproduced. A useful research replication package identifies ingredients (source data), timing (event dates), transformations (cleaning code), and steps (estimation commands). In a multi-country event study, timestamps, adjusted prices, currency handling and missing trading days belong in the recipe.

> *"If another researcher cannot recreate your number, your number has a missing instruction manual.*

Multiple-choice checkpoint

1. **[Foundation]** A script estimates factor betas using days $[-100, -11]$ and ARs using $[-1, +1]$. The first interval is the
A. event window B. estimation window C. post-treatment period only D. control group.

2. **[Challenge]** Four companies have negative CAR around the same announcement. Which concern is important for inference?
A. Their outcomes may share event-level shocks B. Their outcomes are necessarily independent
C. A factor model proves policy causality D. There can be no firm-specific confounders.

Answers and explanations: Appendix A, Chapter 24.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Which file best supports replication of an event study?

A. A raw data file and code with documented transformations **B.** Only a final coefficient screenshot

C. Only an abstract **D.** Only the largest p-value

P2. Foundation. If a policy announcement is on a non-trading day, what must the event study establish?

A. A well-defined next information-response trading date **B.** Every exchange has identical hours

C. The announcement never matters **D.** The model should use calendar days as return observations

P3. Application. An estimation window ends one day before the announcement but a market rumor began a week earlier. Which change is most defensible?

A. Assess possible information leakage and reconsider the uncontaminated estimation and event periods **B.** Ignore the rumor by definition

C. Replace all missing returns with zero **D.** Choose whichever window makes significance largest

P4. Challenge. Four Japanese firms and seven European firms share announcement dates. What is a serious threat to naive independent-firm inference?

A. Event-common shocks create cross-firm dependence **B.** There are too many countries for regression

C. Every firm has an identical abnormal return **D.** Currency conversion always removes dependence

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 25

From Estimates to a Defensible Research Paper

Intuition

The end goal is not to compute coefficients as quickly as possible. It is to make a research claim whose estimand, assumptions, uncertainty and limitations are sufficiently explicit that another researcher can evaluate or reproduce it.

25.1 Ask an estimable question

"Did sanctions damage semiconductor firms?" is too broad: sanctions differ in jurisdiction and scope; firms play different supply-chain roles; expected and realized effects differ; and "damage" could mean returns, revenue or investment. A narrow alternative is: "Around a pre-specified set of 2018–2026 export-control announcement dates, what abnormal-return patterns were observed for selected Japanese and European equipment and materials firms, under transparent benchmark and event-window definitions?" This is primarily a *descriptive event-study* question. Causal language needs additional design justification.

25.2 Separate short-run and long-run estimands

An announcement-window outcome captures stock-market information updating at event time, reflecting anticipated future cash flows and discount rates, not necessarily immediate lost sales. An annual revenue or investment model captures outcomes over long horizons with many additional confounders. These should not be described as interchangeable measures of one effect.

25.3 A methods-section template

A defensible methods section should answer the following, in order:

1. **Units and population**: which firms, countries, exchange listings and dates are in scope?
2. **Policy events**: how are announcements distinguished from implementation and proposals? Who codes the events?
3. **Target**: is the claim descriptive, predictive or causal? Which contrast would identify the target?

4. **Outcome**: how are returns measured, including splits, dividends, currency, timezone and trading sessions?
5. **Benchmark**: what expected-return model is estimated, with what factors and estimation window?
6. **Estimator**: which AR, CAR, group contrasts or panel coefficients are reported?
7. **Uncertainty**: which level is an independent unit, and how are event/date/firm correlations handled?
8. **Threats**: anticipation, overlapping events, earnings, policy selection, multiple testing and missing data.
9. **Robustness**: justified alternative windows, benchmarks, subgroups and exclusion rules.
10. **Reproducibility**: files, code, transforms, seeds and independently reproducible tables.

25.4 An interpretive example

Suppose a descriptive study estimates mean three-day CAR -1.8 percentage points, with a dependence-aware 95% confidence interval $[-3.1, -0.5]$ percentage points for the specified event population. One might write: "The selected firms exhibited a negative average benchmark-adjusted return around the pre-specified policy announcements, with uncertainty assessed using the stated clustering procedure." One *should not* write "The regulation caused a 1.8% reduction in companies' long-run value" without an identification argument, and even then must distinguish a *return difference* from a share-price level or cumulative wealth effect.

25.5 Your mathematical exit criteria

Before finishing the book, independently solve or explain the following tasks: differentiate a multivariable loss function; calculate a 2×2 inverse and verify it; derive scalar OLS and interpret its matrix analogue; derive the σ^2/n sample-mean variance under iid sampling; identify direction of omitted-variable bias; state why HC3 or clustering changes uncertainty but not basic causal identification; design a transparent policy event file; compute and explain AR/-CAR; articulate DiD parallel trends and why it might fail in your application; and reproduce the simulated analysis in R.

Research caution

You do not need to force an advanced causal model into a project if credible identification is unavailable. A careful descriptive paper with a strong event database, transparent benchmark, appropriate uncertainty, and restraint in claims can be academically valuable.

REAL LIFE / A skeptical seminar question

Imagine your adviser asks: "Could something else have happened that day? A mature empirical argument answers with event logs, sensitivity analyses, identification limits and a clear distinction between short-run returns and long-run investment. A precise, limited claim is often stronger than a sweeping causal claim supported only by a significant coefficient.

> *"A good paper is not a victory speech for a coefficient; it is a transparent argument about evidence.*

Multiple-choice checkpoint

1. [**Foundation**] A negative event-window CAR proves long-run revenue losses.
A. Yes, with three-day windows B. Yes, if $p < .05$ C. No: the outcomes and estimands differ D. Yes, with firm fixed effects.

2. [**Challenge**] A methods section reports the best-performing event window but hides 19 other tested windows. The main problem is
A. overlapping hypotheses / selective reporting B. excessive matrix algebra C. lack of a sample mean D. overuse of an intercept.

Answers and explanations: Appendix A, Chapter 25.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Which outcome directly measures an event study's short-run market response?

A. Cumulative abnormal return **B.** Annual patent stock

C. Government cabinet tenure **D.** Five-year capital expenditure

P2. Foundation. Which description of a confidence interval is defensible?

A. The procedure has a specified repeated-sample coverage under assumptions **B.** It is the probability a fixed parameter moves daily

C. It proves causality **D.** Its width does not depend on uncertainty

P3. Application. A researcher finds a -2 percentage-point CAR with a 95% interval $[-3.5, -0.5]$. What is supported?

A. The interval excludes zero for that estimator and inferential procedure **B.** Every firm loses exactly 2% in share price

C. Policy causation is proven **D.** Five-year revenue falls by 2%

P4. Challenge. A researcher reports only the most significant of 12 event windows and claims an exact policy effect. Which change best improves credibility?

A. Pre-specify or transparently report all evaluated windows and discuss confounders and multiplicity **B.** Report the smallest p-value only

C. Eliminate alternative window choices from code history **D.** Interpret significance as mechanism proof

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review F: Research Design and Replication Review

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. Which item best documents a company's return variable construction?

A. A formula, source series and data-processing code **B.** Only a company logo

C. Only the regression p-value **D.** Only a meeting note

R2. Foundation. Which claim is strongest when CAR is statistically negative but identification is weak?

A. A negative conditional market reaction is observed under the chosen benchmark **B.** Long-run employment must fall

C. The policy caused the entire loss **D.** Every affected country had the same response

R3. Application. A company has raw stock return 0.8% and risk-free return 0.1%. What is excess return?

A. 0.7% **B.** 0.8%

C. 0.9% **D.** 8%

R4. Application. An estimated effect is -1.0 pp with a 95% CI $[-2.1, 0.1]$. What can be concluded at corresponding two-sided 5% level?

A. Reject zero unequivocally **B.** Interval includes zero; cannot reject zero at 5%

C. Policy causes positive returns **D.** Standard errors must be zero

R5. Challenge. A researcher changes sample filters until a desired p-value appears. What improves the analysis?

A. Transparent, justified filters and full sensitivity reporting **B.** Hiding failed specifications

C. Reducing available data arbitrarily **D.** Labeling all estimates causal

R6. Challenge. What is the primary purpose of placebo event dates?

A. Examine whether the design produces apparent effects when the hypothesized treatment is absent **B.** Increase official event count

C. Make prices normal **D.** Prove that all parallel trends assumptions are true

Answer key: see worked solutions for Review F.

Part VI

Supplement: Game Theory from Zero

CHAPTER 26

Game Theory I: Players, Strategies, and Payoffs

Intuition

You stated that you had not studied game theory. This supplement therefore makes *no prior-knowledge assumptions*. Game theory studies choices whose consequences depend on what other decision-makers do. It is not a qualitative method merely because the setting involves politics: formal models use precise assumptions and mathematical equilibrium conditions.

26.1 A strategic situation has three ingredients

A *player* chooses from a set of *strategies*; a *payoff* assigns a numerical preference to each combination of strategies. In a simultaneous game, each player chooses without observing the other's contemporaneous action. This does not require decisions to happen at exactly the same clock time; it requires the relevant information to be unavailable at choice time.

Let player 1 be a government considering an export control and player 2 an allied government considering cooperation. Use labels C, N for player 1 and J, R for player 2. Payoffs are ordered (u_1, u_2):

	J	R
C	$(3, 3)$	$(1, 4)$
N	$(4, 1)$	$(2, 2)$

.

These numbers are fictional utilities, not dollars or historical measurements. Utility comparisons for a player determine choices; adding a constant to all of one player's payoffs does not change strategic preferences.

26.2 Strategy versus action and utility versus social welfare

An action is a choice at a decision point. A strategy specifies what a player would do at every decision point where that player might move. In a one-shot simultaneous game, strategy and action coincide. A high combined payoff does not itself prove an outcome is stable: each player may prefer a unilateral deviation.

Worked example

At (C, J) the first player's payoff is 3. If the second player stays at J and the first switches to N, the first payoff becomes 4. Hence (C, J) is not stable against the first player's unilateral deviation.

Game theory in plain language

Fast lane: a useful calculation trick

To read any game, ask four questions: **who moves, what can they choose, what do they know, and what do they prefer?** Do not start by hunting for an equilibrium formula. Payoff numbers are rankings of outcomes for a player unless the model explicitly gives them a cardinal interpretation.

Semiconductor research connection

The numerical examples are invented strategic exercises. They are not factual claims about US, Japanese, or European officials' preferences.

REAL LIFE / Choosing a seat on a train

You would like a quiet seat, but so does another traveler. Your satisfaction depends on which seat you choose and which seat the other person chooses. That dependency, rather than sophisticated equations, is the heart of game theory. A payoff table simply makes the possible outcomes explicit.

"Game theory begins when your best choice depends on somebody else's choice.

Multiple-choice checkpoint

1. [**Foundation**] In a two-player simultaneous game, one cell $(2, 5)$ in a payoff matrix conventionally records
A. player 1 payoff 2, player 2 payoff 5 B. total payoff 2 C. probability 2/5 D. five possible actions.

2. [**Challenge**] A strategy in a sequential game means
A. only the action taken on the realized path B. a complete contingent plan across the player's decision situations C. a random sample D. maximum joint welfare.

Answers and explanations: Appendix A, Game Theory I.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. In a payoff pair $(4, 1)$, how much does player 2 receive by convention?

A. 1 **B.** 4

C. 5 **D.** $1/4$

P2. Foundation. Which is an example of strategic interdependence?

A. Choosing a commuting route when travel time depends on everyone else's routes **B.** Reading a fixed timetable alone

C. Calculating $2 + 2$ **D.** Measuring a book's length

P3. Application. Player 1 may choose A or B, player 2 chooses L or R. How many possible pure action profiles exist?

A. 2 **B.** 4

C. 6 **D.** 8

P4. Challenge. In a sequential game, a player moves at two different decision points, choosing L or R at each. How many pure strategies specify a complete plan?

A. 2 **B.** 3

C. 4 **D.** 8

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 27

Game Theory II: Best Responses and Dominance

Intuition

To find a player's best response, freeze the opponent's strategy and choose the highest available payoff for the player. Repeat for every opponent choice.

27.1 Best-response correspondence

Let player i have strategy set S_i and payoff $u_i(s_i, s_{-i})$. A best response to opponents' strategies s_{-i} satisfies

$$BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

arg max returns the strategy or strategies attaining the maximum, not the maximum payoff number. Ties can create multiple best responses.

27.2 A prisoner's-dilemma-style example

Two regulators choose Cooperate (C) or Defect (D). The invented payoffs are

	C	D
C	$(3,3)$	$(0,4)$
D	$(4,0)$	$(1,1)$

.

If the other regulator chooses C, one's payoff from D is $4 > 3$. If the other chooses D, one's payoff from D is $1 > 0$. So D is a *strictly dominant* strategy for each player. Their mutually optimal response is (D, D) with payoffs $(1, 1)$, although (C, C) would yield $(3, 3)$ to each. That contrast demonstrates why individual optimization need not maximize joint welfare.

27.3 Dominated strategies

A strategy s_i is strictly dominated by s_i' when $u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})$ for every possible opponents' strategy. A weakly dominated strategy has at least-as-good payoffs everywhere and strictly better somewhere. Eliminating dominated strategies can simplify a game, but weak dominance requires more care than strict dominance.

Worked example

In the displayed matrix, player 1 always prefers D to C. Player 2 also always prefers D to C. Dominance resolves the example without needing probabilities. In a game with no dominant action, best responses still permit equilibrium analysis.

REAL LIFE / The queue at the coffee shop

Two coffee counters are available. One is usually fast but becomes slow when everyone rushes there. Your best choice depends on which counter others choose. A best response is conditional: keep their choices fixed, and identify which option gives you the highest payoff.

"Best is not a property of your move alone; best depends on the moves around it.

Multiple-choice checkpoint

1. [**Foundation**] In the displayed dilemma, the dominant strategy of each player is
   A. Cooperate B. Defect C. mix 50–50 D. no strategy.
2. [**Challenge**] If an action is a best response, it is necessarily socially optimal.
   A. Always B. Only if payoff sums are positive C. Not necessarily D. Only in international politics.

Answers and explanations: Appendix A, Game Theory II.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. If player 1 receives payoff 2 by choosing A and 5 by choosing B when player 2 chooses L, what is player 1's best response to L?

A. A **B.** B

C. Both **D.** Neither

P2. Foundation. A strictly dominant strategy is one that is best against which opponent choices?

A. Some opponent choices **B.** Only the observed choice

C. Every possible opponent strategy **D.** Only an equilibrium strategy

P3. Application. If A gives payoffs $(3, 4)$ to player 1 against L and R, while B gives $(2, 5)$, is A strictly dominant?

A. Yes **B.** No; A loses against R

C. Yes if player 2 prefers L **D.** No because all payoffs must be negative

P4. Challenge. In a coordination game, each player prefers matching the other's choice. Why may there be more than one Nash equilibrium?

A. Each matched outcome can be stable against unilateral deviation **B.** All players must mix

C. The game has no strategies **D.** Players must maximize total welfare

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 28

Game Theory III: Nash and Mixed-Strategy Equilibrium

Intuition

A Nash equilibrium is a set of strategies where each player's strategy is a best response to everyone else's. It does not require fairness, cooperation or maximization of total payoff. You initially confused equilibrium with the jointly best outcome; that mistake is now resolved directly.

28.1 Definition and pure equilibria

A profile $s^* = (s_1^*, \dots, s_n^*)$ is a Nash equilibrium if

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for every } s_i \in S_i \text{ and every } i.$$

No player can improve by deviating *alone*, holding the other choices fixed. In the dilemma in Chapter 27, (D, D) is the unique pure Nash equilibrium.

28.2 Coordination example with two equilibria

Consider

	A	B
A	$(2,2)$	$(0,0)$
B	$(0,0)$	$(1,1)$

.

At (A, A), changing alone to B reduces payoff from 2 to 0; similarly, at (B, B) deviating alone reduces payoff from 1 to 0. Thus both (A, A) and (B, B) are pure Nash equilibria. Nash equilibrium need not be unique.

28.3 Mixed strategies and indifference

A mixed strategy is a probability distribution across pure strategies. Consider matching pennies with player 1 rewarded for matching, player 2 for mismatching:

	H	T
H	$(1,-1)$	$(-1,1)$
T	$(-1,1)$	$(1,-1)$

.

Let player 2 play H with probability q. Player 1's expected payoff from H is $q(1)+(1-q)(-1) = 2q-1$, and from T is $q(-1)+(1-q)(1) = 1-2q$. If player 1 uses both actions with positive probability in equilibrium, indifference requires $2q-1 = 1-2q$, so $q = 1/2$. Symmetrically, player 2 is indifferent only when player 1 uses H with probability $p = 1/2$. The unique mixed equilibrium is $(p, q) = (1/2, 1/2)$.

28.4 Expected payoffs

If outcome j has probability p_j and player utility u_j, expected utility is $\sum_j p_j u_j$. Mixed strategies do not mean players are uncertain about arithmetic; they represent randomized choice or, in some interpretations, beliefs that make opponents indifferent.

REAL LIFE / Nobody wants to change seats

At a table, everyone may be content with their seat given where the others are sitting. That is strategic stability, not necessarily the most comfortable arrangement for the group. Nash equilibrium checks whether one person alone would gain by moving. Joint improvement may still be possible if several people coordinate.

"Stable does not mean wonderful; it means nobody wants to move alone.

Multiple-choice checkpoint

1. [**Foundation**] In the coordination matrix, how many pure Nash equilibria exist?
A. 0 B. 1 C. 2 D. 4.

2. [**Challenge**] In matching pennies as displayed, the equilibrium probability that player 1 chooses H is
A. 0 B. 1/4 C. 1/2 D. 1.

Answers and explanations: Appendix A, Game Theory III.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Which condition defines a pure Nash equilibrium?

A. No player can gain by changing strategy alone, holding others fixed **B.** Total payoff is maximized

C. Everyone earns equal money **D.** The game ends after one move

P2. Foundation. In the matrix where player 1 payoffs are $\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}$ for rows A,B and columns L,R, which row is best against L?

A. A **B.** B

C. Both **D.** Neither

P3. Application. If both players choose independently between H and T in matching pennies and the opponent mixes equally, what is the expected probability of matching?

A. 0 **B.** 1/4

C. 1/2 **D.** 1

P4. Challenge. In matching pennies, why is an exactly 50–50 mixture an equilibrium for each player?

A. It leaves the opponent indifferent between their pure actions **B.** It makes every outcome identical

C. It maximizes joint payoff **D.** It guarantees each player wins every round

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 29

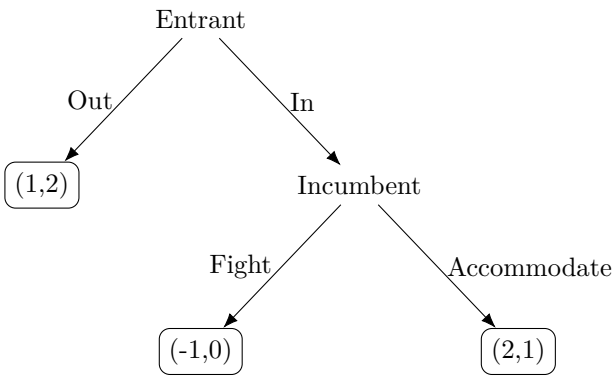
Game Theory IV: Sequential Moves and Backward Induction

Intuition

Some choices unfold in stages. A government acts, another observes, then responds. A game tree records which choices occur when and what players know. Start solving at the end, where incentives are simplest.

29.1 A small entry game

An entrant chooses *Out* or *In*. If Out, payoffs are $(1, 2)$. If In, an incumbent chooses *Fight* yielding $(-1, 0)$, or *Accommodate* yielding $(2, 1)$. The incumbent prefers Accommodate because $1 > 0$. Knowing this, entrant prefers In because $2 > 1$. Backward induction therefore selects *In, then Accommodate*.



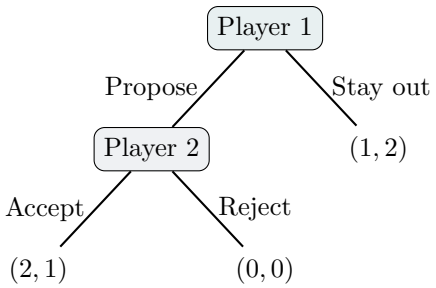
29.2 A strategy must prescribe off-path behavior

The incumbent's strategy is Fight or Accommodate after In even if the entrant actually chooses Out. A full strategy profile for this game specifies the entrant's initial choice and the incumbent's contingent response. This is why a statement of realized actions alone can miss strategically relevant threats.

29.3 Information sets

If a player moves without observing a prior hidden action, several decision nodes can belong to a single *information set*. The player must select the same decision rule at all nodes of that information set. Backward induction works straightforwardly in perfect-information trees but requires more careful belief-based reasoning under incomplete or imperfect information.

Backward induction as a visual recipe



Fast lane: a useful calculation trick

Start at the leaves, not the root. Player 2 would choose Accept over Reject ($1 > 0$). Knowing that, Player 1 compares Propose (payoff 2) with Stay out (1), and proposes. The player who moves first is not necessarily the player whose decision you should calculate first.

REAL LIFE / The chess threat

You threaten to make an expensive move in chess if your opponent advances. But if the moment actually arrives, would that move still benefit you? Sequential models ask what each player will choose *when their turn really comes*. Backward induction starts with later decisions because earlier players anticipate them.

> *"A threat is credible only when tomorrow's self wants to carry it out.*

Multiple-choice checkpoint

1. **[Foundation]** In the entry game, the incumbent's best response after In is
   A. Fight B. Accommodate C. Out D. Indifferent.
2. **[Challenge]** Backward induction solves a perfect-information finite game by
   A. starting at terminal decision points B. choosing largest joint payoff first C. ignoring future players D. guessing random probabilities.

Answers and explanations: Appendix A, Game Theory IV.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. In a perfect-information sequential game, what is the basic method for solving from final decisions backward?

A. Backward induction **B.** Linear regression

C. Differencing **D.** Bootstrap

P2. Foundation. A responder chooses Accept for payoff 2 or Reject for payoff -1. What will a payoff-maximizing responder choose at that node?

A. Accept **B.** Reject

C. Mix equally necessarily **D.** The decision cannot be modeled

P3. Application. Entrant chooses Enter or Stay Out. After Enter, incumbent gets 1 by Accommodate and 0 by Fight. Which continuation action is credible under payoff maximization?

A. Fight **B.** Accommodate

C. Both always **D.** Neither

P4. Challenge. Why must a sequential strategy specify what a player would do at unreached nodes?

A. To assess incentives and off-path credibility **B.** To increase the number of observations

C. To define stock return **D.** Only to simplify a diagram

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 30

Game Theory V: Credible Threats and Subgame Perfection

Intuition

A Nash equilibrium can sometimes rely on promises or threats that would not be optimal if the threatened decision point were actually reached. Subgame-perfect equilibrium eliminates such noncredible continuation plans in subgames.

30.1 Why Nash can be too permissive

Revisit the entry game. Consider the strategy profile (Out, Fight). If the entrant stays Out, the incumbent never moves. Given an incumbent Fight strategy, entrant gets 1 from Out versus -1 from In, so Out is optimal. Given Out, the incumbent's contingent Fight versus Accommodate does not change current payoffs, so both are best responses in strategic-form Nash terms. Hence (Out, Fight) is a Nash equilibrium. But Fight is not sequentially sensible if In happens: incumbent receives 0 from Fight versus 1 from Accommodate. The threat fails when tested in the continuation game.

30.2 Subgames and SPNE

A subgame starts at a singleton information set and includes all its successor nodes without cutting through any information set. A *subgame-perfect Nash equilibrium* is a strategy profile inducing Nash equilibrium in every subgame. In the entry game the only subgame-perfect profile is (In, Accommodate). It includes actions even off the realized path, and every such action is optimal at its subgame.

30.3 Political application, carefully defined

If an allied regulator says it will punish noncompliance but actual punishment would impose large domestic costs, a model may show the threat is not credible absent commitment devices, reputational incentives, or altered future payoffs. This is a conditional theoretical result of a specified game, not a claim that a named real-world government will necessarily behave this way.

30.4 Two solution checks

First, solve each terminal choice. Second, ask whether the earlier player would deviate given those credible choices. Do not assume that any outcome with high payoffs or politically compelling rhetoric must be subgame perfect.

REAL LIFE / The bluff that unravels

A shop may threaten to start a ruinous price war against a newcomer. If fighting would hurt the incumbent more than accommodating entry, the threat may deter entry in a simple strategic-form Nash description yet fail subgame perfection. The subgame-perfect concept checks credibility wherever later play begins.

"Backward induction asks the awkward question: would you really do that?

Multiple-choice checkpoint

1. [**Foundation**] In the entry game, (Out, Fight) is
   A. SPNE only B. Nash but not SPNE C. both Nash and SPNE D. neither Nash nor SPNE.
2. [**Challenge**] SPNE strengthens Nash by requiring Nash behavior
   A. only on the actual path B. only in simultaneous moves C. in every subgame D. only if payoffs are equal.

Answers and explanations: Appendix A, Game Theory V.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. What extra requirement distinguishes subgame-perfect equilibrium from Nash equilibrium in an extensive-form game?

A. Nash equilibrium in every subgame **B.** Equal payoffs for all

C. At least three players **D.** Normally distributed strategies

P2. Foundation. Can a game have a Nash equilibrium that is not subgame-perfect?

A. Yes **B.** No

C. Only with random payoffs **D.** Only if players are identical

P3. Application. A responder's continuation payoff from L is 4 and from R is 1. Which response is consistent with subgame perfection when that subgame is reached?

A. R **B.** L

C. Either without restriction **D.** Neither

P4. Challenge. Which claim is correct for a finite perfect-information game without ties?

A. Backward induction yields a subgame-perfect equilibrium **B.** Nash equilibrium never exists

C. Every Nash equilibrium is automatically subgame-perfect **D.** Every player must randomize

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 31

Game Theory VI: Bayesian Types and Beliefs

Intuition

Players often know their own capabilities or domestic costs better than outsiders do. Bayesian games model such hidden information as player *types* and probabilities over those types. Conditional probability from Chapter 8 becomes strategically useful.

31.1 Types and beliefs

Let the alliance partner have type $\theta \in \{H, L\}$, which the other government does not observe. Suppose $\mathbb{P}(H) = p$. Payoffs can depend on type even for identical observed actions. A strategy for the informed player maps each possible type to a choice; a strategy for the uninformed player is chosen based on its information and beliefs.

31.2 Expected utility under incomplete information

Suppose an uninformed government can Cooperate or Refuse. Cooperation yields utility 3 if the partner is type H and -1 if type L; refusal yields 0 regardless of type. Expected cooperation payoff is

$$EU(C) = 3p + (-1)(1 - p) = 4p - 1.$$

Refusal yields $EU(R) = 0$. Cooperate exactly when $4p - 1 \geq 0$, i.e. $p \geq 1/4$, with indifference at equality. If $p = .3$, expected cooperation payoff is $.2 > 0$.

31.3 Bayes updating with a message

Suppose an observed message m is sent with probabilities $\mathbb{P}(m \mid H) = .8$ and $\mathbb{P}(m \mid L) = .2$. With prior $p = .3$, posterior belief is

$$\mathbb{P}(H \mid m) = \frac{.8(.3)}{.8(.3) + .2(.7)} = \frac{.24}{.38} \approx .632.$$

The message changes expected payoff, but the receiver should update only according to the specified messaging strategies and information structure. Observing a rare signal can be informative, but not necessarily truthful.

31.4 Bayes–Nash equilibrium

A Bayesian strategy profile specifies a best response for each type, given beliefs and opponents' type-dependent strategies. The typical solution procedure calculates expected payoffs for all relevant types and verifies no profitable deviation. This differs from selecting the highest total payoff or guessing the most likely type.

REAL LIFE / The umbrella signal

You see several people carrying umbrellas and wonder whether it will rain. Your prior belief comes from the forecast; the observed umbrellas supply new information, but how useful they are depends on how likely people are to carry umbrellas when it is dry. Bayesian updating multiplies prior odds by evidence strength.

"A signal changes your beliefs only as much as its likelihood deserves.

Multiple-choice checkpoint

1. **[Foundation]** In the cooperation example, the cutoff for cooperation is $p =$
A. 0.1 B. 0.25 C. 0.5 D. 0.75.

2. **[Challenge]** If prior $P(H) = .5$ and a message is twice as likely under H as under L, the posterior $P(H \mid m)$ equals
A. 1/3 B. 1/2 C. 2/3 D. 1.

Answers and explanations: Appendix A, Game Theory VI.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Before seeing new evidence, a player's belief about an opponent being a high-cost type is called what?

A. Prior **B.** Residual

C. Coefficient **D.** Event window

P2. Foundation. A player has beliefs $P(H) = 0.3$ and $P(L) = 0.7$. What is $P(H) + P(L)$ when H,L exhaust possibilities?

A. 0.3 **B.** 0.7

C. 1 **D.** 1.3

P3. Application. A decision yields payoff 4 with probability 0.25 and payoff 0 otherwise. What is expected payoff?

A. 0.25 **B.** 1

C. 2 **D.** 4

P4. Challenge. A signal is twice as likely under H as L, and prior odds H:L equal 1:1. What are posterior odds H:L after the signal?

A. 1 : 2 **B.** 1 : 1

C. 2 : 1 **D.** 4 : 1

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 32

Game Theory VII: Signals, Pooling, and Separation

Intuition

A signal is an action that can reveal information because it is differently costly or beneficial for different types. A credible signal must survive the incentives of types that might imitate it.

32.1 Basic signaling sequence

Nature assigns sender type H or L, known to the sender but not the receiver. Sender chooses Signal S or No signal N. Receiver observes the choice and decides Accept A or Reject R. Receiver gains $+2$ from accepting H, -2 from accepting L, and zero from rejecting either. Sender receives 4 if type H is accepted, 2 if type L is accepted, and zero if rejected, less a signal cost ($c_H = 1$, $c_L = 3$); No signal costs zero.

32.2 Propose a separating strategy

Consider strategy: H sends S, L chooses N. If both types have positive prior probabilities, Bayes' rule then yields $P(H \mid S) = 1$, $P(H \mid N) = 0$. Receiver accepts after S and rejects after N, which is optimal at those beliefs.

32.3 Check both senders for deviations

Type H earns $4 - 1 = 3$ by sending S and receiving acceptance, versus 0 by choosing N and receiving rejection. Hence H does not deviate. Type L earns 0 by choosing N and rejection, versus $2 - 3 = -1$ by imitating H with S and acceptance. Hence L does not deviate. The candidate separating profile satisfies sender optimality and receiver optimality, with beliefs given by Bayes on each observed signal; it is a separating Perfect Bayesian Equilibrium in this simple model.

32.4 Pooling and off-path beliefs

A *pooling* strategy has both types choose the same signal, so observing that signal cannot distinguish them. Bayes' rule determines the on-path posterior at the pooled signal but does

not determine beliefs at an unobserved off-path signal. Depending on the equilibrium concept and refinements, different off-path beliefs may support or eliminate candidates. Do not assert that any proposed signal necessarily separates types without checking all deviations.

Semiconductor research connection

In export-control diplomacy, symbolic cooperation might serve as a signal of alignment only if its costs, payoffs, and information structure make imitation unattractive to less-aligned governments. Empirical measures of those costs are a separate research question.

REAL LIFE / The costly warranty

A seller knows a product's quality better than you do and might offer a costly warranty. If warranties are cheaper for high-quality sellers to honor, the action can reveal information. But if low-quality sellers can profitably imitate, a proposed separating pattern unravels. The game's logic lives in those incentive comparisons.

"A signal is informative only when different types have different reasons to send it.

Multiple-choice checkpoint

1. [**Foundation**] In the described separating equilibrium, the low type's payoff from imitating the high type is
   A. 2 B. 1 C. 0 D. -1.
2. [**Challenge**] Pooling means
   A. all sender types choose the same observed action B. each type chooses a distinct signal C. the receiver has full information by definition D. Bayes' rule is never applicable.

Answers and explanations: Appendix A, Game Theory VII.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. In a signaling game, who typically knows their own type before choosing a signal?

A. Sender **B.** Receiver only

C. Neither player **D.** An outside econometrician only

P2. Foundation. If both sender types choose the same signal in equilibrium, what is the pattern called?

A. Separating **B.** Pooling

C. Dominance **D.** Difference-in-differences

P3. Application. A signal gives benefit 5 after a favorable response but costs 2 for a high type. What is that type's net payoff?

A. 2 **B.** 3

C. 5 **D.** 7

P4. Challenge. A proposed separating equilibrium has the low type imitate the high type and earn 6 rather than its assigned equilibrium payoff 4. What fails?

A. The low type's incentive compatibility **B.** The definition of a matrix

C. The sample-size calculation **D.** The existence of an event window

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

CHAPTER 33

Game Theory VIII: Perfect Bayesian Equilibrium

Intuition

Perfect Bayesian Equilibrium (PBE) combines a contingent strategy for every player with beliefs at information sets. Strategies must be sequentially optimal given beliefs, and beliefs must be consistent with Bayes' rule wherever Bayes' rule can be applied to reached information sets.

33.1 A formal solution checklist

For a finite dynamic incomplete-information game:

1. Specify players, type sets, information sets, available actions and payoffs.
2. State strategies as functions of each player's information; include off-path decisions.
3. Assign beliefs to the relevant information sets.
4. Verify receiver and sender *sequential rationality* at every information set and type.
5. Apply Bayesian updating at information sets reached with positive probability under the candidate strategies.
6. Check any restrictions or refinements the course requires for off-path beliefs.

Do not confuse PBE with merely assigning probabilities to actions. Bayes' rule links types to posterior beliefs; optimization links beliefs to best responses.

33.2 Derive the receiver's acceptance threshold

With receiver utility $+2$ for accepting H and -2 for accepting L, and posterior $\mu = P(H \mid m)$ after observing message m,

$$EU(A \mid m) = 2\mu - 2(1-\mu) = 4\mu - 2, \qquad EU(R \mid m) = 0.$$

The receiver accepts when $\mu \geq 1/2$ (indifferent at equality). This follows from conditional expected utility, not from the actual type the receiver cannot observe.

33.3 Solve the separating candidate completely

Return to Chapter 32's sender costs and payoffs. Strategies: $H \mapsto S$, $L \mapsto N$; receiver $S \mapsto A$, $N \mapsto R$. Beliefs are $\mu(S) = 1$ and $\mu(N) = 0$ on the path because each signal is sent by exactly one type with positive prior probability. The receiver's acceptance rule is optimal: accept at $\mu(S) = 1$, reject at $\mu(N) = 0$. Sender H receives 3 rather than 0 by using S; sender L receives 0 rather than -1 by using N. No player or type has a profitable deviation. This verifies a PBE.

33.4 A parameter sensitivity exercise

If the low-type signal cost decreased from 3 to 1, low-type imitation would yield $2 - 1 = 1 > 0$ instead of 0 from N. The earlier separating profile would fail because L now prefers to deviate. If instead the high type's cost increased above its acceptance benefit 4, the high type would also prefer not to signal. Equilibrium predictions depend sharply on payoff assumptions.

33.5 From a formal model to data

A theoretical model creates conditional implications: more costly signaling should deter imitation, given particular primitives. Empirical analysis still needs measurable types or proxies, observables, timing and variation. You might test whether observed cooperation announcements predict subsequent restrictions, but that correlation alone does not reveal equilibrium selection or causation. Formal equilibrium analysis and quantitative data analysis are complementary, not interchangeable.

Research caution

There are multiple PBE conventions for off-path belief consistency and refinements. A course may require stronger restrictions, including consistency limits or intuitive-criterion arguments. Always follow the explicit equilibrium definition used in the assigned readings.

REAL LIFE / Beliefs meet incentives

A receiver observes a message, updates beliefs about who sent it, then chooses whether to accept. An equilibrium must tell a consistent story twice: what the receiver believes after observing each relevant signal, and whether sender and receiver actions are optimal given those beliefs. Perfect Bayesian Equilibrium tries to make stories and incentives fit together.

> *"Good strategic reasoning requires two witnesses: beliefs that make sense and choices that follow them.*

Multiple-choice checkpoint

1. **[Foundation]** Receiver accepts type H for $+2$, type L for -2, and rejects for 0. Acceptance is optimal when posterior $P(H \mid m)$ is at least
   A. 1/4 B. 1/3 C. 1/2 D. 3/4.

2. **[Challenge]** If low-type imitation payoff changes from -1 to $+1$ while staying with N gives 0, the previous separating candidate
A. remains optimal for low type B. fails the low-type incentive constraint C. necessarily remains a PBE D. implies full information before signaling.

Answers and explanations: Appendix A, Game Theory VIII.

Practice Studio / Additional Problems

Four new problems. Choose one answer per question. Write calculations in the open space, even when using the fast lane. Each question has a difficulty label.

P1. Foundation. Perfect Bayesian Equilibrium combines strategies with what other central objects?

A. Beliefs and sequential rationality **B.** A stock-return index

C. Random sampling only **D.** A regression residual

P2. Foundation. If a receiver believes H with probability $\mu = 0.25$, what probability does it assign to L in a two-type exhaustive model?

A. 0.25 **B.** 0.50

C. 0.75 **D.** 1.25

P3. Application. Receiver's payoff from Accept equals $3\mu - 1$ and Reject yields 0. When does Accept weakly dominate Reject at that information set?

A. $\mu \geq 1/3$ **B.** $\mu \geq 1/2$

C. $\mu \leq 1/3$ **D.** For every μ

P4. Challenge. A proposed PBE specifies sender strategies and on-path Bayesian beliefs but prescribes a receiver action that yields -2 while an alternative yields 0. What fails?

A. Sequential rationality **B.** Matrix invertibility

C. Conditional probability laws **D.** An event-study assumption

Your working / why not the other answers?

Do not peek: worked explanations are in the solution companion, indexed under this chapter.

Review G: Game Theory Review: From Zero to Beliefs

How this assessment works

This is a **six-question cumulative review**. Suggested time: 20–30 minutes. Closed book on the first attempt. Foundation, Application and Challenge labels indicate increasing difficulty. After finishing, use the worked solutions, and revisit any weak chapter.

R1. Foundation. A payoff pair $(1, 7)$ assigns what payoff to player 2?

A. 1 **B.** 7

C. 8 **D.** 1/7

R2. Foundation. A Nash equilibrium requires no profitable deviation by which actors?

A. Only the first player **B.** Only a referee

C. Each player individually, holding others' strategies fixed **D.** All players acting as a coalition

R3. Application. Player 2 can obtain payoff 6 from Accept and 2 from Reject at a given information set. What is a sequentially rational response?

A. Reject **B.** Accept

C. Either with no justification **D.** Change the game's rules

R4. Application. Prior odds H:L are $1 : 3$. A signal is three times as likely under H. What are posterior odds H:L?

A. $1 : 9$ **B.** $1 : 3$

C. $1 : 1$ **D.** $3 : 1$

R5. Challenge. An equilibrium claim requires a low-cost type to refrain from imitating, yet imitation raises its payoff from 2 to 4. Which condition fails?

A. Incentive compatibility **B.** Matrix multiplication

C. Random sampling **D.** Independence of observations

R6. Challenge. A receiver believes H with probability 0.4 and earns $5\mu - 1$ from accepting, 0 from rejecting. What is the rational choice?

A. Reject **B.** Accept

C. Always mix 50:50 **D.** Cannot be evaluated

Answer key: see worked solutions for Review G.

Cumulative Final Examination

How this assessment works

24 questions total: Questions 1–20 are the core mathematics and quantitative-political-science examination; Questions 21–24 cover the optional game theory supplement. Suggested time: 75–90 minutes (core only: 60–70 minutes). No notes on your first attempt; a calculator and scratch paper are allowed. Give yourself one point per correct answer and zero for a wrong or omitted response. Keep the game-theory score separate if you have not yet studied that section. **Do not equate the raw score with graduate-research readiness:** the exam is a diagnostic, and independent derivations remain crucial.

Core examination: Questions 1–20

F1. Foundation. Solve $3x - 4 = 11$.

A. 3 **B.** 4
C. 5 **D.** 7

F2. Foundation. Differentiate $f(x) = 2x^3 + e^x$.

A. $6x^2 + e^x$ **B.** $6x^3 + e^x$
C. $2x^2 + xe^x$ **D.** $3x^2 + e^x$

F3. Application. Evaluate $\int_0^1 (4x + 2)\,dx$.

A. 2 **B.** 3
C. 4 **D.** 6

F4. Application. For $g(x, y) = x^2y + xy^2$, find $\partial g/\partial x$ at $(x, y) = (2, 3)$.

A. 12 **B.** 18
C. 21 **D.** 27

F5. Challenge. A retailer's profit is $\pi(q) = 20q - 2q^2 - 8$ with $q \geq 0$. What is its maximum profit?

A. 34 **B.** 42
C. 50 **D.** 92

F6. Foundation. Compute $(1,3)^\top \cdot (2,-1)^\top$.

A. -1 **B.** 1
C. 5 **D.** 6

F7. Application. If $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, what is $\det A$?

A. 0 **B.** 1
C. 2 **D.** 3

F8. Challenge. If $X^\top X = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$ and $X^\top y = (8,6)^\top$, what is the OLS vector?

A. $(2,3)^\top$ **B.** $(4,12)^\top$
C. $(1/2, 1/3)^\top$ **D.** $(8,6)^\top$

F9. Foundation. A committee has $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.1$. Find probability of A or B.

A. 0.6 **B.** 0.7
C. 0.8 **D.** 0.9

F10. Application. A variable takes 1 with probability 0.6 and 4 with probability 0.4. Find $E[X]$.

A. 1.6 **B.** 2.2
C. 2.5 **D.** 4

F11. Application. With independent observations, $\sigma = 18$ and $n = 81$. What is sample-mean standard error?

A. 1 **B.** 2
C. 9 **D.** 18

F12. Challenge. An estimate is -2.2 with normal SE 1. Which two-sided normal-test conclusion about a zero effect is correct?

A. Reject at 1% and 5% **B.** Reject at 5% but not 1%
C. Reject at neither **D.** The null has probability 2.2%

F13. Application. Through-origin OLS has $\sum_i x_i y_i = 42$ and $\sum_i x_i^2 = 14$. Find estimated slope.

A. $1/3$ **B.** 3
C. 14 **D.** 28

F14. Challenge. True model contains $Y = \beta X + \gamma Z + u$ with $\gamma = -2$ and $\mathrm{Cov}(X, Z)/\mathrm{Var}(X) = -0.5$. What is bias from omitting Z?

A. -1 **B.** 0

C. $+1$ **D.** $+4$

F15. Challenge. A researcher applies heteroskedasticity-robust standard errors to the same OLS regression. Which statement holds?

A. Coefficients need not change but confidence intervals may **B.** Coefficients necessarily double

C. Observations become independent by construction **D.** Robust SE establish causality

F16. Application. A market return is -0.5%, a stock's market beta is 1.2, and the market-model alpha is 0.1%. What is predicted return?

A. -0.7% **B.** -0.5%

C. $+0.5\%$ **D.** $+0.7\%$

F17. Application. Realized stock return is -3%, predicted return is -0.5%. What is AR?

A. -3.5 pp **B.** -2.5 pp

C. $+2.5$ pp **D.** $+3.5$ pp

F18. Challenge. Treated group changes from 10 to 8, control group from 5 to 6. What is two-period DiD?

A. -3 **B.** -2

C. $+1$ **D.** $+3$

F19. Challenge. A firm's policy exposure never changes over time. In a firm-fixed-effects regression, what happens to its standalone exposure slope?

A. Firm effects absorb the constant exposure **B.** The slope must equal one

C. It becomes a market index **D.** Robust SE make the slope identifiable

F20. Challenge. A company issues unexpected earnings news simultaneously with an export-control announcement. Which conclusion is most defensible from a negative CAR alone?

A. The entire CAR is proven to be due to export controls **B.** Earnings news is automatically absorbed by market beta

C. Market response combines contemporaneous information and causal attribution is unresolved **D.** Returns cannot be negative on announcement days

Optional game-theory section: Questions 21–24

F21. Foundation. In a two-player game, each can select A or B. How many pure action profiles exist?

A. 2 **B.** 4

C. 6 **D.** 8

F22. Application. Against opponent strategy L, your payoffs from A and B are respectively 1 and 3. Which is your best response?

A. A **B.** B

C. Both are equally best **D.** Neither

F23. Challenge. A private-type sender has prior odds H:L of $2:1$. An observed signal is half as likely under H as under L. What are the posterior odds?

A. $1:1$ **B.** $2:1$

C. $4:1$ **D.** $1:2$

F24. Challenge. A receiver earns $4\mu - 1$ from accepting and zero from rejecting, where $\mu = P(H \mid s)$. For what beliefs is accepting optimal?

A. $\mu \geq 0.10$ **B.** $\mu \geq 0.25$

C. $\mu \geq 0.50$ **D.** $\mu = 1$ only

SELF-ASSESSMENT / BEFORE YOU OPEN THE KEY

Core result: _____/20 Optional game-theory result: _____/4.
Mark weak domains: ☐ Calculus ☐ Matrices ☐ Probability/Inference ☐ OLS ☐ Identification ☐ Applied research.
Find every worked solution in the final-exam explanation key.

CHAPTER A

Checkpoint Answers with Explanations

The answers are placed here so that you can complete the multiple-choice checkpoints before viewing feedback. Answer letters alone are not evidence of mastery; explain the reasoning in your own words and try a new numerical example if you were guessing.

Core textbook: Chapters 1–7

Chapter 1: 1 C, 2 B, 3 C. (1) $3x - 6 = x + 4$ gives $2x = 10$, so $x = 5$. (2) The exact semi-log binary comparison is $100(e^{0.10} - 1) \approx 10.52\%$. (3) Mean is 3; squared deviations $4 + 0 + 4 = 8$.

Chapter 2: 1 C, 2 B, 3 C. (1) Product rule differentiates both x^3 and e^x. (2) Chain rule gives $(1 + 2x^2)^{-1}(4x)$. (3) $C'(q) = 3 + 2q = 11$ at $q = 4$.

Chapter 3: 1 B, 2 C, 3 A. (1) Power rule gives x^3. (2) Antiderivative is x^2; $1 - 0 = 1$. (3) Derivative of e^{3x} is $3e^{3x}$.

Chapter 4: 1 A, 2 B, 3 C. (1) Holding y fixed gives $2x + 3y$. (2) $8 - 2q = 0 \Rightarrow q = 4$, with second derivative -2. (3) A positive second derivative at a stationary point ensures a strict local minimum.

Chapter 5: 1 B, 2 C, 3 C. (1) Inner dimensions 3 cancel, leaving 4×2. (2) Dot product $2(1)+3(4) = 14$. (3) An intercept plus three slopes is four columns.

Chapter 6: 1 A, 2 B, 3 C. (1) Determinant $1(4) - 2(2) = 0$. (2) Linear dependence implies singular $X^\top X$. (3) $(2, 3)$ satisfies $4 + 3 = 7$ and $2 + 9 = 11$.

Chapter 7: 1 A, 2 B. (1) OLS normal equations imply sample orthogonality $X^\top e = 0$. This does not prove population conditional mean zero. (2) $M_X y = (I - P_X)y = y - \hat{y}$.

Core textbook: Chapters 8–12

Chapter 8: 1 C, 2 A. (1) Inclusion–exclusion gives $.6 + .5 - .2 = .9$. (2) $P(A \mid B) = P(A)$ for positive-probability B is independence.

Chapter 9: 1 B, 2 C, 3 B. (1) $\mathrm{Var}(4 + 2X) = 2^2 \mathrm{Var}(X) = 12$. (2) $\binom{3}{2}(1/2)^3 = 3/8$. (3) Negative covariance describes a negative average cross-deviation product, not causality.

Chapter 10: 1 B, 2 B. (1) $SE = 10/\sqrt{100} = 1$. (2) $SE \propto n^{-1/2}$; a fourfold n halves SE.

Chapter 11: 1 B, 2 C, 3 A. (1) $1.96 < 2.3 < 2.576$, so reject at 5% but not 1% in a two-sided normal test. (2) Confidence describes repeated coverage of the method. (3) $2 \pm 1.96(.5) = 2 \pm .98$.

Chapter 12: 1 B, 2 C. (1) Best linear slope is $-1/2 = -.5$. (2) Statistical significance does not establish credible causal identification.

Core textbook: Chapters 13–17

Chapter 13: 1 B, 2 A, 3 C. (1) $12/8 = 1.5$. (2) Including a constant in OLS guarantees residuals sum to zero in sample. (3) A conditional-zero-mean error is a sufficient condition for conditional unbiasedness in the stated linear setting.

Chapter 14: 1 B, 2 A, 3 C. (1) A matrix with four columns makes $X^\mathsf{T}X$ 4×4. (2) The score equation is $X^\mathsf{T}(y - X\hat{\beta}) = 0$. (3) Full column rank makes the Gram matrix positive definite and invertible.

Chapter 15: 1 B, 2 A, 3 C. (1) Two positive OVB ingredients yield upward distortion. (2) $E[u \mid X] = 0$ is the key conditional mean restriction. (3) Neither p-values nor robust SE establish exogenous variation.

Chapter 16: 1 C, 2 A. (1) The same OLS coefficients receive a different estimated covariance matrix. (2) Under conditional homoskedastic independent errors, the familiar formula is $\sigma^2(X^\mathsf{T}X)^{-1}$.

Chapter 17: 1 C, 2 B. (1) Fit quality does not imply causal identification. (2) Highly correlated regressors provide little independent variation for each partial slope, often inflating variance.

Core textbook: Chapters 18–23

Chapter 18: 1 B, 2 A. (1) First plausible tradable information-response session should define market-specific event time. (2) Estimation periods provide observations for fitting normal-return models.

Chapter 19: 1 B, 2 B. (1) $-4\% - (-1\%) = -3$ percentage points. (2) SMB is a time-varying size factor return in the factor model.

Chapter 20: 1 B, 2 A. (1) $-1 - 2 + 0.5 = -2.5$ percentage points. (2) Positive dependence increases variance of sums/averages when omitted covariance terms would otherwise be set to zero.

Chapter 21: 1 B, 2 C, 3 A. (1) Within-firm demeaning removes a stable regressor. (2) Date dummies absorb date-common shocks. (3) Firm clustering estimates sampling uncertainty allowing within-firm residual dependence under its assumptions.

Chapter 22: 1 B, 2 C, 3 B. (1) Treated change is 3, control change is 1, so DiD is 2. (2) Parallel trends is a restriction on untreated *counterfactual changes*, not equal levels. (3) Failure to reject a pre-trend difference cannot prove an unobserved assumption.

Chapter 23: 1 B, 2 C. (1) AR(1) autocorrelation at lag two is $0.6^2 = 0.36$. (2) Positive autocovariances add positive terms to the mean's variance.

Core textbook: Chapters 24–25

Chapter 24: 1 B, 2 A. (1) The estimation interval is used to fit benchmark parameters. (2) Firms reacting to the same announcement may share a policy shock and correlated residuals.

Chapter 25: 1 C, 2 A. (1) CAR is a market-return outcome, not an observed change in long-run revenue. (2) Reporting only the most significant window among many investigated windows risks selective reporting and inflated false positives.

Game theory supplement: Chapters 26–33

Game Theory I: 1 A, 2 B. A payoff pair is ordered by player, and a sequential strategy describes contingent actions even where a node is not reached.

Game Theory II: 1 B, 2 C. Defect gives a strictly greater individual payoff against either other action. Best responses need not maximize aggregate payoffs.

Game Theory III: 1 C, 2 C. Both (A, A) and (B, B) are self-enforcing, giving two pure equilibria. Matching pennies equalizes expected payoffs only at half-and-half mixing.

Game Theory IV: 1 B, 2 A. After entry, accommodation yields 1 instead of 0 to incumbent. Backward induction begins with the final choice.

Game Theory V: 1 B, 2 C. (Out,Fight) is Nash in strategic form because the threat keeps entrant out, but Fight is suboptimal in the continuation subgame. SPNE requires equilibrium at every subgame.

Game Theory VI: 1 B, 2 C. $4p - 1 \geq 0$ implies $p \geq 1/4$. Prior odds of 1:1 become posterior odds of 2:1 if the observed message's likelihood ratio is 2, giving probability 2/3.

Game Theory VII: 1 D, 2 A. Low type earns $2 - 3 = -1$ if it signals and is accepted. Pooling means different types make the same observed signaling choice.

Game Theory VIII: 1 C, 2 B. Acceptance payoff $4\mu - 2$ becomes nonnegative at $\mu = .5$. Positive gain from low-type imitation violates the proposed separating strategy's incentive compatibility.

CHAPTER B

Worked Solutions: Practice, Reviews and Final Exam

HOW TO USE THIS ANSWER COMPANION

Answers without calculations are useful only for guessing games. Each solution below identifies the correct answer and explains the calculation, assumptions or interpretation. **Foundation** responses may be short; **Challenge** explanations go further. After correcting a mistake, close the book and solve an analogous example from scratch.

B.1 01 Foundations

Question 1 [Foundation] Answer B

Problem: Solve $5(x-1) = 2x + 10$ for x.

Solution: Expand: $5x - 5 = 2x + 10$. Subtract $2x$ and add 5: $3x = 15$, hence $x = 5$. Check: both sides equal 20.

Question 2 [Foundation] Answer C

Problem: A commuter cycles 8 km in 32 minutes. What is average speed in km/h?

Solution: Convert minutes to hours: 32/60 h. Then $8/(32/60) = 15$ km/h. Track units: dividing km by hours produces km/h.

Question 3 [Application] Answer B

Problem: Solve $\ln x + \ln(x-1) = \ln 6$ over the real numbers.

Solution: The logarithm requires $x > 1$. Combining logarithms gives $x(x-1) = 6$, or $(x-3)(x+2) = 0$. The negative root is outside the original domain; $x = 3$ is the only solution.

Question 4 [Challenge] Answer B

Problem: A model predicts $\widehat{Y} = 5 + 0.3X$, where Y is measured in thousands of yen and X in hours. Increasing X by 4 hours changes the prediction by how much?

Solution: The change in a linear prediction is slope times change in input: $\Delta\widehat{Y} = 0.3(4) = 1.2$ thousand yen. The intercept cancels. Interpretation: the prediction changes by 1,200 yen; this is not necessarily causal.

B.2 02 Derivatives

Question 1 [Foundation] Answer A

Problem: Differentiate $f(x) = 7x^3 - 2x + 4$.

Solution: Apply the power rule term by term: $(7x^3)' = 21x^2$, $(-2x)' = -2$, and the derivative of the constant is zero.

Question 2 [Foundation] Answer C

Problem: A ride costs $C(q) = 100 + 15q + q^2$ yen. What is marginal cost at $q = 5$?

Solution: Marginal cost is $C'(q) = 15 + 2q$. At $q = 5$ it is 25 yen per additional unit, interpreted locally.

Question 3 [Application] Answer B

Problem: Differentiate $g(x) = (2x + 1)^5$.

Solution: Use the chain rule: outer derivative $5u^4$, inner derivative $u' = 2$. Multiply: $g'(x) = 10(2x + 1)^4$.

Question 4 [Challenge] Answer C

Problem: Differentiate $h(x) = x^2 e^{3x}$; which expression is correct?

Solution: Apply the product rule to $u = x^2$ and $v = e^{3x}$. Then $u' = 2x$, $v' = 3e^{3x}$, so $h' = u'v + uv' = e^{3x}(2x + 3x^2)$. A common trap is differentiating only one factor.

B.3 03 Integrals

Question 1 [Foundation] Answer A

Problem: Find an antiderivative of $6x^2 - 4x$.

Solution: Integrate each power using $\int x^n \, dx = x^{n+1}/(n+1) + C$. Check by differentiating $2x^3 - 2x^2 + C$.

Question 2 [Foundation] Answer B

Problem: Evaluate $\int_0^2 2x \, dx$.

Solution: An antiderivative is x^2; the definite integral is $x^2|_0^2 = 4 - 0 = 4$. Geometrically this is triangular area under $y = 2x$.

Question 3 [Application] Answer B

Problem: If $f(x) = e^{2x}$, what is $\int_0^1 f(x) \, dx$?

Solution: Reverse the chain rule: $\int e^{2x} dx = \frac{1}{2} e^{2x} + C$. Evaluate at bounds: $(e^2 - e^0)/2$.

Question 4 [Challenge] Answer C

Problem: A bicycle's speed is $v(t) = 3 + 2t$ meters per second for $0 \le t \le 4$ seconds. How far does it travel?

Solution: Distance is the area under the velocity curve: $\int_0^4 (3+2t)dt = [3t+t^2]_0^4 = 12+16 = 28$ meters. The units of speed times time are meters.

B.4 04 Optimization

Question 1 [Foundation] Answer A

Problem: For $f(x, y) = 2x^2 + 5xy + y^2$, calculate $\partial f/\partial x$.

Solution: Treat y as constant: derivative of $2x^2$ is $4x$, of $5xy$ is $5y$, and of y^2 is zero.

Question 2 [Foundation] Answer B

Problem: Which value of $q \ge 0$ maximizes $\pi(q) = 12q - q^2$?

Solution: Differentiate: $\pi'(q) = 12 - 2q$. Solve $\pi' = 0$ to get $q = 6$. The second derivative $-2 < 0$ confirms a strict maximum.

Question 3 [Application] Answer B

Problem: Minimize $L(b) = (4 - 2b)^2 + (1 - b)^2$ over b.

Solution: Differentiate: $L'(b) = -4(4 - 2b) - 2(1 - b) = -18 + 10b$. Set to zero: $b = 18/10 = 9/5$. Since $L''(b) = 10 > 0$, it is the unique minimizer.

Question 4 [Challenge] Answer A

Problem: A researcher minimizes $L(a, b) = \sum_i (y_i - a - bx_i)^2$. Which pair of equalities must hold at an interior optimum?

Solution: Differentiate with respect to a and b: $\partial L/\partial a = -2\sum_i e_i$ and $\partial L/\partial b = -2\sum_i x_i e_i$. Set both to zero. These are the normal equations; they do not imply each residual equals zero or that the estimates are causal.

B.5 05 Vectors

Question 1 [Foundation] Answer B

Problem: What is $(2, -1, 3) \cdot (4, 2, 1)$?

Solution: Multiply entries and add: $2(4) + (-1)(2) + 3(1) = 8 - 2 + 3 = 9$.

Question 2 [Foundation] Answer B

Problem: If A is 3×2 and B is 2×5, what are the dimensions of AB?

Solution: Inner dimensions 2 and 2 match, so multiplication is defined. The result keeps the outer dimensions, 3×5.

Question 3 [Application] Answer A

Problem: With $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, find Av.

Solution: Row one: $1(2) + 2(1) = 4$. Row two: $0(2) + 3(1) = 3$. Assemble $(4, 3)^\top$.

Question 4 [Challenge] Answer A

Problem: In regression $y = X\beta + u$, X has $n = 100$ observations and $k = 4$ columns. What dimensions must β and $X^\top y$ have?

Solution: $X\beta$ must be a 100×1 fitted vector, so β is 4×1. The transpose $X^\top$ has size 4×100; multiplied by y of size 100×1 it gives 4×1.

B.6 06 Inverse

Question 1 [Foundation] Answer B

Problem: What is the determinant of $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$?

Solution: For a 2×2 matrix, $\det A = ad - bc = 3(2) - 1(2) = 4$.

Question 2 [Foundation] Answer A

Problem: Which matrix is the inverse of $\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$?

Solution: A diagonal matrix is inverted by reciprocating nonzero diagonal entries. Check: $5(1/5) = 1$.

Question 3 [Application] Answer B

Problem: Solve $2x + y = 8$ and $x - y = 1$.

Solution: The second equation gives $y = x - 1$. Substitute: $2x + (x - 1) = 8$, hence $x = 3$, $y = 2$. Both equations check: 8 and 1.

Question 4 [Challenge] Answer B

Problem: A design matrix has two identical columns. Which consequence follows for ordinary $(X^\top X)^{-1}$?

Solution: Identical columns imply a nonzero vector v with $Xv = 0$. Then $X^\top Xv = 0$, so the Gram matrix has a nontrivial nullspace and is singular. We must remove redundancy, reparameterize, or use other carefully defined estimation methods.

B.7 07 Projection

Question 1 [Foundation] Answer A

Problem: If $v = (3, 4)^\top$, what is its Euclidean norm?

Solution: $\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. It is the familiar right-triangle distance.

Question 2 [Foundation] Answer B

Problem: If fitted values are $\hat{y} = (2, 3)^\top$ and observed values are $y = (4, 4)^\top$, what is the residual vector?

Solution: Residual equals observed minus fitted: $e = y - \hat{y} = (4 - 2, 4 - 3) = (2, 1)^\top$.

Question 3 [Application] Answer B

Problem: For OLS with full-column-rank X, what is the geometric relationship between residuals e and every column of X?

Solution: The normal equations imply $X^\top e = 0$. Each component is a dot product between one column of X and e, hence every such dot product is zero.

Question 4 [Challenge] Answer A

Problem: A vector $y = (3, 4)^\top$ is projected onto the horizontal axis. Which projected vector and residual are correct?

Solution: Projection onto the horizontal axis preserves the horizontal component and sets the vertical component to zero. Thus $\hat{y} = (3, 0)$ and $e = y - \hat{y} = (0, 4)$. The residual is perpendicular to the projection subspace.

B.8 08 Probability

Question 1 [Foundation] Answer C

Problem: In a survey, $P(A) = 0.4$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.

Solution: Inclusion–exclusion gives $0.4 + 0.6 - 0.2 = 0.8$. Subtract the overlap once so it is not counted twice.

Question 2 [Foundation] Answer C

Problem: If 30 of 50 students use Python and 12 of those 30 also use R, what is $P(R \mid \text{Python})$?

Solution: Conditioning shrinks the denominator to 30 Python users. The probability is $12/30 = 0.40$, not $12/50$.

Question 3 [Application] Answer A

Problem: Two independent events have probabilities 0.2 and 0.3. Find the probability that both occur.

Solution: Independence permits multiplication: $P(A \cap B) = P(A)P(B) = 0.2(0.3) = 0.06$.

Question 4 [Challenge] Answer B

Problem: A screening system flags 90% of truly relevant articles and 10% of irrelevant articles; only 5% of all articles are relevant. What proportion of flagged articles is relevant (approximately)?

Solution: Use Bayes: numerator $0.9(0.05) = 0.045$; overall flag probability $0.045 + 0.1(0.95) = 0.14$. Posterior $0.045/0.14 = 0.3214$, about 32%. A strong detector still produces many false alarms when relevant articles are rare.

B.9 09 Random Variables

Question 1 [Foundation] Answer C

Problem: A variable equals 0 with probability 0.25 and 4 with probability 0.75. What is its expectation?

Solution: Weighted average: $E[X] = 0(0.25) + 4(0.75) = 3$.

Question 2 [Foundation] Answer C

Problem: If $\text{Var}(X) = 9$, calculate $\text{Var}(2X + 5)$.

Solution: Adding 5 changes location but not spread. Multiplying by 2 scales variance by 2^2: $4(9) = 36$.

Question 3 [Application] Answer C

Problem: A fair coin is flipped 3 times. What is the probability of exactly 2 heads?

Solution: There are $\binom{3}{2} = 3$ ways to position two heads among three flips, each with probability $(1/2)^3 = 1/8$. Total $3/8$.

Question 4 [Challenge] Answer B

Problem: Suppose $E[X] = 2$, $E[Y] = 5$, and $\text{Cov}(X, Y) = -1$. Find $E[3X - 2Y + 7]$.

Solution: Linearity of expectation gives $3E[X] - 2E[Y] + 7 = 3(2) - 2(5) + 7 = 3$. Covariance is irrelevant for the mean of a linear combination; it matters for variance.

B.10 10 Sampling

Question 1 [Foundation] Answer B

Problem: A population standard deviation is 15 and a simple random sample contains 25 independent observations. What is $SE(\bar{X})$?

Solution: $SE(\bar{X}) = \sigma/\sqrt{n} = 15/5 = 3$. The sample mean varies less than an individual observation.

Question 2 [Foundation] Answer C

Problem: If sample size quadruples under independent identical sampling, what happens to $SE(\bar{X})$?

Solution: Standard error decreases as $1/\sqrt{n}$. Quadrupling n multiplies SE by $1/\sqrt{4} = 1/2$.

Question 3 [Application] Answer A

Problem: In 100 independent Bernoulli trials with success probability 0.4, what is $\text{Var}(\bar{X})$?

Solution: The Bernoulli variance is $p(1-p) = 0.4(0.6) = 0.24$. For the sample mean, divide by $n = 100$: 0.0024.

Question 4 [Challenge] Answer B

Problem: Two firms' daily returns are strongly correlated because they react to the same news. A researcher treats 200 firm-day observations as independent. What is the major danger?

Solution: Positive cross-observation covariance contributes extra terms to estimator variance. Independence-based uncertainty calculations can understate variation and overstate significance. Use a dependence-aware design and covariance estimator suitable for the sampling structure.

B.11 11 Inference

Question 1 [Foundation] Answer A

Problem: A normal z statistic equals 2.1 for a two-sided test. At the 5% level, what decision uses the standard normal reference?

Solution: Compare $|z| = 2.1$ with the two-sided 5% critical value 1.96. It exceeds the threshold. This does not prove the alternative or causality.

Question 2 [Foundation] Answer B

Problem: For an estimate 5 with standard error 1, what is the approximate 95% normal confidence interval?

Solution: Use estimate plus/minus 1.96 standard errors: $5 \pm 1.96 = [3.04, 6.96]$.

Question 3 [Application] Answer B

Problem: A two-sided normal-test p-value is 0.03. Which statement is correct?

Solution: Compare p to significance cutoffs: $0.03 < 0.05$ but $0.03 > 0.01$. A p-value is a tail probability under the null and model assumptions, not a posterior probability of the null.

Question 4 [Challenge] Answer B

Problem: A researcher tries 20 event windows and reports only the smallest p-value, without disclosing the search. What is the primary concern?

Solution: Searching many hypotheses increases the probability of at least one apparently significant result by chance. For independent tests at 5%, the probability of one or more false rejections is $1 - 0.95^{20} \approx 0.64$. Dependence changes that numerical illustration, but the selection concern remains.

B.12 12 Conditional

Question 1 [Foundation] Answer B

Problem: Suppose $E[Y \mid X = x] = 2 + 0.5x$. What is the conditional mean when $x = 6$?

Solution: Substitute $x = 6$: $2 + 0.5(6) = 5$. This is the predicted conditional average, not an individual observation.

Question 2 [Foundation] Answer B

Problem: If $\mathrm{Cov}(X, Y) = 6$ and $\mathrm{Var}(X) = 3$, what is the slope of the best linear predictor of Y from X?

Solution: The population best-linear-predictor slope is $\mathrm{Cov}(X, Y)/\mathrm{Var}(X) = 6/3 = 2$, provided $\mathrm{Var}(X) > 0$.

Question 3 [Application] Answer A

Problem: Which condition is stronger than $\mathrm{Cov}(X, u) = 0$ and commonly used for the exogeneity discussion?

Solution: Conditional mean zero implies zero covariance (assuming moments exist), because $E[Xu] = E[XE(u \mid X)] = 0$. The converse generally fails.

Question 4 [Challenge] Answer B

Problem: A regression predicts higher returns for high-exposure firms, but exposure is chosen based on expected future demand. Which interpretation is justified by the regression alone?

Solution: A conditional regression is descriptive without design-based identification. If expected demand affects both exposure and returns, coefficient interpretation may mix the effect of exposure with demand selection.

B.13 13 Scalar Ols

Question 1 [Foundation] Answer C

Problem: Observed $y = (2, 4)$ and fitted values $\hat{y} = (1, 5)$. What is the residual sum of squares?

Solution: Residuals are $(2 - 1, 4 - 5) = (1, -1)$. Squaring and summing gives $1^2 + (-1)^2 = 2$; positive and negative errors do not cancel.

Question 2 [Foundation] Answer B

Problem: In the through-origin model $y_i = \beta x_i + u_i$, suppose $\sum x_i y_i = 18$ and $\sum x_i^2 = 9$. Find $\hat{\beta}$.

Solution: Minimize $\sum_i (y_i - \beta x_i)^2$: its derivative is $-2\sum x_i y_i + 2\beta \sum x_i^2$. Setting it to zero yields $\hat{\beta} = 18/9 = 2$.

Question 3 [Application] Answer A

Problem: In an OLS regression with an intercept, $\bar{x} = 3$, $\bar{y} = 10$, and $\hat{\beta}_1 = 2$. Find the intercept.

Solution: With an intercept, the fitted regression line passes through $(\bar{x}, \bar{y})$. Therefore $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 10 - 2(3) = 4$.

Question 4 [Challenge] Answer C

Problem: With data $(x, y) = (1, 1), (2, 3), (3, 5)$, what is the simple OLS slope with an intercept?

Solution: Compute means $\bar{x} = 2$, $\bar{y} = 3$. Numerator $\sum (x_i - 2)(y_i - 3) = (-1)(-2) + 0 + (1)(2) = 4$. Denominator $\sum (x_i - 2)^2 = 1 + 0 + 1 = 2$. Slope $= 4/2 = 2$, intercept $= 3 - 2(2) = -1$. The points lie exactly on $y = -1 + 2x$.

B.14 14 Matrix Ols

Question 1 [Foundation] Answer B

Problem: If X has n rows and $k = 3$ columns, what are the dimensions of $X^\top X$?

Solution: $X^\top$ is $3 \times n$, so $X^\top X$ is $(3 \times n)(n \times 3) = 3 \times 3$.

Question 2 [Foundation] Answer A

Problem: The first-order condition for minimizing $(y - Xb)^\top (y - Xb)$ is which equation?

Solution: Differentiate the quadratic objective: $\nabla_b L = -2X^\top y + 2X^\top X b$. Set to zero and rearrange to the normal equations $X^\top (y - X\hat{b}) = 0$.

Question 3 [Application] Answer A

Problem: If $X^\top X = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ and $X^\top y = (6, 8)^\top$, find $\hat{\beta}$.

Solution: Multiply by the inverse $\text{diag}(1/2, 1/4)$: $\hat{\beta} = (6/2, 8/4)^\top = (3, 2)^\top$. Verify $(X^\top X)\hat{\beta} = X^\top y$.

Question 4 [Challenge] Answer B

Problem: When can the usual OLS formula $(X^\top X)^{-1} X^\top y$ be used with a unique coefficient solution?

Solution: Full column rank ensures $v^\top X^\top X v = \|Xv\|^2 > 0$ for every nonzero v. Therefore $X^\top X$ is positive definite and invertible. Normality of y is not necessary for the algebraic minimizer.

B.15 15 Endogeneity

Question 1 [Foundation] Answer C

Problem: True coefficient on X is 2, but omitting a positively related confounder creates bias $+0.5$. What is the resulting population regression slope?

Solution: Omitted-variable bias adds to the structural slope: $2 + 0.5 = 2.5$. This is the population projection effect under the stated relationship, not a guarantee for one finite sample estimate.

Question 2 [Foundation] Answer A

Problem: Which description matches endogeneity in a simple regression?

Solution: Endogeneity means information in X is related to the unobserved part of the outcome, often written $\text{Cov}(X, u) \neq 0$. It can distort a causal interpretation.

Question 3 [Application] Answer A

Problem: In the omitted-variable expression, $\beta_Z = -3$ and $\text{Cov}(X, Z)/\text{Var}(X) = 2$. What is the bias from omitting Z?

Solution: The omitted-variable-bias term is $\beta_Z \, \text{Cov}(X, Z)/\text{Var}(X) = (-3)(2) = -6$. The sign depends on both the omitted effect and the association with the included regressor.

Question 4 [Challenge] Answer B

Problem: A firm announces earnings unexpectedly on the same day a government announces export controls. Why may a negative event-study return not isolate the policy effect?

Solution: Returns combine responses to all new information. If earnings news and controls arrive simultaneously, observed abnormal return mixes their components. Better event selection, timing, controls, or alternative identification is needed; robust standard errors alone do not separate effects.

B.16 16 Uncertainty

Question 1 [Foundation] Answer C

Problem: A coefficient estimate is 0.6 with standard error 0.2. What is its t statistic for testing zero?

Solution: Calculate estimate minus null divided by standard error: $(0.6 - 0)/0.2 = 3$.

Question 2 [Foundation] Answer C

Problem: Replacing conventional OLS standard errors with heteroskedasticity-robust ones, while keeping the same sample and OLS estimator, changes which element?

Solution: The residual-based covariance formula changes; the minimization objective and hence the OLS coefficient vector do not. Confidence intervals and p-values may change.

Question 3 [Application] Answer A

Problem: Under independent homoskedastic errors with variance σ^2, the conditional variance of OLS is which matrix?

Solution: From $\hat{\beta} - \beta = (X^\top X)^{-1} X^\top u$, take conditional variance and use $\mathrm{Var}(u \mid X) = \sigma^2 I$. Matrix products simplify to $\sigma^2 (X^\top X)^{-1}$.

Question 4 [Challenge] Answer B

Problem: A policy event affects many firms on one date, causing correlated residuals across firms. Which concern is most directly raised?

Solution: The variance of a sum contains cross-covariance terms. Treating correlated firm responses as independent omits these terms and can misstate uncertainty (often understate it with positive covariance). Select inference methods suited to common-event dependence; clustering by firm alone may not capture event-level correlation.

B.17 17 Diagnostics

Question 1 [Foundation] Answer B

Problem: A model achieves $R^2 = 0.95$. What is justified?

Solution: $R^2 = 1 - SSR/SST = 0.95$ reports fitted variation relative to a constant-only benchmark. It does not establish causal identification, zero errors, or individual coefficient significance.

Question 2 [Foundation] Answer C

Problem: If observed $y = (2, 6)$ and predictions are $\hat{y} = (3, 5)$, compute the sum of squared residuals.

Solution: Residuals are $(-1, 1)$. Squared residual sum $1 + 1 = 2$.

Question 3 [Application] Answer A

Problem: Two regressors are nearly linearly dependent. What typically happens to estimates of their separate slopes?

Solution: Near multicollinearity means weak independent variation for distinguishing the partial coefficients; $(X^\top X)^{-1}$ can become unstable, inflating uncertainty.

Question 4 [Challenge] Answer A

Problem: A model fits the estimation sample exceptionally well but predicts poorly in later periods. What is a defensible interpretation?

Solution: Fit on the same data used to estimate parameters can reflect overfitting or unstable relationships. Evaluate out-of-sample prediction, data shifts, and model purpose separately from identification and causal interpretation.

B.18 18 Events

Question 1 [Foundation] Answer B

Problem: Which interval normally estimates a firm's expected-return model before an event?

Solution: Pre-event estimation-window returns help fit benchmark parameters. The event window contains the returns evaluated around the announcement.

Question 2 [Foundation] Answer B

Problem: An announcement happens after Tokyo Stock Exchange trading closes on Monday. Which is the earliest plausible regular trading session for the market reaction?

Solution: If market participants could not trade after the Monday close on that exchange, Tuesday is ordinarily the first regular session that can reflect the new public information. Precise timestamps and market holidays still matter.

Question 3 [Application] Answer A

Problem: A market model predicts $+1\%$ and a firm's realized return is -2%. What is abnormal return?

Solution: $AR = R - \widehat{R} = -2\% - (+1\%) = -3$ percentage points. Subtract the counterfactual benchmark, preserving signs.

Question 4 [Challenge] Answer A

Problem: Why might an anticipated policy announcement generate little abnormal return on its official date?

Solution: Market prices can react to earlier leaks, proposals or public probabilities. The official-date surprise may be small even when the policy is important. Explore information timing and anticipation windows rather than interpreting a near-zero announcement-day AR as no policy impact.

B.19 19 Factors

Question 1 [Foundation] Answer A

Problem: The risk-free daily return is 0.01% and firm return is 0.31%. What is the firm's excess return?

Solution: Excess return subtracts the risk-free rate: $0.31\% - 0.01\% = 0.30\%$ (a difference of 0.30 percentage points).

Question 2 [Foundation] Answer A

Problem: In a three-factor model, what does SMB typically represent?

Solution: SMB (small minus big) is a constructed size-related factor portfolio return. A firm's estimated coefficient is exposure to it, not the firm's own size value.

Question 3 [Application] Answer B

Problem: Predicted excess return equals $\alpha + \beta_m MKT + s\,SMB + h\,HML$. Let $\alpha = 0.1\%$, $\beta_m = 1$, $MKT = -1\%$, $s = 0$, $h = 0$. What is predicted excess return?

Solution: Substitute: $0.1\% + 1(-1\%) + 0 + 0 = -0.9\%$. Distinguish excess return from total return; add the risk-free rate to obtain the latter.

Question 4 [Challenge] Answer B

Problem: A factor model fits past Japanese equity returns well. Why might its abnormal-return estimate still be distorted on an export-control announcement date?

Solution: The factors explain broad systematic variation but generally do not control for every information shock, such as earnings guidance or a unique supply contract. Event selection and information timing remain part of identification.

B.20 20 Car

Question 1 [Foundation] Answer A

Problem: Abnormal returns over three days are $1\%, -2\%, 0.5\%$. Find cumulative abnormal return.

Solution: Add the abnormal returns across event days: $1 - 2 + 0.5 = -0.5\%$. CAR is a sum of returns expressed here in percentage-point units.

Question 2 [Foundation] Answer C

Problem: Which operation defines an average cumulative abnormal return over N firms?

Solution: Average over firms means add their CAR values and divide by the number of firms N.

Question 3 [Application] Answer B

Problem: Two event days have AR standard deviations 1 and 2 percentage points, with zero covariance. What is the standard deviation of their CAR?

Solution: Variances add when covariance is zero: $\mathrm{Var}(AR_1 + AR_2) = 1^2 + 2^2 = 5$ squared percentage points. Taking the square root gives $\sqrt{5}$ percentage points.

Question 4 [Challenge] Answer C

Problem: Two abnormal returns have covariance +0.6 in squared percentage-point units. How does the covariance enter variance of their two-day CAR?

Solution: $\text{Var}(AR_1 + AR_2) = \text{Var}(AR_1) + \text{Var}(AR_2) + 2\,\text{Cov}(AR_1, AR_2)$. With covariance 0.6, the contribution is $2(0.6) = 1.2$. Ignoring positive dependence understates the sum's uncertainty.

B.21 21 Panel

Question 1 [Foundation] Answer B

Problem: A firm has outcomes $(4, 6)$ in two years. What are its within-firm demeaned outcomes?

Solution: The firm-specific mean is 5. Subtract it from both observations: $(4 - 5, 6 - 5) = (-1, 1)$.

Question 2 [Foundation] Answer A

Problem: A firm-fixed-effect regression includes firm dummy variables. Which variation directly identifies a time-varying slope?

Solution: Firm fixed effects absorb time-invariant firm attributes. Estimation of a time-varying regressor uses within-firm movements after accounting for included controls.

Question 3 [Application] Answer A

Problem: A year fixed effect for 2025 absorbs which shock most directly?

Solution: A 2025 year indicator shifts predictions for every firm that year. It captures common time shocks, but not firm-specific changes occurring in that year.

Question 4 [Challenge] Answer B

Problem: A country's export-control exposure indicator is constant for each firm throughout the panel. What happens when you include firm fixed effects?

Solution: For each firm, exposure is a fixed characteristic; subtracting firm means makes the regressor zero. The firm intercept absorbs the variation, so a separate coefficient cannot be estimated without additional time variation or structure.

B.22 22 Did

Question 1 [Foundation] Answer A

Problem: Treated outcomes move from 8 to 13; comparison outcomes move from 4 to 7. Find two-period DiD.

Solution: Treated change is $13 - 8 = 5$ and control change is $7 - 4 = 3$. Subtract: $5 - 3 = 2$.

Question 2 [Foundation] Answer B

Problem: Does parallel trends require equal pre-treatment outcome levels?

Solution: Parallel trends concerns how the two groups would have changed absent treatment. They can begin at different levels.

Question 3 [Application] Answer D

Problem: In the model $Y = \alpha + \beta T + \gamma Post + \delta(T \times Post) + u$, which coefficient gives the simple two-group, two-period DiD contrast?

Solution: The interaction coefficient δ captures the treated group's extra post-period change relative to the comparison group's change; the main effects absorb baseline group and time differences.

Question 4 [Challenge] Answer A

Problem: Treatment is announced months before implementation and firms adjust in advance. What threatens a conventional DiD interpretation using implementation as the treatment date?

Solution: Firms may change behavior during the pre-implementation interval because of the announcement. The supposed untreated pre-period is contaminated, so timing, treatment definition, and pre-trend/event-study evidence require reconsideration.

B.23 23 Timeseries

Question 1 [Foundation] Answer B

Problem: In $Y_t = 0.7Y_{t-1} + u_t$, what is the autoregressive coefficient?

Solution: The coefficient multiplying the previous period's value is $\phi = 0.7$. It governs persistence in a simple AR(1) representation.

Question 2 [Foundation] Answer A

Problem: For a stationary AR(1) with $\phi = 0.5$, what is the theoretical lag-two autocorrelation?

Solution: Autocorrelation at lag k is ϕ^k in stationary AR(1). At two lags it is $(0.5)^2 = 0.25$.

Question 3 [Application] Answer A

Problem: For daily observations, why might serially correlated regression errors matter for inference?

Solution: Correlation across time adds off-diagonal terms to estimator variance. Treating observations as independent can misstate standard errors, even when coefficient calculations remain algebraically unchanged.

Question 4 [Challenge] Answer A

Problem: A price level trends over time, but daily returns fluctuate around a stable mean. Why might returns be preferable to raw prices for a simple short-window event study?

Solution: Event studies usually ask whether the realized change over an interval departs from an expected change. Returns are scale-comparable and often closer to weak stationarity than asset prices. This does not automatically solve anticipation or simultaneous-news problems.

B.24 24 Replication

Question 1 [Foundation] Answer A

Problem: Which file best supports replication of an event study?

Solution: Replication requires data provenance, reproducible code and a record of transformations. A screenshot cannot reconstruct estimation windows, filters, or standard errors.

Question 2 [Foundation] Answer A

Problem: If a policy announcement is on a non-trading day, what must the event study establish?

Solution: Event time follows tradable sessions, not necessarily calendar days. Map policy timestamps to the exchange and verify holidays and close times.

Question 3 [Application] Answer A

Problem: An estimation window ends one day before the announcement but a market rumor began a week earlier. Which change is most defensible?

Solution: Benchmarks should be estimated on periods plausibly unaffected by the event. Anticipation can influence returns and estimated parameters, so justify windows transparently and report robustness.

Question 4 [Challenge] Answer A

Problem: Four Japanese firms and seven European firms share announcement dates. What is a serious threat to naive independent-firm inference?

Solution: Firms exposed to the same announcement can have correlated abnormal returns. A naive variance calculation may treat common information as independent evidence. Consider event-level inference, suitable clustering, and sample-size limits.

B.25 25 Research

Question 1 [Foundation] Answer A

Problem: Which outcome directly measures an event study's short-run market response?

Solution: CAR sums deviations of returns from a benchmark over an event window. Long-run investment and revenue require different data and research designs.

Question 2 [Foundation] Answer A

Problem: Which description of a confidence interval is defensible?

Solution: A standard confidence interval's nominal level refers to long-run coverage of the procedure, assuming the model and sampling conditions. It is not a causal certification.

Question 3 [Application] Answer A

Problem: A researcher finds a -2 percentage-point CAR with a 95% interval $[-3.5, -0.5]$. What is supported?

Solution: The interval excludes zero, indicating statistical significance at the corresponding two-sided 5% level when interval-test equivalence applies. Its relevance still depends on benchmark construction, event overlap and identification.

Question 4 [Challenge] Answer A

Problem: A researcher reports only the most significant of 12 event windows and claims an exact policy effect. Which change best improves credibility?

Solution: Window selection can inflate false discoveries and create a misleading impression of precision. A credible design discloses selection rules, robustness windows, simultaneous announcements and the uncertainty associated with searching multiple specifications.

B.26 G1 Players

Question 1 [Foundation] Answer A

Problem: In a payoff pair $(4, 1)$, how much does player 2 receive by convention?

Solution: Ordered payoff pairs give the first number to player 1 and second to player 2. These may be utility numbers, not money.

Question 2 [Foundation] Answer A

Problem: Which is an example of strategic interdependence?

Solution: One person's route affects congestion encountered by others. When payoffs depend on others' choices, a strategic interaction arises.

Question 3 [Application] Answer B

Problem: Player 1 may choose A or B, player 2 chooses L or R. How many possible pure action profiles exist?

Solution: Multiply the number of actions: 2 choices for player 1 times 2 for player 2 gives 4 pairs: $(A, L), (A, R), (B, L), (B, R)$.

Question 4 [Challenge] Answer C

Problem: In a sequential game, a player moves at two different decision points, choosing L or R at each. How many pure strategies specify a complete plan?

Solution: A pure strategy must describe what the player would do at both contingencies, even if one never occurs. Thus $2 \times 2 = 4$ complete plans: LL, LR, RL, RR.

B.27 G2 Best Response

Question 1 [Foundation] Answer B

Problem: If player 1 receives payoff 2 by choosing A and 5 by choosing B when player 2 chooses L, what is player 1's best response to L?

Solution: With opponent action L fixed, compare only player 1's payoffs: $5 > 2$, so B yields greater utility.

Question 2 [Foundation] Answer C

Problem: A strictly dominant strategy is one that is best against which opponent choices?

Solution: Strict dominance requires a strategy's payoff to exceed alternatives for every possible opponent strategy.

Question 3 [Application] Answer B

Problem: If A gives payoffs $(3, 4)$ to player 1 against L and R, while B gives $(2, 5)$, is A strictly dominant?

Solution: Against L, A gives $3 > 2$; against R, B gives $5 > 4$. Therefore A is not best against every opponent choice and cannot strictly dominate B.

Question 4 [Challenge] Answer A

Problem: In a coordination game, each player prefers matching the other's choice. Why may there be more than one Nash equilibrium?

Solution: If both choose A, deviating alone creates mismatch and hurts the deviator. The same logic may hold at (B,B). Stability can occur at multiple profiles without a unique socially best outcome.

B.28 G3 Nash

Question 1 [Foundation] Answer A

Problem: Which condition defines a pure Nash equilibrium?

Solution: Nash equilibrium is a mutual best-response profile. It need not maximize total welfare or equalize payoffs.

Question 2 [Foundation] Answer A

Problem: In the matrix where player 1 payoffs are $\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}$ for rows A,B and columns L,R, which row is best against L?

Solution: Against column L, player 1 compares 2 (row A) with 1 (row B). Because $2 > 1$, A is the best response.

Question 3 [Application] Answer C

Problem: If both players choose independently between H and T in matching pennies and the opponent mixes equally, what is the expected probability of matching?

Solution: Against an even 50–50 mix, any pure choice matches half the time. Each of the two matching cells has probability $1/4$, summing to $1/2$.

Question 4 [Challenge] Answer A

Problem: In matching pennies, why is an exactly 50–50 mixture an equilibrium for each player?

Solution: When one player mixes half-and-half, the expected payoff from either pure response is equal. Neither can improve unilaterally by changing mixing probabilities, yielding the mixed-strategy equilibrium.

B.29 G4 Sequential

Question 1 [Foundation] Answer A

Problem: In a perfect-information sequential game, what is the basic method for solving from final decisions backward?

Solution: Start at the last decision nodes, choose optimal actions there, and then work backward to earlier players anticipating those choices.

Question 2 [Foundation] Answer A

Problem: A responder chooses Accept for payoff 2 or Reject for payoff -1. What will a payoff-maximizing responder choose at that node?

Solution: Comparing the responder's own payoffs gives $2 > -1$, so Accept is preferred at that information set.

Question 3 [Application] Answer B

Problem: Entrant chooses Enter or Stay Out. After Enter, incumbent gets 1 by Accommodate and 0 by Fight. Which continuation action is credible under payoff maximization?

Solution: At the incumbent's actual post-entry decision, $1 > 0$, so accommodating maximizes its continuation payoff. A threat to fight is not credible when it is worse at the moment of choice.

Question 4 [Challenge] Answer A

Problem: Why must a sequential strategy specify what a player would do at unreached nodes?

Solution: Equilibrium requires contingency plans. Even if a node is not reached under proposed play, the strategy at that node affects whether earlier threats or promises are credible.

B.30 G5 Spe

Question 1 [Foundation] Answer A

Problem: What extra requirement distinguishes subgame-perfect equilibrium from Nash equilibrium in an extensive-form game?

Solution: Subgame perfection requires continuation play to form Nash equilibria in every subgame, ruling out many noncredible threats.

Question 2 [Foundation] Answer A

Problem: Can a game have a Nash equilibrium that is not subgame-perfect?

Solution: In sequential entry games, a threat can deter entry in Nash form even if carrying it out would be irrational in the subgame reached after entry.

Question 3 [Application] Answer B

Problem: A responder's continuation payoff from L is 4 and from R is 1. Which response is consistent with subgame perfection when that subgame is reached?

Solution: The responder must choose L because $4 > 1$; a subgame-perfect plan cannot prescribe a strictly inferior response at this continuation point.

Question 4 [Challenge] Answer A

Problem: Which claim is correct for a finite perfect-information game without ties?

Solution: Solve each terminal decision node optimally and move backward; the resulting strategy profile is subgame-perfect. The no-ties condition helps make choices definite but is not required for existence in finite perfect-information games.

B.31 G6 Bayesian

Question 1 [Foundation] Answer A

Problem: Before seeing new evidence, a player's belief about an opponent being a high-cost type is called what?

Solution: A prior is the probability assigned before observing the new signal. New evidence can update it to a posterior.

Question 2 [Foundation] Answer C

Problem: A player has beliefs $P(H) = 0.3$ and $P(L) = 0.7$. What is $P(H) + P(L)$ when H,L exhaust possibilities?

Solution: Mutually exclusive exhaustive types form a partition. Their prior probabilities add to one: $0.3 + 0.7 = 1$.

Question 3 [Application] Answer B

Problem: A decision yields payoff 4 with probability 0.25 and payoff 0 otherwise. What is expected payoff?

Solution: Take probability-weighted payoffs: $0.25(4)+0.75(0) = 1$. Expected utility compares lotteries under the model's assumptions.

Question 4 [Challenge] Answer C

Problem: A signal is twice as likely under H as L, and prior odds H:L equal 1:1. What are posterior odds H:L after the signal?

Solution: Posterior odds equal prior odds multiplied by likelihood ratio: $(1:1) \times 2 = 2:1$. Normalize to posterior probability $P(H \mid s) = 2/(2+1) = 2/3$.

B.32 G7 Signaling

Question 1 [Foundation] Answer A

Problem: In a signaling game, who typically knows their own type before choosing a signal?

Solution: The sender privately knows its type in standard signaling models, then chooses an observable message or action.

Question 2 [Foundation] Answer B

Problem: If both sender types choose the same signal in equilibrium, what is the pattern called?

Solution: Pooling means different private types generate the same observed signal. The receiver cannot infer type solely from that signal.

Question 3 [Application] Answer B

Problem: A signal gives benefit 5 after a favorable response but costs 2 for a high type. What is that type's net payoff?

Solution: Utility net of signaling cost is $5 - 2 = 3$. Incentive compatibility compares this amount to what the same type earns by deviating.

Question 4 [Challenge] Answer A

Problem: A proposed separating equilibrium has the low type imitate the high type and earn 6 rather than its assigned equilibrium payoff 4. What fails?

Solution: A separating profile cannot be an equilibrium if the low type profits by imitating the high type. The deviation earns $6 > 4$, violating the low type's incentive constraint, even if receiver beliefs seem plausible.

B.33 G8 Pbe

Question 1 [Foundation] Answer A

Problem: Perfect Bayesian Equilibrium combines strategies with what other central objects?

Solution: PBE requires beliefs about unobserved types or histories, strategy optimality given those beliefs, and Bayesian consistency wherever Bayes' rule applies.

Question 2 [Foundation] Answer C

Problem: If a receiver believes H with probability $\mu = 0.25$, what probability does it assign to L in a two-type exhaustive model?

Solution: Beliefs over exhaustive types sum to one: $P(L) = 1 - P(H) = 0.75$.

Question 3 [Application] Answer A

Problem: Receiver's payoff from Accept equals $3\mu - 1$ and Reject yields 0. When does Accept weakly dominate Reject at that information set?

Solution: Compare expected payoffs: Accept is optimal when $3\mu - 1 \geq 0$. Rearrange to $\mu \geq 1/3$ (indifferent at equality).

Question 4 [Challenge] Answer A

Problem: A proposed PBE specifies sender strategies and on-path Bayesian beliefs but prescribes a receiver action that yields -2 while an alternative yields 0. What fails?

Solution: At that information set, the receiver has a profitable deviation because $0 > -2$. Sequential rationality requires each chosen action to maximize expected payoff given beliefs, regardless of how those beliefs were constructed.

B.34 Review A: Calculus and Algebra Checkpoint

Question 1 [Foundation] Answer B

Problem: Solve $4x + 9 = 2x + 17$.

Solution: Move terms: $4x - 2x = 17 - 9$, so $2x = 8$ and $x = 4$. Substitute to verify $25 = 25$.

Question 2 [Foundation] Answer A

Problem: Differentiate $5x^4 - 3x^2$.

Solution: Differentiate each monomial: $5(4)x^3 - 3(2)x = 20x^3 - 6x$.

Question 3 [Application] Answer C

Problem: Evaluate $\int_0^3 2x\,dx$.

Solution: An antiderivative is x^2, so $\int_0^3 2x\,dx = 3^2 - 0^2 = 9$.

Question 4 [Application] Answer B

Problem: Find $\partial(x^2y + 3y^2)/\partial y$.

Solution: Hold x fixed: derivative of x^2y is x^2 and derivative of $3y^2$ is $6y$.

Question 5 [Challenge] Answer C

Problem: For $f(q) = 18q - 3q^2$, find the maximizing q over $q \geq 0$.

Solution: Set derivative $f' = 18 - 6q = 0$, giving $q = 3$. Because $f'' = -6 < 0$, it is the unique global maximum on $q \geq 0$.

Question 6 [Challenge] Answer B

Problem: Differentiate xe^{2x} at $x = 0$.

Solution: Product and chain rules give $(xe^{2x})' = e^{2x} + 2xe^{2x} = e^{2x}(1 + 2x)$. At $x = 0$, it equals 1.

B.35 Review B: Matrix Fluency Checkpoint

Question 1 [Foundation] Answer C

Problem: Find $(1, 2, 1) \cdot (3, 0, 4)$.

Solution: Multiply matching entries: $1(3) + 2(0) + 1(4) = 7$.

Question 2 [Foundation] Answer B

Problem: If X is 80×5, what is the shape of $X^\top y$ when y is 80×1?

Solution: $X^\top$ is 5×80; multiplying by 80×1 yields 5×1.

Question 3 [Application] Answer B

Problem: Compute determinant of $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Solution: Use $ad - bc = 4(3) - 1(2) = 10$.

Question 4 [Application] Answer B

Problem: Solve $x + y = 7$ and $x - y = 1$.

Solution: Add the equations: $2x = 8$, so $x = 4$ and $y = 3$. Both equations hold.

Question 5 [Challenge] Answer B

Problem: A least-squares residual is orthogonal to the regressor $x = (1, 1)^\top$. Which residual could satisfy this?

Solution: Orthogonality means dot product zero. $x^\top(1, -1) = 1 - 1 = 0$. The other candidates have nonzero component sums.

Question 6 [Challenge] Answer A

Problem: Suppose $X^\top X = \mathrm{diag}(1, 9)$ and $X^\top y = (3, 18)^\top$. Find OLS $\hat{\beta}$.

Solution: The normal equations are diagonal: $1\beta_1 = 3$ and $9\beta_2 = 18$. Hence $\hat{\beta} = (3, 2)^\top$.

B.36 Review C: Probability and Statistical Inference Review

Question 1 [Foundation] Answer B

Problem: Suppose $P(A) = 0.3$, $P(B) = 0.4$, $P(A \cap B) = 0.1$. What is $P(A \cup B)$?

Solution: Inclusion–exclusion: $0.3 + 0.4 - 0.1 = 0.6$.

Question 2 [Foundation] Answer B

Problem: A random variable takes values 1 and 5 with equal probability. What is $E[X]$?

Solution: The expectation is $0.5(1) + 0.5(5) = 3$.

Question 3 [Application] Answer B

Problem: With standard deviation $\sigma = 20$ and independent sample size $n = 100$, what is $SE(\bar{X})$?

Solution: Divide σ by $\sqrt{n}$: $20/10 = 2$.

Question 4 [Application] Answer B

Problem: An estimate is 10 with normal standard error 2. Which is the approximate 95% confidence interval?

Solution: Use $10 \pm 1.96(2) = 10 \pm 3.92$, producing $[6.08, 13.92]$.

Question 5 [Challenge] Answer C

Problem: An event has prevalence 20%, a detector is positive for 90% of events and 10% of nonevents. Find $P(\text{event} \mid +)$.

Solution: Numerator $0.9(0.2) = 0.18$; denominator $0.18 + 0.1(0.8) = 0.26$; posterior $0.18/0.26 \approx 0.6923$.

Question 6 [Challenge] Answer B

Problem: An estimate is 2.4 with SE 1.0. Under a two-sided standard normal test of zero, which decision is appropriate?

Solution: $z = 2.4$. It exceeds 1.96 but is smaller than 2.576, so reject at 5% but not 1%. The statistic is not a posterior probability.

B.37 Review D: Regression and Identification Review

Question 1 [Foundation] Answer B

Problem: In a through-origin regression, $\sum x_i y_i = 30$ and $\sum x_i^2 = 15$. What is OLS slope?

Solution: Differentiate the squared-error objective and set its derivative to zero, yielding $\hat{\beta} = \sum xy / \sum x^2 = 30/15 = 2$.

Question 2 [Foundation] Answer B

Problem: A dummy regression coefficient is -0.03 for outcome returns in decimals. What is its conditional predicted difference?

Solution: A difference of -0.03 in decimal returns equals -3 percentage points. The conditional association is not automatically causal.

Question 3 [Application] Answer B

Problem: If the omitted-variable coefficient is $+2$ and its association ratio is -0.4, what is the bias?

Solution: Bias equals $2(-0.4) = -0.8$, a downward distortion of the included regressor's population slope.

Question 4 [Application] Answer A

Problem: A coefficient is -0.8 with standard error 0.4. What is its two-sided normal z statistic testing zero?

Solution: Divide by the estimated SE: $(-0.8)/0.4 = -2$. For a two-sided test the relevant magnitude is 2, not its sign.

Question 5 [Challenge] Answer A

Problem: Why doesn't a low residual sum of squares prove exogeneity?

Solution: OLS solves a numerical optimization problem. Exogeneity is an additional data-generating-process condition about unobserved components; the algebra of least squares does not impose that condition.

Question 6 [Challenge] Answer A

Problem: A regressor is exactly a sum of two other design-matrix columns. What is the algebraic problem?

Solution: There exists a nonzero vector v with $Xv = 0$. Then $X^\top Xv = 0$, giving singularity, so the conventional inverse formula is not defined for a unique coefficient vector.

B.38 Review E: Event Studies, Panels and Time Series

Question 1 [Foundation] Answer D

Problem: Realized return is $+2\%$ and benchmark predicts -1%. What is AR?

Solution: Abnormal return is actual minus predicted: $2\% - (-1\%) = +3$ percentage points.

Question 2 [Foundation] Answer A

Problem: Three-day ARs are $-0.5\%, -1\%, +0.25\%$. What is CAR?

Solution: Add them: $-0.5 - 1 + 0.25 = -1.25$ percentage points.

Question 3 [Application] Answer B

Problem: Treated firms change from 5 to 9; controls change from 6 to 8. What is two-period DiD?

Solution: Treated change $9 - 5 = 4$; control change $8 - 6 = 2$. Difference $4 - 2 = 2$.

Question 4 [Application] Answer A

Problem: A regressor stays unchanged within each firm for all years. Which statement is correct in a firm fixed-effects model?

Solution: Firm demeaning subtracts the same constant within each firm, leaving zero. The time-invariant regressor is absorbed by firm effects.

Question 5 [Challenge] Answer A

Problem: Returns of 10 firms react together to a single policy news release. Which uncertainty issue deserves attention?

Solution: Same-event shocks induce contemporaneous dependence. A standard error that treats each firm's response as an independent event can overstate effective information.

Question 6 [Challenge] Answer A

Problem: Why should a researcher not use an implementation date automatically as event day when policy was announced earlier?

Solution: Event studies identify market reactions to new information. Prices can incorporate expectations when the announcement occurs, not necessarily when rules later take effect.

B.39 Review F: Research Design and Replication Review

Question 1 [Foundation] Answer A

Problem: Which item best documents a company's return variable construction?

Solution: Reproducibility requires enough detail to recreate observations, including adjustments, calendars, corporate actions and code.

Question 2 [Foundation] Answer A

Problem: Which claim is strongest when CAR is statistically negative but identification is weak?

Solution: A negative CAR indicates deviation from an estimated return benchmark. Causation depends on timing, omitted concurrent news and other design assumptions.

Question 3 [Application] Answer A

Problem: A company has raw stock return 0.8% and risk-free return 0.1%. What is excess return?

Solution: Subtract risk-free return: $0.8 - 0.1 = 0.7$ percentage points.

Question 4 [Application] Answer B

Problem: An estimated effect is -1.0 pp with a 95% CI $[-2.1, 0.1]$. What can be concluded at corresponding two-sided 5% level?

Solution: Zero lies inside $[-2.1, 0.1]$. The corresponding two-sided 5% test does not reject when interval-test correspondence holds. This does not prove no effect.

Question 5 [Challenge] Answer A

Problem: A researcher changes sample filters until a desired p-value appears. What improves the analysis?

Solution: Specification search increases selective reporting risk. Document filters in code, defend exclusions, and disclose sensitivity across reasonable alternatives.

Question 6 [Challenge] Answer A

Problem: What is the primary purpose of placebo event dates?

Solution: A placebo checks how often methods generate effects at dates without the hypothesized shock. It can reveal fragility, but passing placebo checks does not establish causal identification on its own.

B.40 Review G: Game Theory Review: From Zero to Beliefs

Question 1 [Foundation] Answer B

Problem: A payoff pair $(1, 7)$ assigns what payoff to player 2?

Solution: In the ordinary ordered-pair convention, the second entry belongs to player 2.

Question 2 [Foundation] Answer C

Problem: A Nash equilibrium requires no profitable deviation by which actors?

Solution: Nash equilibrium is unilateral stability for every player simultaneously. It need not maximize joint welfare.

Question 3 [Application] Answer B

Problem: Player 2 can obtain payoff 6 from Accept and 2 from Reject at a given information set. What is a sequentially rational response?

Solution: Sequential rationality compares expected continuation payoffs conditional on available information: $6 > 2$, so choose Accept.

Question 4 [Application] Answer C

Problem: Prior odds H:L are $1 : 3$. A signal is three times as likely under H. What are posterior odds H:L?

Solution: Multiply prior odds by likelihood ratio: $(1/3) \times 3 = 1$. Thus posterior odds are $1 : 1$.

Question 5 [Challenge] Answer A

Problem: An equilibrium claim requires a low-cost type to refrain from imitating, yet imitation raises its payoff from 2 to 4. Which condition fails?

Solution: The proposed type has a profitable strategy deviation; $4 > 2$. It is not optimal to follow the claimed equilibrium signal, so the proposed separating equilibrium fails.

Question 6 [Challenge] Answer B

Problem: A receiver believes H with probability 0.4 and earns $5\mu - 1$ from accepting, 0 from rejecting. What is the rational choice?

Solution: At $\mu = 0.4$, accept gives $5(0.4) - 1 = 1$, exceeding reject's 0. Therefore Accept is the optimal response under the stated payoffs.

B.41 Cumulative Final Exam: All Worked Solutions

Question 1 [Foundation] Answer C

Problem: Solve $3x - 4 = 11$.

Solution: Add four to both sides: $3x = 15$, so $x = 5$. Substitution gives $15 - 4 = 11$.

Question 2 [Foundation] Answer A

Problem: Differentiate $f(x) = 2x^3 + e^x$.

Solution: Power rule gives $6x^2$ and derivative of e^x is e^x. Add the results.

Question 3 [Application] Answer C

Problem: Evaluate $\int_0^1 (4x + 2)\,dx$.

Solution: Antiderivative $2x^2 + 2x$; at $x = 1$ minus at 0 is $2 + 2 = 4$.

Question 4 [Application] Answer C

Problem: For $g(x, y) = x^2y + xy^2$, find $\partial g/\partial x$ at $(x, y) = (2, 3)$.

Solution: Holding y constant, $g_x = 2xy + y^2$. Evaluate $2(2)(3) + 3^2 = 12 + 9 = 21$.

Question 5 [Challenge] Answer B

Problem: A retailer's profit is $\pi(q) = 20q - 2q^2 - 8$ with $q \geq 0$. What is its maximum profit?

Solution: Optimize $\pi'(q) = 20 - 4q = 0$ giving $q^* = 5$. Second derivative $-4 < 0$, so a maximum. Evaluate $20(5) - 2(25) - 8 = 100 - 50 - 8 = 42$. Check nonnegative-domain boundary $q = 0$ gives -8.

Question 6 [Foundation] Answer A

Problem: Compute $(1, 3)^\top \cdot (2, -1)^\top$.

Solution: Dot product $1(2) + 3(-1) = 2 - 3 = -1$.

Question 7 [Application] Answer B

Problem: If $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, what is $\det A$?

Solution: Determinant is $ad - bc = 2(1) - 1(1) = 1$, so this matrix is invertible.

Question 8 [Challenge] Answer A

Problem: If $X^\top X = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$ and $X^\top y = (8, 6)^\top$, what is the OLS vector?

Solution: Solve the normal equations: $4\hat{\beta}_1 = 8$, $2\hat{\beta}_2 = 6$. Therefore $\hat{\beta} = (2, 3)^\top$. This is $\hat{\beta} = (X^\top X)^{-1} X^\top y$.

Question 9 [Foundation] Answer C

Problem: A committee has $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.1$. Find probability of A or B.

Solution: Union formula: $P(A \cup B) = 0.5 + 0.4 - 0.1 = 0.8$. The overlap was double-counted in simple addition.

Question 10 [Application] Answer B

Problem: A variable takes 1 with probability 0.6 and 4 with probability 0.4. Find $E[X]$.

Solution: Expectation is weighted average $1(0.6) + 4(0.4) = 0.6 + 1.6 = 2.2$.

Question 11 [Application] Answer B

Problem: With independent observations, $\sigma = 18$ and $n = 81$. What is sample-mean standard error?

Solution: Use $\sigma/\sqrt{n} = 18/9 = 2$. It measures sampling fluctuation of the mean, not population dispersion.

Question 12 [Challenge] Answer B

Problem: An estimate is -2.2 with normal SE 1. Which two-sided normal-test conclusion about a zero effect is correct?

Solution: $z = -2.2$; $|z| = 2.2$ exceeds 1.96 but not 2.576. Hence two-sided rejection at 5%, not at 1%. The sign determines effect direction, not critical-value magnitude.

Question 13 [Application] Answer B

Problem: Through-origin OLS has $\sum_i x_i y_i = 42$ and $\sum_i x_i^2 = 14$. Find estimated slope.

Solution: Differentiating $\sum(y_i - bx_i)^2$ gives $-2\sum xy + 2b\sum x^2 = 0$, hence $\hat{b} = 42/14 = 3$.

Question 14 [Challenge] Answer C

Problem: True model contains $Y = \beta X + \gamma Z + u$ with $\gamma = -2$ and $\mathrm{Cov}(X, Z)/\mathrm{Var}(X) = -0.5$. What is bias from omitting Z?

Solution: The simple omitted-variable bias formula is $\gamma\,\mathrm{Cov}(X, Z)/\mathrm{Var}(X) = (-2)(-0.5) = +1$. Double-negative signs matter. Minimizing squared errors does not eliminate the bias.

Question 15 [Challenge] Answer A

Problem: A researcher applies heteroskedasticity-robust standard errors to the same OLS regression. Which statement holds?

Solution: Robust covariance estimation keeps the OLS optimizing coefficients fixed and changes the estimated variance. This may alter test statistics and intervals. Robustness to heteroskedasticity does not solve endogeneity or event confounding.

Question 16 [Application] Answer B

Problem: A market return is -0.5%, a stock's market beta is 1.2, and the market-model alpha is 0.1%. What is predicted return?

Solution: Calculate $\widehat{R} = \alpha + \beta R_m = 0.1\% + 1.2(-0.5\%) = 0.1\% - 0.6\% = -0.5\%$.

Question 17 [Application] Answer B

Problem: Realized stock return is -3%, predicted return is -0.5%. What is AR?

Solution: $AR = R - \widehat{R} = -3\% - (-0.5\%) = -2.5$ percentage points.

Question 18 [Challenge] Answer A

Problem: Treated group changes from 10 to 8, control group from 5 to 6. What is two-period DiD?

Solution: Treated change $8-10 = -2$; comparison change $6-5 = +1$. DiD $= -2-(+1) = -3$. Whether it is causal depends on identifying assumptions such as parallel untreated trends and no anticipation.

Question 19 [Challenge] Answer A

Problem: A firm's policy exposure never changes over time. In a firm-fixed-effects regression, what happens to its standalone exposure slope?

Solution: Within-firm transformation subtracts each firm's time average. For fixed exposure X_i, $X_{it} - \bar{X}_i = 0$ at every date, so no independent within-firm exposure variation remains.

Question 20 [Challenge] Answer C

Problem: A company issues unexpected earnings news simultaneously with an export-control announcement. Which conclusion is most defensible from a negative CAR alone?

Solution: Event studies estimate deviations from benchmark expected returns. Simultaneous firm news can generate part or all of the abnormal return; no ordinary return model automatically separates the shocks. Identification needs timing evidence or further design.

Optional game theory

Question 21 [Foundation] Answer B

Problem: In a two-player game, each can select A or B. How many pure action profiles exist?

Solution: Multiply possible actions: two for player 1 times two for player 2 yields four profiles.

Question 22 [Application] Answer B

Problem: Against opponent strategy L, your payoffs from A and B are respectively 1 and 3. Which is your best response?

Solution: Hold the opponent action L fixed and compare your payoffs: $3 > 1$, hence B.

Question 23 [Challenge] Answer A

Problem: A private-type sender has prior odds H:L of $2:1$. An observed signal is half as likely under H as under L. What are the posterior odds?

Solution: Bayesian odds update: posterior odds = prior odds times likelihood ratio, so $2 \times (1/2) = 1$. Thus odds H:L become $1:1$, corresponding to posterior $P(H \mid s) = 1/2$.

Question 24 [Challenge] Answer B

Problem: A receiver earns $4\mu - 1$ from accepting and zero from rejecting, where $\mu = P(H \mid s)$. For what beliefs is accepting optimal?

Solution: Accept is (weakly) optimal when $4\mu - 1 \geq 0$. Solve $4\mu \geq 1$ to obtain $\mu \geq 1/4$. Sequential rationality compares expected continuation payoffs given beliefs.

B.42 Six deeper solution workshops for earlier checkpoints

1. Why is the derivative of a squared residual negative? Let $e_i(b) = y_i - bx_i$. The chain rule says $\frac{d}{db} e_i(b)^2 = 2e_i(b)(-x_i)$. Thus the objective derivative is $-2\sum_i x_i(y_i - bx_i)$, not $+2\sum_i x_i(y_i - bx_i)$. Set to zero and solve only after differentiating correctly.

2. Why can $X^\top X$ be inverted? If X has full column rank, then for nonzero v, $v^\top X^\top X v = \|Xv\|^2 > 0$. A positive-definite matrix has a nonsingular inverse. If a regressor duplicates another one, some nonzero v gives $Xv = 0$; the argument fails and so does ordinary inversion.

3. Where does omitted-variable bias come from? Let $Y = \beta_1 X + \beta_2 Z + u$ with $\mathrm{Cov}(X, u) = 0$. The population slope from regressing Y only on X equals $\mathrm{Cov}(X, Y)/\mathrm{Var}(X)$. Substitute the true model: $\mathrm{Cov}(X, Y) = \beta_1 \mathrm{Var}(X) + \beta_2 \mathrm{Cov}(X, Z)$; dividing through yields $\beta_1 + \beta_2 \mathrm{Cov}(X, Z)/\mathrm{Var}(X)$. That second term is bias relative to β_1. Small residuals do not remove it.

4. Why can two-sided significance change between 5% and 1%? An observed standard-normal statistic $|z| = 2.5$ crosses the 5% two-sided critical value 1.96 but not the 1% value 2.576. Its p-value is $2[1 - \Phi(2.5)] \approx 0.0124$. So reject at 5%, fail to reject at 1%. The shaded tails of the Z table explain the result.

5. Why does difference-in-differences subtract *two changes*? Write treated post-minus-pre as the observed treatment-associated movement plus a background time movement; use controls to approximate that background time movement. The expression $(Y_{T,post} - Y_{T,pre}) - (Y_{C,post} - Y_{C,pre})$ removes the latter only when the counterfactual parallel-trends assumption is credible, with suitable design conditions. Equal initial group levels are not required.

6. Why doesn't "separating" automatically give PBE? A separating proposal needs a signal rule for each type, receiver beliefs after signals, receiver optimality given beliefs, and a demonstration that no sender type would profitably mimic the other. A high-type signal costing 2 may be acceptable to that type, yet if a low type earns 6 by mimicking and only 4 by staying put, the claimed equilibrium fails incentive compatibility. Equilibrium requires checking the entire deviation menu, not just drawing two branches.

CHAPTER C

Formula Reference, Glossary, and Reading Guide

C.1 Algebra, calculus, and matrices

$$\frac{d}{dx}(uv) = u'v + uv', \qquad \frac{d}{dx}g(h(x)) = g'(h(x))h'(x),$$

$$\int x^p dx = \frac{x^{p+1}}{p+1} + C \ (p \neq -1), \qquad \int_a^b f(x)dx = F(b) - F(a),$$

$$(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}, \qquad \nabla_\beta(\beta^\mathsf{T}A\beta) = 2A\beta \quad (A = A^\mathsf{T}).$$

C.2 Probability and statistical inference

$$\begin{aligned}
\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \\
\mathbb{P}(A \mid B) &= \mathbb{P}(A \cap B)/\mathbb{P}(B), \\
\operatorname{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}X)^2, \\
\operatorname{Var}(X+Y) &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y), \\
SE(\bar{X}) &= \sigma/\sqrt{n} \quad \text{(iid sampling)}, \\
CI_{95}(\theta) &\approx \hat{\theta} \pm 1.96\,\widehat{SE}(\hat{\theta}) \quad \text{(normal approximation)}.
\end{aligned}$$

C.3 Econometrics and research design

$$\begin{aligned}
\hat{\beta}_{OLS} &= (X^\mathsf{T}X)^{-1}X^\mathsf{T}y, \\
\operatorname{Var}(\hat{\beta} \mid X) &= \sigma^2(X^\mathsf{T}X)^{-1} \quad \text{(homoskedastic, independent errors)}, \\
\text{OVB} &= \beta_Z \frac{\operatorname{Cov}(X,Z)}{\operatorname{Var}(X)} \quad \text{(simple omitted regressor setting)}, \\
AR_{it} &= R_{it} - \widehat{R}_{it}, \\
CAR_i(a,b) &= \sum_{\tau=a}^{b} AR_{i\tau}, \\
\hat{\delta}_{DiD} &= (\bar{Y}_{T1} - \bar{Y}_{T0}) - (\bar{Y}_{C1} - \bar{Y}_{C0}).
\end{aligned}$$

C.4 A compact glossary

Estimator A rule mapping sample observations to an estimate of a population parameter.

Estimand The precisely defined population quantity the study intends to estimate.

Standard error Estimated standard deviation of an estimator's sampling distribution.

Exogeneity A restriction linking regressors to disturbances or potential outcomes that supports an identification or estimation argument.

Endogeneity A failure of a relevant exogeneity condition; in simple regression discussions often signaled by $\operatorname{Cov}(X, u) \neq 0$.

Heteroskedasticity Conditional disturbance variance differs across observations or covariate values.

Fixed effect Unit- or period-specific intercept absorbing designated stable or common components.

Counterfactual An outcome that would have occurred under a different treatment regime.

Event window A prespecified set of trading sessions around the information event.

Parallel trends Equality of untreated expected changes for treated and comparison groups over relevant periods.

Nash equilibrium A strategy profile in which no player profits from unilateral deviation.

PBE Strategies and beliefs consistent with sequential rationality and Bayesian updating where applicable.

C.5 Further reading, once the chapter mathematics is comfortable

Use the following as companion references rather than prerequisites: Jeffrey M. Wooldridge, *Introductory Econometrics*; James H. Stock and Mark W. Watson, *Introduction to Econometrics*; Joshua D. Angrist and Jörn-Steffen Pischke, *Mostly Harmless Econometrics*; Scott Cunningham, *Causal Inference: The Mixtape*; Martin J. Osborne, *An Introduction to Game Theory*. Match any chosen edition's terminology and assumptions with your course, as definitions and coverage vary. For event-study research, read contemporary field-specific articles and replication material critically, especially about overlapping policy announcements and cross-sectional dependence.

C.6 A closing note about your learning trajectory

Your assessments establish a useful starting point, not a fixed ceiling. Your strongest demonstrated skills are probability intuition and interpretation of basic statistical and event-study

expressions. Your highest-payoff practice is *reconstructing proofs with pencil and paper*: differentiating a loss, checking a matrix inverse, deriving a variance formula, and explicitly stating what a causal comparison assumes. Revisit a missed derivation a week later without peeking, even if all multiple-choice answers were correct.

CHAPTER D

The Formula Atlas: Everything in One Place

Intuition

This is the place to return when your brain says, "I have definitely seen this equation before—where did it go?" Every formula comes with a meaning, an assumption where necessary, and an easy mental check. The atlas is a *reference*, not a replacement for the derivations in the chapters.

Find the formula you need

- Algebra and sums
- Derivatives and optima
- Integration
- Matrix operations
- Probability and Bayes
- Expectations and distributions
- SEs, confidence intervals, tests
- Printable Z-score chart
- OLS estimators
- Bias and robust covariance
- Factors, AR and CAR
- Fixed effects and DiD
- Time series and lags
- Game theory and beliefs

D.1 Symbols and how to read a paper quickly

Symbol	Meaning and reading rule
i, t, n, k	observation or firm index, time index, sample size, and number of regressors (including intercept if fitted).
$\bar{x}, \hat{y}, \hat{\beta}$	sample mean, fitted value, and estimated parameter. A hat represents a sample-based object.
$\mathbb{E}[X], \mathrm{Var}(X), \mathrm{Cov}(X, Y)$	expectation, variance and covariance; these require an explicitly defined distribution.
$\mathbb{P}(A \mid B)$	probability of A in the restricted world where B is known.
$X^{\mathsf{T}}, X^{-1}, \mathrm{rank}(X)$	transpose, inverse where defined, and matrix rank.
ε, u, e	commonly error/disturbance/residual symbols; whether observed and what is assumed depends on the model.

Symbol	Meaning and reading rule
R_{it}, R_{ft}, R_{mt}	firm return, risk-free return and market return, expressed on compatible time intervals.
$D_{it}, \alpha_i, \lambda_t$	treatment indicator, unit effect and time effect in a panel model.
$O_p(\cdot), \xrightarrow{p}, \xrightarrow{d}$	stochastic order, convergence in probability and convergence in distribution.

Fast lane: a useful calculation trick

A formula-reading ritual: (1) inspect indices, (2) inspect units, (3) identify what is observed, estimated or assumed, (4) state the conditions under which the equals sign is valid, and (5) ask what would make the interpretation fail. This five-step habit carries across almost every quantitative paper.

D.2 Algebra, functions, and summation

$$\sum_{i=1}^{n} c = nc, \qquad \sum_i (ax_i + by_i) = a\sum_i x_i + b\sum_i y_i, \tag{D.1}$$

$$\bar{x} = \frac{1}{n}\sum_i x_i, \qquad \sum_i (x_i - \bar{x}) = 0, \tag{D.2}$$

$$S_{xx} = \sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2, \tag{D.3}$$

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y}. \tag{D.4}$$

For positive a, b, and for real expressions where defined,

$$a^m a^n = a^{m+n}, \qquad (a^m)^n = a^{mn}, \tag{D.5}$$

$$\ln(ab) = \ln a + \ln b, \qquad \ln(a^c) = c \ln a, \tag{D.6}$$

$$\ln(e^x) = x, \qquad e^{\ln x} = x \quad (x > 0). \tag{D.7}$$

Useful unit conversion: 0.01 of a return equals one *percentage point* and 100 basis points. A percent change in share *price* is a separate concept.

D.3 Differentiation, differentials, and optimization

$$\frac{d}{dx} c = 0, \qquad \frac{d}{dx} x^a = a x^{a-1}, \tag{D.8}$$

$$(af + bg)' = af' + bg', \qquad (fg)' = f'g + fg', \tag{D.9}$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \qquad \frac{d}{dx} f(g(x)) = f'(g(x))g'(x), \tag{D.10}$$

$$\frac{d}{dx} e^{g(x)} = e^{g(x)} g'(x), \qquad \frac{d}{dx} \ln g(x) = \frac{g'(x)}{g(x)} \quad (g(x) > 0), \tag{D.11}$$

$$\frac{d}{dx} \ln |g(x)| = \frac{g'(x)}{g(x)} \quad (g(x) \neq 0), \qquad \frac{d}{dx} \log_a x = \frac{1}{x \ln a} \quad (a > 0, a \neq 1). \tag{D.12}$$

$$\frac{\partial}{\partial x} x^a y^b = a x^{a-1} y^b, \qquad \nabla f(\boldsymbol{\theta}) = \begin{pmatrix} \partial f/\partial \theta_1 \\ \vdots \\ \partial f/\partial \theta_k \end{pmatrix}, \tag{D.13}$$

interior first-order condition: $$\nabla f(\theta^*) = 0, \tag{D.14}$$

strict local minimum sufficient condition: $$\nabla^2 f(\theta^*) \succ 0, \tag{D.15}$$

strict local maximum sufficient condition: $$\nabla^2 f(\theta^*) \prec 0. \tag{D.16}$$

The Hessian conditions concern a stationary point and require the stated smoothness and definiteness. They do not automatically resolve boundaries or global optima.

Fast lane: a useful calculation trick

For $f(x) = ax^2 + bx + c$ with $a < 0$, the vertex maximum is $x^* = -b/(2a)$. Differentiate to see why, and always check feasible bounds on x.

D.4 Integration and accumulation

$$\int x^a \, dx = \frac{x^{a+1}}{a+1} + C \quad (a \neq -1), \tag{D.17}$$

$$\int \frac{1}{x} \, dx = \ln |x| + C \quad (x \neq 0), \tag{D.18}$$

$$\int e^{ax+b} \, dx = \frac{e^{ax+b}}{a} + C \quad (a \neq 0), \tag{D.19}$$

$$\int_a^b f(x) \, dx = F(b) - F(a), \quad F' = f, \tag{D.20}$$

$$\int f(g(x)) g'(x) \, dx = \int f(u) \, du \quad (u = g(x)), \tag{D.21}$$

$$\int u(x) v'(x) \, dx = u(x)v(x) - \int u'(x) v(x) \, dx. \tag{D.22}$$

Last identity: integration by parts, whenever the integrals and derivatives exist. **Check:** take the derivative of an antiderivative and see if you recover the integrand. A definite integral of a nonnegative function must be nonnegative.

D.5 Vectors and matrices

Let A be $m \times k$, B be $k \times n$, $x \in \mathbb{R}^k$. Then:

$$(AB)_{ij} = \sum_{\ell=1}^{k} a_{i\ell} b_{\ell j}, \tag{D.23}$$

$$(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}, \qquad (A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}, \tag{D.24}$$

$$(A^{-1})^{\mathsf{T}} = (A^{\mathsf{T}})^{-1} \quad \text{when } A^{-1} \text{ exists}, \tag{D.25}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0, \tag{D.26}$$

$$x^{\mathsf{T}} x = \sum_{j} x_j^2 \geq 0, \tag{D.27}$$

$$\nabla_\beta (y - X\beta)^{\mathsf{T}} (y - X\beta) = -2X^{\mathsf{T}}(y - X\beta), \tag{D.28}$$

$$\nabla_\beta (\beta^{\mathsf{T}} A \beta) = (A + A^{\mathsf{T}})\beta. \tag{D.29}$$

For symmetric A, the last derivative equals $2A\beta$. **Shape check:** $(n \times k)^{\mathsf{T}}(n \times 1)$ gives a $k \times 1$ vector. In numerical code, solve linear systems rather than explicitly inverting large matrices.

D.6 Probability, Bayes, and counting

For events with $\mathbb{P}(B) > 0$:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \tag{D.30}$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \tag{D.31}$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \tag{D.32}$$

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B), \tag{D.33}$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}, \tag{D.34}$$

$$\mathbb{P}(B) = \sum_{j} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j) \quad (\{A_j\} \text{ partitions the sample space}), \tag{D.35}$$

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \quad \text{if } A, B \text{ independent}, \tag{D.36}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{D.37}$$

Fast lane: a useful calculation trick

Bayes: *prior* × *likelihood, then normalize.* Binomial counts: the coefficient $\binom{n}{k}$ is the number of possible placements of the k successes.

D.7 Random variables, expectation, and variance

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x) \quad \text{(discrete)}, \tag{D.38}$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx \quad \text{(continuous)}, \tag{D.39}$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \tag{D.40}$$

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2, \tag{D.41}$$

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X), \tag{D.42}$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \tag{D.43}$$

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y), \tag{D.44}$$

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} \quad \text{if variances positive}, \tag{D.45}$$

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Z]], \tag{D.46}$$

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Z)] + \operatorname{Var}(\mathbb{E}[X \mid Z]). \tag{D.47}$$

$$X \sim \text{Bernoulli}(p): \quad \mathbb{E}X = p, \quad \operatorname{Var}(X) = p(1-p), \tag{D.48}$$

$$X \sim \text{Binomial}(n, p): \quad \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \tag{D.49}$$

$$\mathbb{E}X = np, \quad \operatorname{Var}(X) = np(1-p), \tag{D.50}$$

$$X \sim N(\mu, \sigma^2): \quad \frac{X - \mu}{\sigma} \sim N(0, 1) \qquad (\sigma > 0). \tag{D.51}$$

Linearity of expectation does *not* need independence. Adding variances without covariance terms generally *does*.

D.8 Sampling, standard errors, and confidence intervals

For iid observations with finite population variance σ^2:

$$\bar{X} = \frac{1}{n}\sum_i X_i, \qquad \mathbb{E}[\bar{X}] = \mu, \tag{D.52}$$

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \tag{D.53}$$

$$s^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2, \tag{D.54}$$

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \quad \text{(iid CLT, under suitable conditions)}, \tag{D.55}$$

$$z = \frac{\hat{\theta} - \theta_0}{\widehat{SE}(\hat{\theta})}, \tag{D.56}$$

$$CI_{1-\alpha} \approx \hat{\theta} \pm z_{1-\alpha/2}\widehat{SE}(\hat{\theta}) \quad \text{(normal approximation)}, \tag{D.57}$$

$$p_{\text{two-sided}} = 2\big(1 - \Phi(|z|)\big) \quad \text{(two-sided } N(0,1) \text{ test)}. \tag{D.58}$$

Approximate two-sided level	10%	5%	1%
Normal critical $\lvert z\rvert$	1.645	1.960	2.576

Fast lane: a useful calculation trick

Doubling sample size multiplies an iid mean's standard error by $1/\sqrt{2}$; quadrupling it halves the SE. When the data are dependent, derive or estimate an appropriate variance instead of applying the iid formula mechanically.

YOUR TWO-SECOND NAVIGATION TRICK

When a test says $z = 2.5$, use the printed **standard normal chart** to read $\Phi(2.50) \approx 0.9938$, then double the upper tail for a two-sided $p \approx 0.0124$. To explore what happens when z changes continuously, open the bundled `interactive/Z_Score_Lab.html` in a browser (offline).

D.9 Linear regression and OLS derivations

With an intercept, full column rank, and suitable defined moments:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \tag{D.59}$$

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \tag{D.60}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \tag{D.61}$$

$$\hat{\beta}_{\text{origin}} = \frac{\sum_i X_i Y_i}{\sum_i X_i^2} \quad \text{(slope through origin)}, \tag{D.62}$$

$$\hat{\beta} = \arg\min_b (y - Xb)^\mathsf{T}(y - Xb), \tag{D.63}$$

$$\nabla_b L(b) = -2X^\mathsf{T}(y - Xb), \tag{D.64}$$

$$X^\mathsf{T}X\hat{\beta} = X^\mathsf{T}y, \tag{D.65}$$

$$\hat{\beta} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y, \tag{D.66}$$

$$\hat{y} = X\hat{\beta}, \qquad \hat{e} = y - \hat{y}, \tag{D.67}$$

$$X^\mathsf{T}\hat{e} = 0, \tag{D.68}$$

$$\hat{y} = P_X y, \quad P_X = X(X^\mathsf{T}X)^{-1}X^\mathsf{T}. \tag{D.69}$$

The inverse formulas require full column rank; otherwise OLS coefficients may not be unique. The least-squares fitted values remain well defined as a projection even when some coefficient representations are not unique.

D.10 Unbiasedness, omission, heteroskedasticity, and identification

$$\hat{\beta} - \beta = (X^\mathsf{T}X)^{-1}X^\mathsf{T}u, \tag{D.70}$$

$$\mathbb{E}[\hat{\beta} \mid X] = \beta \quad \text{if } \mathbb{E}[u \mid X] = 0 \text{ and rank holds}, \tag{D.71}$$

$$\text{Var}(\hat{\beta} \mid X) = \sigma^2(X^\mathsf{T}X)^{-1} \quad \text{if } \text{Var}(u \mid X) = \sigma^2 I, \tag{D.72}$$

$$\text{OVB}_{\text{short}} = \beta_Z \frac{\text{Cov}(X, Z)}{\text{Var}(X)} \quad \text{(simple linear omitted-regressor case)}, \tag{D.73}$$

$$\widehat{\text{Var}}_{HC0}(\hat{\beta}) = (X^\mathsf{T}X)^{-1}\left(\sum_i x_i x_i^\mathsf{T}\hat{e}_i^2\right)(X^\mathsf{T}X)^{-1}. \tag{D.74}$$

In HC0, x_i is a regressor column vector; robust covariance estimation requires an appropriate sampling/asymptotic justification. HC0 is not an automatic solution for within-firm or same-event dependence, and it does not remove omitted-variable bias. HC1–HC3 and clustering employ other corrections/assumptions.

D.11 Event studies and factor models

$$R_{it}^{\text{simple}} = \frac{P_{it} + D_{it} - P_{i,t-1}}{P_{i,t-1}} \quad \text{(single-period total return if terms align)}, \tag{D.75}$$

$$R_{it}^{\log} = \ln(P_{it}/P_{i,t-1}) \quad \text{(price-only unless adjusted)}, \tag{D.76}$$

$$R_{it} = \alpha_i + \beta_i R_{mt} + u_{it} \quad \text{(market model)}, \tag{D.77}$$

$$R_{it} - R_{ft} = \alpha_i + \beta_{M,i}(R_{mt} - R_{ft}) + s_i SMB_t + h_i HML_t + u_{it}, \tag{D.78}$$

$$AR_{it} = R_{it} - \widehat{\mathbb{E}}[R_{it} \mid \mathcal{I}_{t-}], \tag{D.79}$$

$$CAR_i(a,b) = \sum_{t=a}^{b} AR_{it}, \tag{D.80}$$

$$AAR_t = \frac{1}{N}\sum_i AR_{it}, \tag{D.81}$$

$$CAAR(a,b) = \frac{1}{N}\sum_i CAR_i(a,b), \tag{D.82}$$

$$\operatorname{Var}\left(\sum_t AR_t\right) = \sum_t \operatorname{Var}(AR_t) + 2\sum_{s<t} \operatorname{Cov}(AR_s, AR_t). \tag{D.83}$$

In the abnormal-return expression, $\mathcal{I}_{t-}$ is the information set before the event-time return; in practice, estimated benchmarks depend on a chosen estimation sample and method. The displayed factor model is a specification, not a guarantee that factors completely capture expected returns. Returns, factors, currencies, and calendars must be compatible.

Fast lane: a useful calculation trick

CAR = *sum of abnormal returns*. It is **not** a compounded buy-and-hold return. Variances (and covariances), not standard errors, aggregate under summation.

D.12 Panel data and causal inference

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + u_{it}, \tag{D.84}$$

$$\tilde{Y}_{it} = Y_{it} - \bar{Y}_i, \qquad \tilde{X}_{it} = X_{it} - \bar{X}_i, \tag{D.85}$$

$$\hat{\beta}_{\text{within}} = \frac{\sum_{i,t} \tilde{X}_{it}\tilde{Y}_{it}}{\sum_{i,t} \tilde{X}_{it}^2} \quad \text{(one regressor, unit FE, no additional controls)}, \tag{D.86}$$

$$\hat{\delta}_{DiD} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}), \tag{D.87}$$

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1]. \tag{D.88}$$

The two-period two-group DiD identifies the ATT with suitable conditions, particularly appropriate parallel *untreated* trends, no problematic anticipation, and defensible sampling/composition assumptions. A two-way FE coefficient is not universally a valid treatment-effect summary in complex staggered-adoption settings.

D.13 Time series and lagging

$$\Delta Y_t = Y_t - Y_{t-1}, \tag{D.89}$$

$$Y_t = c + \phi Y_{t-1} + u_t \quad \text{(AR(1) illustration)}, \tag{D.90}$$

$$|\phi| < 1 \quad \text{(stationary AR(1) under standard conditions)}, \tag{D.91}$$

$$\operatorname{Cov}(u_t, u_{t-h}) = 0 \quad (h \neq 0) \tag{D.92}$$

$$\text{if errors are serially uncorrelated.} \tag{D.93}$$

Careful: trading-time $t-1$ is not always the previous calendar date. Financial market closures and policy-announcement timestamps demand explicit alignment.

D.14 Game theory formula shelf: from zero to Bayesian updating

$$\text{Expected payoff } U_i(\sigma) = \sum_a \Pr_\sigma(a) u_i(a), \tag{D.94}$$

$$s_i^* \in \arg\max_{s_i} u_i(s_i, s_{-i}^*) \quad \text{(Nash best response condition)}, \tag{D.95}$$

$$\Pr(\theta \mid s) = \frac{\Pr(s \mid \theta)\Pr(\theta)}{\sum_{\theta'} \Pr(s \mid \theta')\Pr(\theta')} \quad \text{if denominator positive.} \tag{D.96}$$

Subgame perfection requires the strategy profile to induce a Nash equilibrium in every subgame. **Perfect Bayesian Equilibrium** combines sequentially rational strategies with beliefs updated by Bayes' rule wherever possible; additional specifications may be required to discipline off-path beliefs. A Nash equilibrium need not maximize joint payoffs.

D.15 A one-page exam and research checklist

1. **Algebra:** simplify units, domain, and parentheses; substitute to check.
2. **Calculus:** identify the outermost rule; differentiate the answer to check an integral.
3. **Matrices:** verify dimensions first, then rank; test an inverse with $AA^{-1} = I$.
4. **Probability:** specify the conditioning set and check all probabilities lie in $[0, 1]$.
5. **Inference:** distinguish coefficient, standard error, p-value, and effect size.
6. **Regression:** state exogeneity and variance assumptions; separate fit from identification.
7. **Event studies:** define announcement time, estimation window, event window, benchmark, and overlap policy.
8. **Panels:** identify within-firm variation and dependence-aware inference.
9. **DiD:** say precisely what *untreated* counterfactual trends must do.
10. **Game theory:** list players, actions, information, payoffs, and beliefs before solving.

Intuition

A formula sheet is most useful when you can reconstruct the important equations without it. When one looks mysterious, jump to its chapter, do a worked problem, and come back. The equation should feel like a compressed story rather than a spell.

CHAPTER E

Standard Normal Z-Score Tables and Visual Guide

Intuition

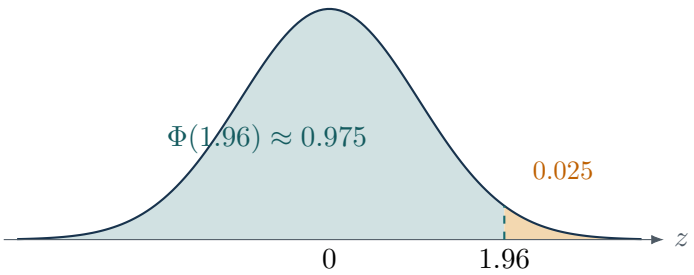
A statistics book without a Z-table feels like a kitchen without a measuring cup. The table tells you the *area under a standard normal curve to the left of a given z-score*. It is not the height of the bell curve, and it is not directly a two-sided p-value. The calculations are trustworthy once you know which area the table reports.

THE THREE OPERATIONS YOU NEED

$$\begin{aligned} \text{Left tail:} \quad & P(Z \le z) = \Phi(z), \\ \text{Right tail:} \quad & P(Z > z) = 1 - \Phi(z), \\ \text{Two-sided:} \quad & P(|Z| \ge |z|) = 2[1 - \Phi(|z|)]. \end{aligned}$$

For negative values, **mirror the number**: $\Phi(-a) = 1 - \Phi(a)$ for $a > 0$. Here $Z \sim N(0, 1)$ and Φ is the standard normal CDF. This chapter prints $z \ge 0$ to two decimals; for negative z, use symmetry.

E.1 Picture the area, not the curve's height



The teal area is the *left-tail cumulative probability*; the orange area is the *right tail*. At $z = 1.96$ they total 1. An upper-tail probability of 0.025 becomes 0.05 when you count both symmetric tails.

E.2 Four lookup examples (with the shortcut)

Worked example

(A) Positive z: $z = 1.23$. Locate row **1.2** and column **.03**: $\Phi(1.23) = 0.8907$.
(B) Negative z: $z = -1.23$. Keep the positive lookup, then take its complement: $\Phi(-1.23) = 1 - 0.8907 = 0.1093$.
(C) Two-sided p: $z = 2.50$. The table gives $\Phi(2.50) = 0.9938$. Thus $p = 2(1-0.9938) \approx 0.0124$. Hence reject at 5%, not at 1%.
(D) Confidence interval: For $\hat{\beta} = 3$ with SE 1.2, the two-sided normal 95% interval is $3 \pm 1.96(1.2) = [0.648, 5.352]$.

E.3 Fast significance landmarks

Two-sided significance	$\lvert z \rvert$ cutoff	Area left $\Phi(z)$	Upper tail
10%	1.645	0.9500	0.0500
5%	1.960	0.9750	0.0250
1%	2.576	0.9950	0.0050

Values 1.645 and 2.576 require interpolation beyond a two-decimal table. This is a *standard normal* reference. For small samples or complex regressions, a t-distribution or other appropriate reference may be needed.

E.4 How the printed chart is organized

Read the row for the first decimal and the column for the hundredth. For example, row 1.2 and column .03 gives $z = 1.23$. The entries below are $\Phi(z) = P(Z \leq z)$ (four decimals), *not* the area between 0 and z.

Z Table A: $0.00 \leq z \leq 1.99$

z	$+.00$	$+.01$	$+.02$	$+.03$	$+.04$	$+.05$	$+.06$	$+.07$	$+.08$	$+.09$
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	**0.8907**	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	**0.9750**	0.9756	0.9761	0.9767

Examples: row 1.2, column .03 gives 0.8907; row 1.9, column .06 gives 0.9750.

Z Table B: $2.00 \leq z \leq 3.99$

z	+.00	+.01	+.02	+.03	+.04	+.05	+.06	+.07	+.08	+.09
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	**0.9938**	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Example: row 2.5, column .00 gives 0.9938. Beyond $z = 3.99$, the left-tail probability rounds to 1.0000 at four decimals.

E.5 A practical way to remember what to subtract

Fast lane: a useful calculation trick

LEFT / RIGHT / BOTH: Left = use the table. Right = subtract the table from 1. Both = double the right tail of the positive absolute z. For $z = -2.50$, symmetry ensures the two-sided result is identical to that for $z = +2.50$.

Semiconductor research connection

Imagine testing whether a hypothetical firm's abnormal return is zero: $\widehat{AR} = -0.03$ and $SE = 0.012$. Then $z = -2.5$, the two-sided p-value is about 0.0124, and the result is statistically significant at 5% but not at 1%. *This does not establish that export controls caused the return.* The event-window design and overlapping news still matter.

GO FURTHER: THE LIVE Z-SCORE LAB

The PDF is designed for reading, searchable lookup, and clickable navigation. For *actual* interactive changes of z, shading, tail direction, and live probabilities, open the standalone `interactive/Z_Score_Lab.html` supplied in the source ZIP (or download the HTML separately). It works in a browser without an internet connection.

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